

Chapter 1

Prevention of Unwanted Proteolysis

Robert J. Beynon

1. Introduction

Inescapably, all cells contain proteases, introducing the possibility that disruption of the tissue can bring together a protease and a protein, with the result that the latter suffers hydrolytic damage. To quote Pringle (1, 2), "Proteolytic artifacts are pervasive, perplexing, persistent and pernicious but with proper precautions, preventable." Autolysis has long been recognized as a problem during protein purification, but methods for its control are still far from perfect. Moreover, there are many circumstances other than during protein purification in which endo- or exopeptidase attack upon a pro-

tein can be at best a frustrating nuisance and at worst an undetected artifact that leads to erroneous conclusions.

The purpose of this chapter is to build upon the excellent papers by Pringle (1,2) and to provide updated information on methods for prevention of unwanted proteolysis. (Few of my colleagues have been impressed by my suggestion that an effective general purpose protease inhibitor is 2M sulfuric acid!) Unfortunately, no global solution to the problem exists, and to a great extent, an *ad hoc* solution depends upon elucidation of some of the properties of the protease(s) that is (are) suspected to be responsible. This chapter may differ from many others in the volume because I cannot present a "method" as much as a philosophy based upon the advice "know thine enemy." Hence, the methods include a sensitive protease assay in addition to a discussion of the handling of protease inhibitors. Largely, I shall restrict the subject matter to proteolytic artifacts that occur *in vitro*. Control of proteolysis of proteins *in vivo* is still difficult, although of increasing importance in studies that aim to express a normal or mutated gene in a foreign cell type.

Critically important but sometimes overlooked is the need to establish that the artifact is truly attributable to proteolysis. Dramatic losses of activity of a protein may be caused by proteases, but may also be caused by, among others, thermal denaturation, dissociation of a cofactor, adsorption onto surfaces, dephosphorylation, or inadvertent modification of the redox status of sulfhydryl/disulfide groups. In a crude homogenate, it may be difficult to assign changes in the properties of a protein to the action of proteases, and often, the only successful approach may require addition of potentially protective protease inhibitors. Limited exoproteolytic attack can combine dramatic changes in the biological

properties of a protein with minimal effects upon physicochemical properties; such modifications are virtually undetectable by analytical methods and are best identified by judicious use of inhibitors in a diagnostic fashion. It is difficult to offer any hard and fast guidelines for circumstances in which proteolytic artifacts are most likely or preventable, but the following should be kept in mind.

- a. Cells differ in the intracellular concentrations of proteases, and unwanted attack upon a protein of interest may be diminished in a cell / tissue in which protease levels are low. In many single-cell systems, mutant strains are available that are defective in the expression of protease-coding genes; this should be considered as an option.
- b. Homogenization of a tissue often allows for complexation of a protease with a pool of (previously isolated) inhibitor. Such enzyme-inhibitor complexes may be dissociated later by inactivation of the inhibitor, however, or the two components may be resolved by a purification step. Thus, proteolysis may manifest itself in later stages of preparation of a protein.
- c. Proteins compete for the active site of proteases, and, therefore, a purification may separate the target protein from contaminants that are protective, particularly if the protease is copurifying with the target protein. Such behavior will manifest itself as proteolytic attack that occurs as the protein become more highly purified.
- d. Many proteins are made more resistant to a variety of denaturing/destabilizing assaults by

complexation with their ligands. Substrate or cofactor-mediated protection of a protein from hydrolytic attack is a common observation, but care should be taken to avoid the alternative of ligand-induced labilization of the target protein.

- e. Proteases are often more stable than their substrates. Thus a denaturing treatment that has the goal of inactivating the protease may have the opposite result of labilizing the target protein to the more resistant protease. This behavior can manifest itself during sample preparation for sodium dodecyl sulfate polyacrylamide gel electrophoresis; the lack of a detectable band may imply that the protease in the preparation was more tolerant than the target protein to the detergent in sample buffer. In these circumstances, the target protein, rendered vulnerable by the detergent, is exposed to a short but effective proteolytic attack.
- f. Proteolytic inactivation of a protein is relatively easy to detect and, thus, to control. Far more difficult to identify and regulate is limited digestion that leads to relatively minor changes in biological properties, but that may introduce microheterogeneity in the final product.

2. Classes of Protease

Proteases can be divided conveniently into endopeptidases and exopeptidases (3). Endopeptidases are further subdivided into five classes based upon the mechanisms that they employ to achieve hydrolysis of the peptide bond. Exopeptidases are classified primarily in terms of the terminal amino acids, dipeptides, or

tripeptides that they remove from the carboxyl or amino termini of a protein (4). I stress that successful control of adventitious proteolysis can best be attained if the reader has some appreciation of the mechanistic class to which the offending protease belongs. Pertinent features are given below.

2.1. Serine Endopeptidases

Serine endopeptidases achieve hydrolysis of the peptide bond by attack upon the carbonyl carbon by a nucleophilic serine residue. The active-site serine residue is a much stronger nucleophile than other serine residues in proteins, and the special properties of this residue are a consequence of electron flow to the serine side chain oxygen atom via a histidine residue. Thus, both the serine and histidine residues are effective targets for many serine endopeptidase inhibitors.

2.2. Cysteine Endopeptidases

Cysteine endopeptidases (previously referred to as thiol proteases) employ a nucleophilic cysteine residue in an analogous fashion to the serine residue above, and again, a histidine residue is implicated in the catalytic mechanism. The special properties of the cysteine residue in particular make it a valuable target for mechanism-based inhibitors.

2.3. Aspartic Endopeptidases

Aspartic endopeptidases (previously known as acid proteases) employ a pair of aspartic residues to labilize and hydrolyze the peptide bond. There are few

effective inhibitors that are directed to aspartic residues, and strategies for inhibition of aspartic endopeptidases usually rely upon tight-binding transition state analogs rather than modification of the active-site residues.

2.4. Metalloendopeptidases

Metalloendopeptidases capitalize upon the electron-withdrawing properties of a metal ion (thus far, always zinc) to weaken the peptide bond. It follows that the target for effective inhibition will be the active site metal ion. It is important to discriminate between true metalloendopeptidases and metal-activated proteases that employ another mechanism, such as the calcium activated cysteine endopeptidase, calpain.

Finally, it is worth noting that a number of the newly discovered endopeptidases do not fall naturally into any one of these classes and may employ totally new hydrolytic mechanisms. These proteases do not respond in a predictable fashion to archetypical class-specific inhibitors.

In most cases, classification of exopeptidases has yet to be formalized in terms of catalytic mechanisms, but it is increasingly apparent that many of them have evolved mechanisms that are similar to the endopeptidases (4). Thus, inhibitor strategies will often be similar for prevention of endopeptidase or exopeptidase attack.

3. Measurement of Endoproteolytic Activity

Protease assays vary from the highly specific, using a substrate deliberately optimized for a single enzyme, to the most general, based on a substrate that is hydrolyzed to a greater or lesser extent by all proteases (5). Assays in the former category are of limited use for

determination of (usually unknown) contaminating proteases. General protease substrates are usually proteins that are intrinsically vulnerable to proteolytic attack, such as casein, denatured proteins, or smaller peptides that do not possess the higher-order structure that confers proteolytic resistance. It is feasible to monitor the hydrolysis of unmodified proteins as the appearance of acid-soluble peptides or amino acids, determined as ultraviolet absorbing material, or capitalizing upon the properties of specific amino acid residues. These assays are relatively insensitive, however. Greater sensitivity and convenience can be attained by labeling the substrate, usually with chromogenic, fluorogenic, or radioactive moieties. Representative labeled substrates are given in Table 1, together with references to the labeling techniques.

Given here is a method for the preparation of a very sensitive radiolabeled substrate; the B-chain of insulin, radioiodinated at its two tyrosine residues (6). Digestion of the substrate releases smaller peptides that are soluble in a trichloroacetic acid concentration that precipitates the undigested substrate. The method is presented in two sections. First, the labeled substrate is not available commercially and, thus, an iodination reaction must be performed. Second, a typical assay based upon this substrate is described, although the conditions (buffer, pH, ionic strength, temperature, substrate concentration, assay volume) can be altered at will provided that certain basic conditions are met.

4. Prevention of Undesired Proteolysis

It is likely that the reader will become aware of endoproteolytic attack much more readily, since the consequences are more tangible. I shall therefore concen-

Table 1
Some Representative General-Purpose Endopeptidase Assays*

Substrate	Product	Detected by	Reference
Azocasein (sulfanilamide-dyed protein)	Dye-peptides	Acid-soluble 340 nm	12, 13
Fluorescamine-casein	Fluorescamine-peptides	Acid-soluble fluorescence 405/475 nm	14, 15
Fluorescein isothiocyanate-casein	FTTC-peptides	Acid-soluble fluorescence 490/525 nm	16
Succinyl casein + leucine aminopeptidase and L-amino acid oxidase	Amino acids > keto acids also generates hydrogen peroxide	Absorbance 550 nm	17
[¹⁴ C]-Collagen	[¹⁴ C]-Peptides	Soluble radioactivity	18
[³ H]-Elastin	[³ H]-Peptides	Acid-soluble radioactivity	19
[¹²⁵ I]-Casein Sepharose	[¹²⁵ I]-Peptides	Released radioactivity	20
Glucosidase-casein-Sepharose	Glucosidase-peptides	Released enzyme activity	21

*This table is far from exhaustive, but illustrates the variety of assay methods that may be employed. It should not prove difficult to identify an assay that has the appropriate degree of sensitivity and is compatible with the specific conditions of the study. Additional assays may be found in ref 22, a valuable reference to mammalian proteases.

trate upon the control of artifactual endopeptidase action. There are in fact two strategies. The first relies upon separation of the substrate from proteases. This may be dependent upon a high resolution and exhaustive purification scheme and may require that the protease activity in addition to the protein of interest is monitored. A more sophisticated approach relies upon a step that is specifically designed as an affinity purification stage for the *protease* in which the unbound material is the fraction of interest. Affinity ligands for proteases abound, but the choice is facilitated if the catalytic mechanism of the protease is known. Such considerations justify a series of experiments to assess the effects of a series of protease inhibitors (such as those in Table 2) upon artifactual proteolysis or upon general proteolytic activity. Good affinity ligands for proteases are provided by proteinaceous inhibitors (Table 3) that are reasonably stable and that can be coupled to insoluble matrices with the minimum of chemistry. An oxirane-derivatized bead, Eupergit C, offers remarkable ease of coupling, reasonable capacities, and good mechanical and flow properties (7), and a method for preparation of an Eupergit C-protein complex is given below. Alternatively, proteinase inhibitors can be successfully coupled to cyanogen-bromide-activated Sepharose (8); some of these are commercially available. Such immobilized inhibitors are also valuable for the removal of proteases after intentional proteolytic attack upon a protein (9,10).

The second strategy permits the protease to remain in the biological sample in an inactive form, attained by judicious addition of inhibitors. The choice of inhibitors is not simple. There is no general-purpose inhibitor that can inhibit all proteases (alpha-2 macroglobulin is closest to this ideal), and thus, a mixture of inhibitors will usually be added. The number of additions is mini-

Table 2
Low Molecular Weight Protease Inhibitors^a

Inhibitor	Specificity	Stock solution/ solvent	Stability	Effective con- centration	Note
<i>Irreversible Inhibitors</i>					
Diosopropylphospho- flouridate (DipF), 184.2 mol. wt.	Serine proteases	200 mM in dry propan-1-ol	Long term at -20°C	0.1–1.0 mM	<i>b</i>
Phenylmethane sul- fonylfluoride (PMSF), 174.2 mol. wt.	Serine proteases	200 mM in dry propan-1-ol or methanol	Long term at -20°C	1.0–10 mM	<i>c</i>
Tosylphenylalanyl- chloro-methyl ketone (TPCK), 351.9 mol. wt	Chymotrypsin-like serine proteases	10 mM in methanol	Stable below pH 7.5	0.1 mM	
Tosyllysylchloromethyl ketone (TLCK), 369.3 mol. wt	Trypsin-like serine proteases	10 mM in aqueous solution	Prepared fresh as needed	0.1 mM	
Iodoacetic acid, 208.0 mol. wt.	Cysteine proteases	100 mM in aqueous solution	Decomposes slowly	0.1–1.0 mM	
E-64, 357.4 mol. wt.	Most cysteine pro- teases	1 mM in aqueous solution	At least 1 mo at -20°C	10 µM	
<i>Reversible Inhibitors</i>					
Leupeptin, 426.6 mol. wt	Trypsin-like serine proteases, cysteine proteases	1mg/mL in aqueous solution	At least 1 wk at -20°C	25 µg/mL	<i>d</i>

Chymostatin, 582.7 mol. wt.	Chymotrypsin-like serine proteases, cysteine proteases	1 mg/mL in DMSO	1 mo at -20°C	25 µg/mL
Elastatinal, 512.6 mol wt.	Elastase-like serine proteases	1 mg/mL in aqueous solution	At least 1 mo at -20°C	1 µg/mL
Pepstatin, 685.9 mol wt.	Aspartic proteases	1 mg/mL in DMSO	Long term at -20°C	0.1 ng/mL
Phosphoramidon, 543.6 mol wt	Metalloendopeptidases	1 mg/mL in water	At least 1 mo at -20°C	25 µg/mL
1,10-Phenanthroline, 198.2 mol wt	Metalloendopeptidases	200 mM in methanol	Long term at 4°C	1-10 mM

*Listed here are the protease inhibitors that are most commonly used to suppress protease action. All of them are commercially available. Suggested working concentrations are provided, together with concentrations that have proved effective toward true members of each class of protease. Further information on many of these inhibitors can be found in refs 22-25

[†]DipF is highly toxic and volatile, and, in my experience, vials have a tendency to develop a positive pressure. Stringent precautions should be taken with this reagent. Once dissolved in propan-1-ol, it can be stored at -20°C in tightly capped vials. Preparations of DipF contain a contaminant that is able to inhibit cysteine proteases and, if necessary, this should be inactivated before use (5)

*PMSF appears to be a less potent inhibitor of serine proteases than DipF, often giving partial inhibition under conditions in which the latter would inhibit completely. Higher concentrations of PMSF are therefore routinely used. Note that both DipF and PMSF are inactivated in aqueous solutions and are only active for a limited time after addition to the sample.

[†]Leupeptin, chymostatin, and, indeed, many protease inhibitors are themselves peptides and are susceptible to peptidase action. Mammalian tissue preparations contain peptidases that inactivate leupeptin and chymostatin (26-28). The reader should be alert to this possibility.

[†]1,10-Phenanthroline has a strong ultraviolet absorbance that can interfere with spectrophotometric measurements.

Table 3
Proteinaceous Protease Suitable for Coupling to an Insoluble Matrix^a

Inhibitor	Target proteases	Reference (to inhibitor)
Soybean trypsin inhibitor	Trypsin and chymotrypsin-like serine proteases	29
Lima bean trypsin inhibitor	Trypsin and chymotrypsin-like serine proteases	29
Egg white ovomucoid	Trypsin and chymotrypsin-like serine proteases (specificity depends on species)	29
Egg white ovostatin	Most metallo-proteinases, including collagenases and gelatinases	30
Egg white cystatin	Cysteine proteases, but not calpain	31
Alpha-2macroglobulin	Virtually all endopeptidases except those with a very restricted substrate specificity	32
Alpha-1 proteinase inhibitor	Trypsin-like serine proteases	32
Pancreatic trypsin inhibitor (aprotinin)	Mammalian serine proteases	33

^aAll of these inhibitors can be coupled to insoluble matrices and used as chromatographic media to separate proteases from samples. It is advisable to determine the binding capacity of the matrix by titration with a pure protease. Capacities of 1–5 mg protease/mL of matrix are readily attainable.

mized by knowledge of the inhibitor-sensitivity profile of the sample. An additional complication may arise if the protease inhibitor also affects the biological activity of the protein of interest, a problem that is particularly prevalent when chelators or thiols/thiol-reactive agents are involved.

Irreversible inhibitors have the advantage that they need only be added at the early stages of the experiment. Reversible inhibitors must be maintained at inhibitory concentrations throughout the experiment. The best criterion for effective inhibition of problematical proteases is the disappearance of the problem, but it may be advisable to check for the presence of active proteases as well. The inhibitors listed in Table 2 (and to some extent, those in Table 3) can be added to samples in appropriate combinations at the recommended concentrations. I stress, however, that the susceptibility of proteases to each of the inhibitors extends over several orders of magnitude, and optimal protection may be attained at considerably higher concentrations.

5. Materials

5.1. Method 1. Preparation of Radioiodinated Insulin B-Chain

The following materials are used for Method 1.

1. 0.5 M Sodium phosphate buffer, pH 7.5.
2. Insulin B-chain (performic acid oxidized or reduced/carboxymethylated).
3. Na^[125]I (100 mCi/mL; 15 mCi/μg).
4. 0.15M KI (25 mg/mL in water).
5. 0.44M Chloramine T (100 mg/mL) prepared just before use.

6. 0.32M Sodium metabisulfite (60 mg/mL).
7. Sephadex G-25™ (medium) column, $V_t = 10-15$ mL, equilibrated with:
8. 6.0 mM KI (1 mg/mL) adjusted with NaOH to pH 9-10.
9. 2% (w/v) Casein.
10. 25% (w/v) Trichloroacetic acid.

5.2. Method 2. Protease Assay Using 125 I-Insulin B-Chain

The following materials are used for Method 2.

1. 0.5 mg/mL 125 I-Insulin B-chain.
2. 0.04M Hepes/0.28M NaCl, pH 7.5.
3. 2% (w/v) Casein.
4. 25% (w/v) Trichloroacetic acid.
5. Siliconized reaction tubes.
6. 1.5-mL Microcentrifuge tubes.

5.3. Method 3. Preparation of Immobilized Protease Inhibitors

The following materials are used for Method 3.

1. Eupergit C™ oxirane acrylic beads (Rohm Pharma GmbH, Weiterstadt, FDR).
2. 1.0M Potassium phosphate buffer, pH 7.5.
3. 0.1M Potassium phosphate buffer, pH 7.5.
4. 2-Mercaptoethanol.
5. Protease inhibitor.

6. Methods

6.1. Preparation of Radioiodinated Insulin B-Chain

The following applies to Method 1.

1. Dissolve 2–3 mg insulin B-chain in 0.25 mL of phosphate buffer. The insulin B-chain is not particularly soluble, and it may be preferable to make the solution slightly alkaline with NaOH to dissolve the peptide fully and then readjust the pH to 7.5 (narrow range pH papers are adequate for measuring the pH of samples).
2. Add 5 μL of 0.15M KI (25 mg/mL) to lower the specific radioactivity of the isotope.
3. Add 2 μL of $\text{Na}^{[125]}\text{I}$ (approximately 200 μCi /7.4 MBq).
4. Add 50 μL of chloramine T and mix completely. Allow reaction to proceed at room temperature for 15 min.
5. Add 20 μL of sodium metabisulfite to terminate the reaction.
6. Add 20 μL of 0.15M KI to act as carrier during chromatography.
7. Apply whole sample to the Sephadex G-25 column and elute with 6.0 mM KI, pH 9–10. Manually collect fractions in disposable tubes marked at approximately 1 mL.
8. For each fraction, transfer 5 μL to 1.0 mL of water. Determine the radioactivity in 100 μL of each dilution by gamma counting. This gives the total radioactivity in 0.5 μL of each original fraction.
9. Remove 100 μL of each dilution and add to 100 μL of 2% (w/v) casein as a coprecipitant. Precipitate

the sample with 200 μL of 25% (w/v) trichloroacetic acid and sediment the precipitated material by centrifugation at 10,000g for 2 min. Determine the radioactivity in 100 μL of the supernatant. This gives the acid-precipitable radioactivity in 0.125 μL of the original fraction.

10. Determine the total radioactivity and acid-precipitable radioactivity of each fraction, and pool those that are more than 95% precipitable (*see* Fig. 1).
11. Determine the concentration of insulin B-chain by measurement of the A_{280} nm ($E1\% = 9.92$). Use an acrylic cuvet; it is sufficiently transparent at 280 nm and is disposable.
12. Dilute the solution to a final concentration of 0.5 mg/mL, adjust the pH to near-neutrality, and store aliquots of the solution at -20°C in a lead-lined box.

6.1.1. Notes on Method 1

1. Unlike many iodinations (e.g., ref. 11), the goal here is not maximal specific activity, but the generation of the maximal amount of substrate labeled to a workable specific activity. Thus, the iodinations are performed in relatively high volumes at high protein concentrations.
2. As supplied by the manufacturer, $\text{Na}^{[125]}\text{I}$ is at a very high specific activity and potentially hazardous. Small microaerosols or droplets caused by careless transfers are very radioactive and can potentially be inhaled. Therefore, conduct all iodinations in a fume hood that has been certified for radioactivity work. Isotopes of iodine have been shown to pass

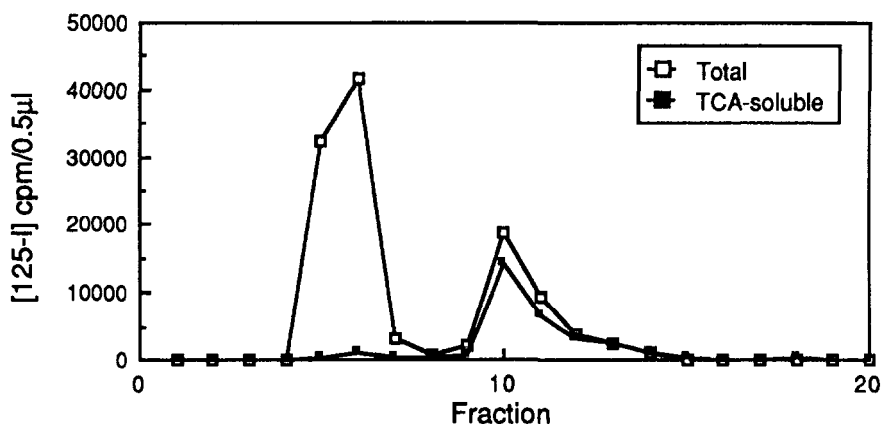


Fig. 1. Separation of radioiodinated insulin B-chain from unincorporated sodium iodide. Insulin B-chain (2–3 mg) was iodinated with 200 μ Ci of sodium iodide and subsequently applied to a 10 mL column of Sephadex G-25. The total and acid soluble radioactivity was determined for an aliquot of each fraction. Fractions 5 and 6 were pooled and used as substrate.

across disposable gloves, and we routinely wear two pairs of gloves when dispensing the iodine; the outer pair is discarded immediately after the radioiodine is added.

3. When the carrier iodide is added (step 6) the solution may turn pale yellow/brown, indicating elemental iodine. If this happens, add a further 20 μ L of sodium metabisulfite to retain the isotope in solution as the less hazardous iodide ion.

6.2. Protease Assay Using [¹²⁵I]-Insulin B-Chain

The following applies to Method 2.

1. For n assays, prepare buffered substrate solution by adding $25n$ – $250n$ μL of $2\times$ buffer.
2. Set up incubation by mixing, in a siliconized tube, 275 μL of buffered substrate with 225 μL of enzyme preparation, diluted with water if required. Start the incubation at the desired temperature.
3. Immediately remove 100 μL of the reaction mixture and add to 100 μL of casein. Add 200 μL of trichloroacetic acid to stop the reaction, and precipitate undigested material with the assistance of the casein coprecipitant.
4. Repeat step 3 at three further time intervals.
5. At the end of the reaction, remove 50 μL of the reaction mixture, add to 100 μL of water, and count directly, without precipitation (100% sample). This figure gives the specific radioactivity of the sample on that day and permits the calculation of extent of solubilization of insulin B-chain = (cpm in supernatant from steps 3 and 4)/(cpm in 100% sample).
6. Calculate the slope of the line (preferable by linear regression) and ensure that the time course is linear. Calculate the rate of solubilization of the insulin B-chain. Remember that the specific radioactivity of the substrate will fall over a few weeks and, thus, measurement of the rate of release of radioactivity is less acceptable.

6.2.1. Notes on Method 2

1. No strict guidelines can be given as to the choice of reaction buffer, time course of assay, need for cofac-

tors, or amount of biological material to be assayed. This will depend upon the particular samples and circumstances. I recommend that some "ranging" assays be performed first to establish reasonable working conditions.

2. The time course of the assay is linear provided that the extent of the solubilization remains below approximately 10%. More extensive digestion will usually manifest itself as a downward curvature in the process curve. For this reason, I recommend a time course assay as superior to a single sampled assay that gives no indication of the linearity of the digestion.
3. Exopeptidases are not very efficient at solubilizing radioactivity in this assay, but they can potentiate activity by attacking the new termini that are generated by endopeptidases.
4. The substrate concentration will undoubtedly be well below the Michaelis constant for most proteases. This too may contribute to nonlinearity if the substrate is depleted significantly. The low concentration probably contributes to the tendency of the substrate to adhere to glass and plastic surfaces. It is advisable to perform the digestion reaction in siliconized, well-washed glass tubes.
5. Always include a protease-free blank to determine the intrinsic precipitability of that particular batch of the substrate (zero time) and the background rate of release of material into the trichloroacetic acid-soluble fraction.
6. With some proteases we have noticed that extensive digestion leads ultimately to a decrease in acid-soluble radioactivity. This may be caused by the formation of short peptides that recover the property of insolubility in acid solution and is a further

justification for minimizing the extent of hydrolysis of the substrate.

7. Most of the radioactivity will be in the precipitated material—this can be disposed of as solid waste.

6.3. Preparation of Immobilized Protease Inhibitors

The following applies to Method 3.

1. Dissolve 10–50 mg of the protease inhibitor in 4.0 mL of 1.0M phosphate buffer.
2. Add the inhibitor solution to 1 g of Eupergit C dry beads and react for 16–72 h at 4°C.
3. Wash the inhibitor-Eupergit beads with five washes of 100 mL of 0.1M phosphate buffer.
4. Add 4.0 mL of 5% (v/v) aqueous 2-mercaptoethanol to block unreacted groups. Leave for 16 h at 4°C.
5. Wash the beads as in step 3.
6. Determine the protease binding capacity of a sample of the beads by titration with a pure protease.
7. Store the inhibitor-Eupergit at 4°C in buffer containing a suitable bacteriostatic agent.

6.3.1. Notes on Method 3

1. The manufacturers recommend 20–25°C for coupling, but this may compromise the stability of the inhibitor. Make sure that the inhibitor preparation does not contain any ammonium salts, since these will react rapidly with the oxirane groups of the matrix. The coupling reaction is best carried out in a wide-necked flask. The manufacturer's literature states that mixing should not be necessary, but we routinely place the mixture on a slow bottle roller.

There is very little free buffer at this ratio of beads-to-protein solution.

2. Blocking of unreacted groups is not always necessary, but is a relatively straightforward step to perform. Mercaptoethanol has the advantage that it does not disturb the hydrophilicity and electro-neutrality of the matrix.
3. The water regain value of the beads is about 2.5 mL/g dry beads.

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Chapter 2

The Bradford Method for Protein Quantitation

***John B. W. Hammond
and Nicholas J. Kruger***

1. Introduction

A rapid and accurate method for the estimation of protein concentration is essential in many fields of protein study. The Lowry method (Chapter 1 in vol. 1 of this series) has been widely used, but is susceptible to interference from a wide range of compounds commonly present in biological extracts. Although interference can be avoided by trichloroacetic acid precipitation of the protein prior to assay, this lengthens the procedure.

The Bradford method (1), by contrast, is not subject to interference by most common reagents or nonprotein

substances likely to be present in biological samples (*see* Note 3 in section 4). It is simpler, faster, and more sensitive than the Lowry method. The assay relies on the binding of the dye Coomassie Blue G250 to the protein molecule. The cationic form of the dye, which predominates in the acidic assay reagent solution, has a λ_{max} of 470 nm. The dye binds to protein as the anionic form, which has a λ_{max} of 595 nm (2). Thus the amount of dye bound to the protein can be quantitated by measuring the absorbance of the solution at 595 nm.

The dye appears to bind most readily to arginine residues (but not to the free amino acid) (2). This can lead to variation in response to different proteins, which is the main drawback of the method. The original Bradford assay shows large variation in response between different proteins, with the common protein standard bovine serum albumin at the higher end of the range (3,7–9). For this reason, the standards should be carefully chosen if absolute protein concentrations are required or pure proteins are being assayed (*see* Note 1 in section 4). Several modifications to the method have been proposed to overcome this problem (*see* Note 2 of section 4). The original method still appears to be the most widely used, however, and is described here. A microassay protocol is also described.

2. Materials

1. Reagent: The assay reagent is made by dissolving 100 mg of Coomassie Blue G250 in 50 mL of 95% ethanol. The solution is then mixed with 100 mL of 85% phosphoric acid and made up to 1 L with distilled water.

The reagent should be filtered through Whatman No. 1 filter paper before storage in an amber bottle at room temperature. It is stable for several weeks, but slow precipitation of the dye will occur, so filtration of the stored reagent is necessary before use.

2. Protein standards (*see* Note 1 in section 4). Ovalbumin at a concentration of 1 mg/mL (100 μ g/mL for the microassay) in distilled water is used as a stock solution. This should be stored frozen.
3. Plastic and glassware used in the assay should be absolutely clean and detergent-free. Quartz (silica) spectrophotometer cuvetts should not be used, since the dye binds to this material. Traces of dye bound to glassware or plastic can be removed with ethanol or detergent solution.

3. Method

3.1. Standard Method

1. Pipet between 10 and 100 μ g of protein in 100 μ L total volume into a test tube. If the approximate sample concentration is unknown, assay a range of dilutions (1, 1/10, 1/100, 1/1000). Duplicate each sample.
2. For the calibration curve, pipet duplicate volumes of 10, 20, 40, 70, and 100 μ L of 1 mg/mL ovalbumin stock solution into test tubes, and make each up to 100 μ L with distilled water. Pipet 100 μ L of distilled water into a further tube for the reagent blank.
3. Add 5 mL of protein reagent to each tube and mix well by inversion or gentle vortexing. Avoid foaming, which will lead to poor reproducibility.

4. Measure the A_{595} of the samples and standards against the reagent blank after at least 2 min and within 1 h of mixing. The 100 μg standard should give an A_{595} value of about 0.2. Since the calibration curve is not linear, it should be determined and plotted for each set of assays.

3.2. Microassay

This form of the assay is more sensitive to protein (and interfering substances) and is thus useful when the amount of the unknown protein is limited.

1. Pipet duplicate samples containing between 1 and 10 μg in a total volume of 100 μL into test tubes or Eppendorf tubes. If the approximate sample concentration is unknown, assay a range of dilutions (1, 1/10, 1/100, 1/1000).
2. For the calibration curve, pipet duplicate volumes of 10, 20, 40, 70, and 100 μL of 100 $\mu\text{g}/\text{mL}$ ovalbumin stock into test tubes, and make up to 100 μL . Pipet 100 μL of distilled water into a tube for the reagent blank.
3. Add 1 mL of protein reagent, mix, and measure the A_{595} as in step 4 of the standard method. The 10 μg standard should give an A_{595} value of about 0.15.

4. Notes

1. The assay technique described here is subject to variation in sensitivity between individual proteins (*see* Table 1). Valid comparisons can be made

between protein content of solutions of similar composition. For critical determination of the concentration of solutions containing one or a few proteins, the initial use of a second method, e.g., Lowry, to check the relative dye binding capacities of the standard protein and the unknown in the Bradford assay, is recommended. Ovalbumin is a more suitable general standard than the commonly used bovine serum albumin since its dye binding capacity is closer to the mean of those proteins that have been compared (Table 1).

2. A number of modifications have been suggested that reduce the variability between proteins (3,4). These rely either on increasing the dye content of the solution or reducing the acid content. They appear to be most effective when a relatively pure form of the dye is used (3). Suppliers' figures for dye purity may not relate well to Coomassie Blue G250 content (5), so caution is necessary in choosing a suitable dye source for these methods.
3. Although the Bradford assay is relatively free from interference by most commonly used biochemical reagents, a few may cause significant alteration in absorbance (2,6). The materials that are most likely to cause problems in biological extracts are detergents and ampholytes (Table 2). These should be removed from the sample solution, for example, by gel filtration or dialysis. Alternatively, they should be included in the reagent blank and calibration standards at the same concentration as in the sample, since such interference may cause a change in the shape of the calibration curve. The presence of a base in the assay increases absorbance by shifting the equilibrium of the free dye toward the anionic form. This may also occur with concentrated basic

buffers (2). Guanidine hydrochloride and sodium ascorbate compete with the dye for protein, leading to underestimation of the protein content (2).

Table 1

Response of a Range of Proteins in the Bradford Assay^a

Protein	Relative A_{595}
Cytochrome c	128
Bovine serum albumin	100
Histone H1	10
H2B	102
H4	89
Carbonic anhydrase	83
Ovalbumin	64
Chymotrypsinogen A	50
Lysozyme	40
Trypsin	24
Pepsin	13
RNAse	12
Immunoglobulin G	10
Gelatin	1

^aFor each protein, the A_{595} is expressed relative to that of the same concentration of bovine serum albumin (data from refs. 3 and 7-9).

Table 2

Effects of Common Reagents and Interfering Compounds on the A_{595} in the Bradford Assay^a

Compound	A_{595}
1M KCl	0.000
2M Tris	0.026
0.1M HEPES	0.010
0.1M EDTA	0.004
1M (NH ₄) ₂ SO ₄	0.000
99% Glycerol	0.012
1M 2-Mercaptoethanol	0.004
1M Dithiothreitol	0.002
1M Sucrose	0.013
5% Phenol	0.046
1% Triton X-100	0.590
1% Sodium dodecyl sulfate	0.495
10 mM Quercetin	0.676
10 mM Apigenin	0.968
10 mM Flavone	0.052
1% Ampholytes (pH, 3.5–10.0)	0.089

^aData from refs. 1, 2, and 6.

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Chapter 3

Amino Acid Analysis by Precolumn Derivatization

E. L. V. Harris

1. Introduction

Amino acid analysis is of central importance to many areas of biological and medical research as a tool for characterization and quantitation of both free amino acids present in biological fluids, beverages, or food and proteins/peptides.

Spackman et al., in 1958 (1), described a method for separation of amino acids by ion exchange chromatography and postcolumn derivatization with ninhydrin. Since then improvements in instrumentation and chromatography matrices have increased both the sensitivity and speed of the technique. In recent years techniques using precolumn derivatization prior to separation by reverse-phase HPLC have become widely

used as alternatives to the traditional post-column derivatization techniques.

Precolumn derivatization offers several advantages:

- Instrumentation is not dedicated to amino acid analysis;
- increased sensitivity, <10 pmol/amino acid, compared to ≥ 500 pmol; and
- decreased analysis time, 20–45 min/run compared to ≥ 1 .

Several derivatization reagents have been used, as shown in Table 1.

The most widely used reagent has been *o*-phthalaldehyde (OPA)/thiol (2,3). This reagent suffers from two disadvantages; the derivatives have a short half-life, and secondary amino acids such as proline are not detected. In order to overcome the instability of the derivatives, autosamplers capable of mixing reagent and sample immediately prior to injection onto the HPLC column are used. Techniques have been described to enable detection of secondary amino acids by oxidation with chloramine T or hypochlorite prior to derivatization with OPA (4,5); these are not widely used, however, because of technical difficulties.

Phenylisothiocyanate (PITC) (6,7) and 9-fluorenylmethyl chloroformate (FMOC) (8) as precolumn derivatization reagents are gaining popularity, since they both allow for determination of secondary amino acids. Automation of derivatization by either of these reagents is not easy, since drying *in vacuo* or solvent extraction is required to remove excess reagent and byproducts. Waters Associates however, sell instrumentation, reagents, solvents, and column for PITC derivatization

Table 1

Comparison of Properties of Amino Acid Derivatives Suitable for Amino Acid Analysis by Precolumn Derivatization

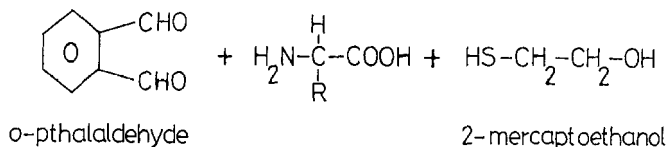
Derivatization reagent	<i>o</i> -Phthalaldehyde (OPA)	Phenylisothiocyanate (PITC)	9-Fluoroenylmethyl chloroformate (FMOC)	Dansyl or dabsyl chloride
Detection of secondary amino acids	No	Yes	Yes	Yes
Stability of derivatives	Poor <10 min	Moderate 1–2 d	Good	Moderate 1–2 d
Detection limits	0.5 pmol	1 pmol	0.5 pmol	10 pmol (dansyl) 0.5 pmol (dabsyl)
Detection method	Fluorescence	UV absorption	Fluorescence* or UV absorption	Fluorescence* or UV absorption
Interference by excess reagent or byproducts	No	Yes	Yes	Yes
Method for removing excess reagent or byproducts	—	Drying <i>in vacuo</i>	Extraction with pentane	None
Ease of automation	Good	Poor/moderate	Moderate	Poor

*Fluorescence is more sensitive than absorption.

under the name Pico-Tag™, and Varian Associates sell a package for FMOC derivatization under the name Amino-Tag™. Both of these packages all for automation of amino acid analysis.

In this chapter the techniques using OPA derivatization and PITC derivatization are described. Figure 1 shows the reactions involved in the two techniques.

a)



b)

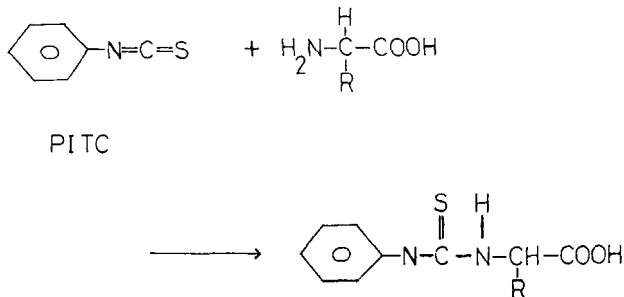


Fig. 1. (a) Reaction of *o*-phthalaldehyde (OPA) and a thiol (2-mercaptoethanol) with an amino acid (b) Reaction of phenylisothiocyanate (PITC) with an amino acid to yield the phenylthiocarbamoyl (PTC) derivative.

2. Materials

2.1. Hydrolysis

1. Constant boiling hydrochloric acid (e.g., Pierce Chemicals) and crystalline phenol.
2. Hydrolysis tubes: Soda glass tubes (about 5 x 0.5 cm) previously treated by baking at 450°C to destroy contaminating proteins/amino acids. Oxygen/butane flame to seal these tubes and diamond tip pen for opening them after hydrolysis.
3. 105°C oven.
4. Freeze-drier or centrifugal concentrator (e.g., Savant Speedvac concentrator) and vacuum pump (e.g., Edwards, to give a vacuum of 50 mtorr; water pumps are not suitable).
5. Amino acid standard. Dissolve the following amino acids together at 5 μ M each in HPLC grade water: aspartic acid, glutamic acid, serine, glycine, histidine, arginine, threonine, alanine, proline, tyrosine, valine, methionine, cysteine, isoleucine, leucine, phenylalanine, and lysine.

2.2. PITC Derivatization

1. Small volume, pointed bottom vials with caps, e.g., 0.5 mL polypropylene centrifuge vials from Sarstedt.
2. Ethanol:water:triethylamine (TEA) (2:2:1) (solution 1). Ethanol, TEA, and water must be HPLC grade.
3. Ethanol:TEA:water:phenylisothiocyanate (PITC) (7:1:1:1) (derivatization reagent). This reagent should be made fresh daily. High-purity PITC

should be used and stored at -20°C to prevent formation of breakdown products.

4. HPLC equipment to run a binary solvent gradient including column-heater, autosampler, detector to monitor for absorbance at 254 nm, and an integrator.
5. C18 reverse-phase column with 3 or 4 μm packing (e.g., Millipore Novapak (15 cm $\times$ 4 mm id) or LKB 3 μm SuperPac (10 cm $\times$ 4 mm id)).
6. 0.14M Sodium acetate, 0.05% TEA titrated to pH 6.35 with glacial acetic acid (buffer A).
7. 60% Acetonitrile in water (buffer B).
8. HPLC grade acetonitrile and water should be used for both buffers A and B. Before use, filter both buffers through 0.2 μm filters (cellulose acetate and nitrate filters for buffer A and polyvinylidene difluoride filters for buffer B) and degas by bubbling helium through for 5 min.

2.3. OPA Derivatization

1. 400 mM Sodium borate buffer. Dissolve 2.47 g of boric acid in HPLC grade water, adjust the pH to 9.5 with sodium hydroxide, and adjust the volume to 100 mL with water.
2. OPA/MCE reagent. Dissolve 50 mg of OPA (e.g., Sepramar grade, BDH Chemicals Ltd.) in 1 mL of methanol (HPLC grade) and dilute to 10 mL with sodium borate buffer. Add 40 μL of 2-mercaptoethanol. Store this reagent in the dark at room temperature for up to 1 wk, adding an additional 4 μL of 2-mercaptoethanol every 3 d. Dilute this reagent 1-in-10 in sodium borate buffer prior to use.
3. Iodoacetic acid reagent. Dissolve 0.74 g of iodoacetic acid and 0.62 g of boric acid in approximately

- 50 mL of HPLC grade water. Adjust the pH to 9.5 with sodium hydroxide and make up to 100 mL with water.
4. HPLC equipment to run a binary solvent gradient, including autosampler, fluorescence detector with excitation wavelength set at 335 nm and emission wavelength at 455 nm, and an integrator.
 5. C-8 or C-18 reverse-phase HPLC column with 5 μ m or smaller packing (e.g., Rainin Microsorb, 15 cm $\times$ 4.5 mm id, or LKB SuperPac 3 μ m, 10 cm $\times$ 4.5 mm id).
 6. 54.3 mM propionic acid/76 mM disodium hydrogen phosphate adjusted to pH 6.5 with sodium hydroxide. Buffer A is made by mixing the propionate/phosphate buffer with acetonitrile in the ratio 1840:160 (v/v), respectively.
 7. Buffer B is water:methanol:acetonitrile:dimethyl sulfoxide (840:600:500:60, v/v/v/v). HPLC grade solvents and water should be used for both buffers A and B. Prior to use, filter both buffers through 0.2 μ m filters and degas by bubbling with helium for 5 min.

3. Methods

3.1. Hydrolysis

1. Place the protein samples to be analyzed (0.5–10 μ g) in hydrolysis tubes and dry down in the centrifugal concentrator or by freeze-drying.
2. When the samples are dry, add 100 μ L of constant boiling HCL and a small crystal of phenol (to give a final concentration of approximately 1%).

3. Seal the tubes by heating in the oxygen/butane flame about 1–1.5 cm from the open end of the tube and twist and draw out the end until sealed.
4. Place in a 105°C oven for 24 h.
5. Crack open the tubes by prescoring with the diamond tip pen.
6. Dry down the samples *in vacuo*, preferably in a centrifugal concentrator to prevent "bumping."

3.2. PITC Derivatization

1. For standards, dry 10 μL of the amino acid mix into a vial.
2. Add 10–20 μL of solution 1 to each of the samples and standards, transfer the samples to vials, and dry *in vacuo*. This step ensures that the pH of the samples is raised to the optimum for derivatization.
3. Add 20 μL of derivatization reagent, cap the vials, and incubate for 20 min at room temperature.
4. Uncap the vials and thoroughly dry the samples and standards *in vacuo*. This should preferably be done overnight to ensure complete removal of excess reagent.

3.3. PTC Separation

1. Redissolve the derivatized samples and standards in a suitable volume of buffer A to allow injection of 5–50 μL onto the HPLC column.
2. The HPLC column should be equilibrated in 90% buffer A:10% buffer B at a flow rate of 1 mL/min at 35°C.
3. The derivatized amino acids are eluted from the HPLC column by a gradient of increasing concen-

tration of buffer B. The exact gradient required will differ from column type to column type; the gradient used with the LKB Super Pac column is shown in Fig. 2.

4. The eluate is monitored for absorbance at 254 nm, and the peaks integrated and quantified by comparison to amino acid standards.

3.4. OPA Derivatization and Separation of Derivatives

1. For standards, dilute the amino acid standard 1:10 in HPLC grade water. Redissolve hydrolysates in 50 μ L of HPLC grade water.
2. Equilibrate the HPLC column in buffer A at a flow rate of 1.5 mL/min for 5 μ m packing, or 1 mL/min for 3 μ m packing.
3. (a) *Manual derivatization.* Mix 20 μ L of sample or standard with 90 μ L of iodoacetic acid reagent. Add 90 μ L of OPA/MCE reagent, mix, and inject onto the column exactly 2 min after addition of the OPA/MCE reagent, or
(b) *Automated derivatization.* Program the auto-sampler to mix 20 μ L of sample or standard with 90 μ L of iodoacetic acid reagent followed by 90 μ L of OPA/MCE reagent. The mixing may be done in a separate vial by bubbling air through the solution or may be mixed in-line by passing through a small column of glass beads. The iodoacetic acid reagent and OPA/MCE reagent may be taken from the same vials for a series of samples and standards.
4. The OPA derivatives are eluted by a gradient of buffer B. The exact gradient will vary from column type to column type; that used with the Rainin Microsorb is shown in Fig. 3.

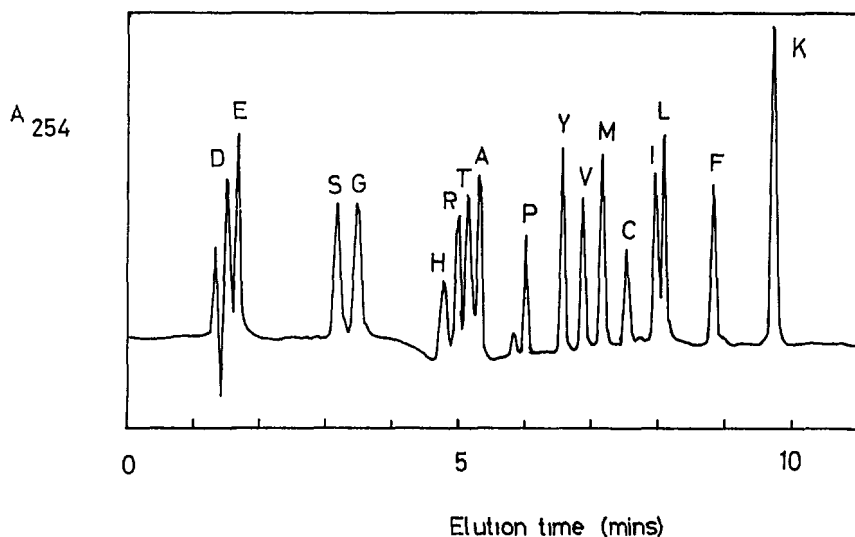


Fig. 2. Amino acid analysis of a standard mix of amino acids using precolumn derivatization with PITC. The column used is a SuperPac 3 μ m (10 cm x 4 mm id) at a flow rate of 1 mL/min at 35°C. The composition of the buffers is described in the text. The gradient used was a linear increase from 10% B to 53% B over 10 min, followed by 53% B to 100% B over the next 2 min. The column was re-equilibrated to initial conditions (i.e., 10% B) for 5 min prior to injection of the next sample. The eluate was monitored for absorbance at 254 nm. The amount injected was 5 pmol of each amino acid. The one-letter amino acid code has been used to label the peaks (D, aspartic acid; E, glutamic acid; S, serine; G, glycine; H, histidine; R, arginine; T, threonine; A, alanine; P, proline; Y, tyrosine; V, valine; M, methionine; C, cysteine; L, isoleucine; I, leucine; F, phenylalanine; K, lysine).

5. The eluate is monitored for fluorescence with an excitation wavelength of 335 nm and an emission wavelength of 455 nm. Amino acid peaks are integrated and quantified by comparison with an amino acid standard.

4. Notes

1. In order to achieve maximum sensitivity, all reagents and solvents should be of the highest grade available and all tubes and vials should be scrupulously clean (glass vials should preferably be baked at 450°C, polypropylene tubes can be used without further treatment provided they are kept free from dust). Plastic, talc-free gloves should also be worn to avoid contamination from proteins/amino acids on the skin. Although detection limits of 0.5 pmol, and less, can be achieved, analysis at these levels is not routine because of the levels of amino acids present in reagents and buffers.
2. Samples for amino acid analysis should be free of salt, ammonia, detergents, and buffers containing primary amines such as glycine or Tris. Thus the final purification step prior to analysis should be dialysis or chromatography [e.g., reverse-phase or gel-permeation (*see* Chapters 2 and 5 in vol. 1 of this series)] into water or a volatile buffer (e.g., 0.1% trifluoroacetic acid in water with a gradient of acetonitrile for reverse-phase chromatography or 10 mM ammonium bicarbonate for dialysis or gel-permeation chromatography). The sample is then dried by freeze-drying or drying *in vacuo* [if ammonium bicarbonate is used as the buffer, the sample should be redissolved in water after drying and

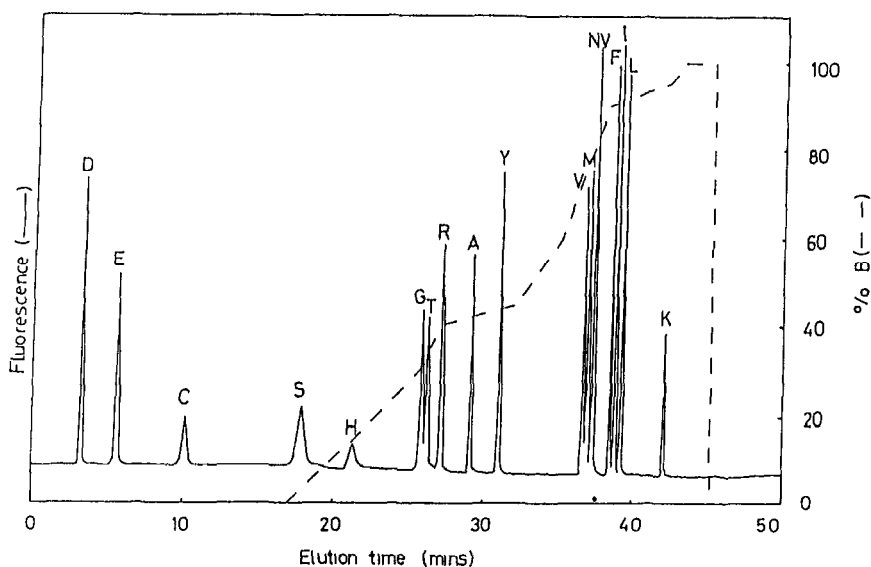


Fig. 3. Amino acid analysis of a standard mix of amino acids using precolumn derivatization with OPA and thiol reagent. The column used is a Rainin Microsorb 5- μ m (15 cm $\times$ 4 mm id) at a flow rate of 1.5 mL/min at ambient temperature. The composition of the buffers is described in the text, and the gradient of buffer B used is shown in the diagram. The eluate was monitored for fluorescence with an excitation wavelength of 335 nm and an emission wavelength of 455 nm. The amount of each amino acid injected was 10 pmol. The standard one-letter code is used to label the peaks (see legend for Fig. 2). NV = norvaline and is included as an internal standard.

then dried a second time to ensure removal of ammonia (ammonium bicarbonate breaks down into ammonia, carbon dioxide, and water *in vacuo*).

3. Hydrolysis in constant boiling HCl destroys tryptophan and converts asparagine and glutamine to their respective amino acids. Cystine, cysteine, methionine, and tyrosine are partially oxidized, though the effect on tyrosine is minimized by the addition of phenol. Quantification of cysteine is most accurate if the sample is carboxymethylated with iodoacetamide prior to hydrolysis (for method, see Chapter 5 in vol. 1 of this series). Hydrolysis for 24-hr results in destruction of approximately 10% of both threonine and serine, and in some cases only partial hydrolysis of bonds between two hydrophobic amino acids (e.g., valine–valine). Thus for accurate quantification of these amino acids the sample is hydrolyzed for different times and the values extrapolated back to zero time to give values for threonine and serine and to infinite time for the hydrophobic amino acids.
4. For analysis of free amino acids in biological fluids, beverages, or food, hydrolysis prior to derivatization is not required. It is preferable, however, to precipitate any proteins present; a suitable protocol is addition of 9 vol of freshly prepared acetone containing 0.2% 2-mercaptoethanol. For analysis by PITC derivatization, a longer shallower gradient may be required to separate all the amino acids that occur in biological fluids. The gradient given in Fig. 3 for separation of OPA derivatives should resolve all the amino acids found in biological fluids.
5. Separation of all the amino acid derivatives may require individual tailoring of the eluant buffers to achieve maximum resolution on different columns

and different HPLC systems. This can often be achieved by alteration of the gradient of organic solvent (i.e., decreasing the slope at a particular point in the separation in order to achieve better resolution). Alternatively the ionic strength or pH of buffer A may be altered. Alteration of the ionic strength by up to 10% will effect the separation of arginine or histidine from the other amino acids. The separation of the acidic amino acids can be altered by increasing or decreasing the pH in 0.05-unit steps.

6. The PTC derivatives are stable for several weeks if stored dry at -20°C or for 3 d in solution at 4°C . At room temperature in solution the response obtained for cysteine, valine, and isoleucine decreases by 5% over 6 h; for the derivatives of the other amino acids, the response falls by 5% over 10 h.
7. The PTC derivatives of the standard amino acids obtained from protein hydrolysates are eluted in the first 12 min of the HPLC separation. Peaks will be observed eluting later than this; these peaks are caused by the reagents and do not interfere with the analysis of the amino acids.

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Chapter 4

Identification of N-Terminal Amino Acids by High-Performance Liquid Chromatography

E. L. V. Harris

1. Introduction

1-Dimethylaminonaphthalene-5-sulfonyl chloride (dansyl chloride, Dns-Cl) has been widely used in protein chemistry for determination of the N-terminal amino acid of peptides. The dansyl chloride reacts with both α - and Σ -amino groups (Fig. 1). After derivatization the peptide is acid hydrolyzed releasing free amino acids, amino acids such as lysine and histidine with the dansyl group attached to their side chains, together with the derivatized N-terminal amino acid. Traditionally

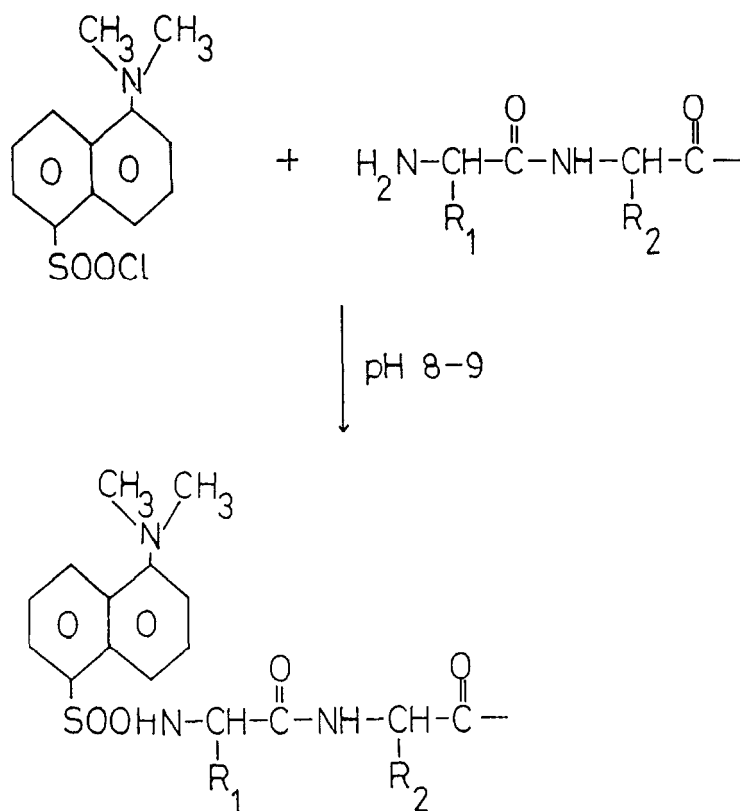


Fig. 1. Reaction of dansyl chloride with the amino terminus of a peptide.

the Dns-amino acids have been separated by thin layer chromatography and viewed by fluorescence in UV light (Chapter 23 in vol. 1 of this series). This procedure is time-consuming, however, and is not easily quantified. More recently, methods have been developed for separation of the Dns-amino acids by reverse-phase HPLC (1,2), thus allowing for automation of the separation and quantitation of the derivatives. In order to minimize the interference by reagent byproducts, the reaction conditions have been modified from those of the traditional method.

2. Materials

1. 2.3 mg/mL of Dansyl chloride in acetonitrile are prepared immediately prior to use from a 100 mg/mL stock in acetone. 100 mg/mL solutions of dansyl chloride in acetone are commercially available (e.g., Pierce Chemicals) and should be stored at 4°C in the dark. Care should be taken when removing aliquots from the stock not to introduce water that hydrolyzes the reagent.
2. 40 mM Lithium carbonate, pH 9.5.
3. 0.1% Ethylammonium chloride.
4. Constant boiling hydrochloric acid (e.g., from Pierce Chemicals).
5. HPLC grade methanol (water-free).
6. Soda glass tubes (approximately 5 x 0.5 cm) with pointed bottoms pretreated by baking at 450°C for 4 h to destroy contaminating amino acids and proteins.
7. Oxygen/butane flame for sealing hydrolysis tubes and a diamond tip for opening them.
8. 105°C Oven.
9. Centrifugal concentrator (e. g., Savant Speedvac) and vacuum pump (e. g., Edwards).
10. HPLC equipment to run a binary solvent gradient with column heater, autosampler, detector for absorption at 254 nm, and integrator. For more sensitive determination, the UV monitor should be substituted with a fluorescence detector with the excitation wavelength at 330 nm and the emission wavelength at 470 nm.
11. Buffer A: 25 mM Trifluoroacetic acid in HPLC-grade water adjusted to pH 7.6 with sodium hydroxide. This sodium trifluoroacetate solution is mixed with HPLC-grade acetonitrile in a ratio of 90:10 (v/v).

12. Buffer B: 30/70 (v/v) sodium trifluoroacetate pH 7.6/acetonitrile. Prior to use, filter both buffers A and B through 0.2- μ m filters and degas by bubbling with helium for 5 min.
13. C-18 reverse-phase HPLC column with 5- μ m, or smaller, packing (e. g., Ultrasphere ODS column, 25 cm x 4.6 mm id, or LKB SuperPac 3- μ m column, 10 cm x 4.6 mm id).
14. Amino acid standard made by dissolving each of the following amino acids at 20 μ M in HPLC grade water: aspartic acid, glutamic acid, cysteine, serine, threonine, glycine, alanine, proline, arginine, valine, methionine, leucine, isoleucine, phenylalanine, lysine, histidine, and tyrosine.

3. Methods

3.1. Derivatization and Hydrolysis

1. For standards, pipet 5 μ L of the standard mix into a tube. For samples, pipet 5–10 μ L of a solution containing 100–200 pmol of peptide into a tube.
2. Dry the standards and samples in the centrifugal concentrator, or if not available, just under vacuum.
3. Redissolve in 5 μ L of HPLC grade water and redry. Repeat this procedure once more.
4. Add 4 μ L of lithium carbonate solution and 2 μ L of a suitable dilution of the 8.7 mM Dns-Cl in acetonitrile to give a 5–10-fold excess of reagent over amino groups (i. e., for the standards, the reagent is not diluted, whereas for 200 pmol of peptide, the reagent is diluted 1 in 10).

5. Centrifuge briefly to ensure that reagents are at the bottom of tube.
6. Cover the tubes with parafilm to minimize evaporation, and incubate at approximately 20°C for 35 min in the dark.
7. Add 2 μ L of ethylammonium chloride solution, and centrifuge briefly.
8. After 1 min incubation, dry in the centrifugal concentrator.
9. Add 50 μ L of HCl and seal the tube in a oxygen/butane flame by twisting and drawing out when hot.
10. Incubate at 105°C for 8 h. Crack open the tubes by prescoring with the diamond tip pen, and dry off the HCl under vacuum.

3.2. Separation of Derivatives

1. Equilibrate the HPLC column at 39°C in 93% buffer A/7% buffer B at a flow rate of 1.2 mL/min for 5 μ m packing or 1.0 mL/min for 3 μ m packings.
2. Add 4 μ L of dry methanol to each of the sample and standard tubes and dry in the centrifugal concentrator. Repeat this procedure once more. This step ensures removal of all volatile contaminants.
3. Redissolve the samples and standards in a suitable volume to enable injection of 5 μ L of solution containing 50–200 pmol of Dns-amino acid(s).
4. The derivatives are eluted with a gradient of buffer B. The exact gradient required will vary from column type to column type; that used for the LKB SuperPac column is shown in Fig. 2.
5. The eluate is monitored for absorbance at 254 nm, or for more sensitive detection for fluorescence

with an excitation wavelength of 330 nm and an emission wavelength of 470 nm. Amino acid peaks are integrated and quantified by comparison with standards.

4. Notes

1. Samples for N-terminal determination should contain no salts, ammonia, or buffers with primary amines such as glycine or Tris. Thus, a preferred final purification step for peptides destined for N-terminal analysis is reverse-phase chromatography in a volatile buffer system such as 0.1% trifluoroacetic acid in water with a gradient of acetonitrile (*see* Chapter 5 in vol. 1 of this series).
2. The procedure described here is suitable for use with peptides of up to 30 amino acids.
3. Hydrolysis in HCl destroys tryptophan and converts asparagine and glutamine to their respective acids. Thus peptides with an N-terminal tryptophan do not yield a Dns derivative, whereas those with an N-terminal asparagine or glutamine will yield Dns-aspartic acid or Dns-glutamic acid, respectively. Several Dns-amino acids are partially destroyed by hydrolysis, e.g., methionine and proline; therefore, for better quantification the amounts obtained should be corrected for losses by comparison to a hydrolyzed Dns-amino acid standard mix. A 4–10 h hydrolysis time will be sufficient for most peptides. The time required for hydrolysis can be shortened further by use of a mix of 5.7M HCl and trifluoroacetic acid (2/1 v/v) at 166°C. Under these conditions, the optimum time of hydrolysis is 50 min.

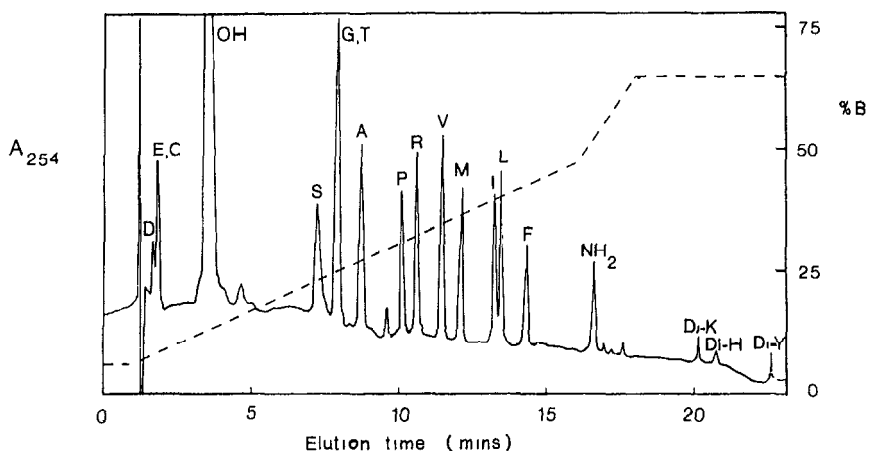


Fig. 2. Separation of a standard mix of amino acids derivatized with dansyl chloride prior to injection onto the HPLC column (LKB SuperPac 3 μ m) (10 cm $\times$ 4 mm id). The flow rate was 1.0 mL/min with a column temperature of 39°C. The compositions of the buffers used are described in the text, and the gradient of buffer B used to elute the derivatives is shown in the figure. The eluate was monitored for absorbance at 254 nm. 50 Picomoles of each amino acid was injected. The standard one-letter code is used to label the amino acid peaks (*see* Chapter 3, Fig. 2: C, cysteic acid; OH, dansyl sulfonic acid; NH₂, dansyl amide; Di, di-dansyl derivative). For the separation achieved on an Ultrasphere ODS 5 μ m column, *see* refs. 1 and 2.

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Chapter 5

Enzymatic Methods for Cleaving Proteins

Bryan John Smith

1. Introduction

There could be various reasons the protein chemist may want to break a protein of interest into fragments, but foremost among them must be the purposes of peptide mapping, primary structure determination, and preparation of particular fragments for studies relating structure and function. Proteinases (or more specifically endo-proteinases) are commonly used to generate peptides for these purposes, but the number of particularly useful and commercially available proteinases is not great. This is because enzyme specificity is an important consideration—the laboratory worker generally requires good yields of clean, pure peptides, but sig-

nificantly less than 100% cleavage at some or all sites of action gives rise to a complex mixture of a large number of polypeptides.

One of the best characterized and widely used proteinases is trypsin (E.C.3.4.21.4), and it is the use of this enzyme that is described below. Other useful enzymes for cleaving proteins are described in section 4. Trypsin is synthesized as trypsinogen by the vertebrate pancreas, but rapidly becomes converted by the removal of the amino-terminal hexapeptide to the active enzyme of molecular weight approximately 23,500. It has a serine at its active site, and therefore belongs to the serine proteinase family. Trypsin displays good specificity, catalyzing the hydrolysis of the peptide bond to the COOH side of lysyl and arginyl residues. If an acidic residue occurs to either side of the basic residue, hydrolysis is slower, and if the residue to the COOH side is proline, hydrolysis is very slow. Polylysine sequences may also be fully cleaved with difficulty. Trypsin is optimally active at about pH 8.

2. Materials (for Trypsin)

1. Trypsin (E.C.3.4.21.4) is available from various commercial suppliers. It is stable for periods of years as a dry solid at -20°C .
2. Stock solutions:
 - a. Trypsin, 1 mg/mL in distilled water. Use fresh or divide into aliquots and store frozen. A stock solution may be thawed and refrozen several times, but for consistent results, thaw only once.
 - b. Ammonium bicarbonate (0.4M) in distilled water, pH ~ 8.5 . May be stored refrigerated or frozen for long periods.

3. Method (for Trypsin)

1. Dissolve the substrate in water to 2 mg/mL. Add an equal volume of 0.4M ammonium bicarbonate solution. Add trypsin solution to an enzyme/substrate ratio of 1/50 (w/w), i.e., to 1 mL of 1 mg/mL substrate solution add 20 μ L of 1 mg/mL of trypsin solution.
2. Incubate at 37°C for 24 h.
3. Terminate digestion by immediate submission to peptide mapping or isolation techniques, acidification, freeze-drying, or addition of specific inhibitor (e.g., *N* α -tosyl-L-lysyl chloromethyl ketone, or "TLCK," in molar excess to the trypsin used). As with other serine proteinases, trypsin may be inactivated by reaction of the serine residue at the active site with phenylmethylsulfonyl fluoride (PMSF). Prepare this agent just before use by dissolving in propan-2-ol to a 1M solution. Dilute 1000-fold to 1 mM in the reaction mixture. It is effective within a few minutes. Beware of the toxic nature of PMSF.

4. Notes

1. Various grades of trypsin are available commercially. Most are from bovine pancreas. That which has been treated with L-1-chloro-3-tosylamido-4-phenylbutan-2-one (or "TPCK"), or similar, is recommended since this treatment specifically inhibits chymotrypsin, which may contaminate trypsin preparations. Enzyme activity may vary, in detail, from source to source or batch to batch. Attention to this is recommended if reproducibility is important (e.g., for peptide mapping).

2. During incubation with substrate, trypsin will also undergo autolysis to produce (among other products) ψ -trypsin, which has chymotrypsin-like activity. This may contribute low-frequency "non-tryptic" cleavages at some tyrosyl, phenylalanyl, or tryptophanyl bonds. The literature claims that low levels of Ca^{2+} (0.1 mM CaCl_2) added to digestion buffers may reduce production of ψ -trypsin by autolysis.
3. The buffer described above is a simple volatile buffer that gives an appropriate pH of approximately 8. Other buffers of pH 8 may be readily substituted, e.g., 0.05M Tris-HCl, pH 8.
4. The method described above is the basic procedure and is subject to great change, according to the requirements of the worker and protein in question. Any one set of experimental conditions will give different results with different substrates, and the reader is encouraged to discover optimal conditions for digestion of a substrate empirically.

For reproducible peptide mapping, when complete digestion is desirable, prolonged digestion may be required (e.g., 48 h at 37°C, pH 8, with a second addition of trypsin, similar to the first at 24 h). On the other hand, for preferential cleavage of particularly sensitive bonds or for production of partial cleavage products (e.g., overlapping peptides for sequencing purposes or production of folded domains), digestion can be limited by use of a low enzyme/substrate ratio, shorter digestion times, and lower incubation temperatures, and/or the buffer may be adjusted to suboptimal pH. High salt conditions (e.g., 0.5M NaCl, in buffer), which favor compact folding of a structured polypeptide chain, may also be used. Such a tightly folded se-

quence will be more resistant to proteolytic attack than nonstructured regions, though not necessarily completely or indefinitely resistant. Thus, by careful adjustment of buffer and incubation conditions, these partially resistant regions may be prepared and used, for example, in studies on their function.

5. The condition of the substrate is important. First, the substrate should be soluble, or as finely divided as possible, in the digestion buffer. If a sample is not readily soluble in water or ammonium bicarbonate solutions, suitable solvents can be used initially and then adjusted by dilution or titration of pH to allow for trypsin action. If the polypeptide remains insoluble, the precipitate should be kept in suspension by stirring. Thus, 8M urea may be used to solubilize a protein or disrupt a tightly folded structure, and then diluted to 2M urea for digestion by trypsin. Trypsin will also function in 2M guanidinium chloride, in the presence of sodium dodecyl sulfate (SDS, e.g., 0.1% w/v), or in the presence of acetonitrile (up to about 50% v/v), so that fractions from reverse-phase HPLC (acetonitrile gradients in water/trifluoroacetic acid, 0.1% v/v) may be readily digested after simple addition of ammonium bicarbonate or other neutralization, and dilution (if necessary to lower the acetonitrile concentration).

Second, a native protein may be tightly folded such as to markedly slow up or inhibit proteolytic attack. To remedy this, the substrate may be denatured and the structure opened out to allow for access of the proteinase. This may be done by boiling in neutral pH solution, or by use of such agents as urea, SDS, or organic solvent, as described above. Low concentrations of these agents (e.g., 5–10% v/v acetonitrile) may give more rapid

digestion than will a buffer without them, but high concentrations (e.g., 50% v/v acetonitrile) will slow the digestion.

An additional, and very common, technique is reduction and carboxymethylation, i.e., permanent disruption of disulfide bonds. This treatment opens out the protein structure to allow for ready digestion and to minimize complications in peptide separation that are caused by pairs (or greater) of peptides remaining connected by S—S bonds. This treatment is carried out as described in Chapter 7. Alternatively, if a sample has already been digested, S—S bonds may be reduced by simple addition of small amounts of solid dithiothreitol and incubation at room temperature (and pH 8) for 30 min or so. This treatment is followed immediately by HPLC to separate the various peptides. The amount of dithiothreitol required (i.e., slight molar excess over S—S bonds) may be calculated accurately if the cystine content of the protein concerned is known.

6. The result of proteolytic cleavage may be monitored, and the resulting peptides purified, by various forms of electrophoresis and chromatography, as described in other chapters in this series. Typical HPLC conditions might be, for example, reverse-phase chromatography on a 4 mm id C-18 or C-8 column, using a gradient of 0–100% v/v acetonitrile in water, 0.1% v/v trifluoroacetic acid, at a flow rate of 1 mL/min at room temperature.
7. During digestion, autolysis produces a low background of peptides from the proteinase itself. If this is to prove a problem, enzyme-only controls are recommended. The primary structure of trypsin is known (e.g., *see* the structure of pig trypsinogen

given in ref. 1), so the products of autolysis should not prove misleading to the protein sequencing fraternity!

8. Lysine is a fairly common constituent of proteins, and digestion with trypsin can generate a large number of peptides of small average size. This is a good point when peptide mapping, but some purposes, such as sequencing, may require longer peptides. The action of trypsin may be modified in an attempt to achieve this. It is done by modification of the side chains of lysyl or arginyl residues as described in ref. 2 and Chapter 7, such that cleavage only occurs at unmodified residues. Perhaps the commonest such method is succinylation of lysyl side chains, leading to tryptic cleavage at arginyl (and any remaining unmodified lysyl) residues (see Chapter 7).

Introduction of additional sites of cleavage by trypsin may be achieved by conversion of cysteinyl residues to aminoethyl cysteinyl residues by reaction with ethyleneimine as described in Chapter 7.

9. The need for substrate modification as described above has been reduced by the commercial advent of proteinases with good specificity for either lysyl or arginyl residues, namely endoproteinases lys C and arg C (see Table 1).

Clostripain is also reported to cleave to the COOH side of arginyl residues, but also may cleave (less frequently) to the COOH side of lysyl residues. It will, however, cleave Arg-Pro bonds.

10. There are a large number of proteinases available commercially. Data on some of the more useful ones are given in Table 1, summarized from personal observation, published data, and suppliers' information. They may be used in much the same

Table 1
Summarized Data on Some Commercially Available Endoproteinases
Suitable for the Generation of Peptides from Proteins

Endoproteinase	E.C. No.	Type of proteinase	Apparent molecular weight	pH optimum	Specificity*
Chymotrypsin	3.4.21.1	Serine	25,000	7-9	-X↓Y- (X = aromatic)
Clostripain	3.4.22.8	-SH	50,000	7.7	-Arg↓Y- (Lys↓Y-, less frequently)
Elastase	3.4.21.11	Serine	25,900	7-9	-X↓Y- (X = uncharged aliphatic)
Endoproteinase Arg-C ("submaxillary gland protease")	3.4.21—	Serine	30,000	7.5-8.5	-Arg↓Y-; -Lys-Lys↓Y-
Endoproteinase Glu-C	3.4.21.19	Serine	27,000	4 7.8	-Glu↓Y- (not if Glu within two or three residues of the N-terminus) -Glu↓Y; -Asp↓Y-
Endoproteinase Lys-C (nonreduced) (reduced)	—	Serine	30,000 33,000	8.5-8.8	-Lys↓Y-
Kallikrein (α Form) (β Form)	3.4.21.8	Serine	27,000	7-8 28,900	-X-Arg↓Y- (preferentially if X = Phe, Leu)
Papain	3.4.22.2	-SH	23,000	6-7	-X↓Y- (preferentially if X = Arg, Lys, Gln, His, Gly, Tyr)
Pepsin	3.4.23.1	Acid	34,500	1.8-2.2	-X↓Y- (Y = Hydrophobic; preferentially if X = Phe, met, Leu, Tryp)
Plasmin	3.4.21.7	Serine	85,000	8.9	-Arg↓Y-, -Lys↓Y-
Thermolysin	3.4.24.4	Metallo- (Ca ²⁺)	37,500	7-9	-Y↓X- (X = Hydrophobic; also less frequently if Gly, Ser, Thre; Not if -Y-X-Pro-)
Thrombin (α Form) (Factor IIa) (β Form)	3.4.21.5	Serine	36,000 38,000	8.2-9	-Arg↓Y
Trypsin	3.4.21.4	Serine	23,500	8	-Arg↓Y-, -Lys↓Y-, Amino ethylcysteine↓Y-

*Y, unspecific. NB, Frequently, susceptibility to proteolysis is reduced or lost if either the potentially labile bond is to proline (e.g., -x-Pro- for chymotrypsin) or is between two like residues (e.g., -Glu-Glu- for endoproteinase Glu-C)

manner as is trypsin, with due attention paid to buffer (pH) conditions and methods of termination of digestion.

Thus, for pepsin (active in the range pH 1–5), the volatile buffer 5% v/v acetic acid in water may be used. To stop digestion, raise the pH beyond pH 5 and/or freeze dry.

Proteinases such as clostripain containing a sulfhydryl group in the active site require activation by inclusion in the buffer of a sulfhydryl agent, such as dithiothreitol (1–10 mM) and may be inactivated by addition of iodoacetic acid in molar excess to the sulfhydryl agent present.

Metalloproteinases require the presence of metal ions. Thus, thermolysin requires Ca^{2+} . Preparations of this enzyme, or of a substrate, may themselves contain sufficient metals ions, but otherwise the Ca^{2+} should be added to the buffer to a concentration of at least 5 mM. Reaction may be ended by addition of EDTA in excess to the metal ion concentration (remembering that sample and enzyme may contribute metal also).

Most of the common proteinases are serine proteinases (*see* Table 1) and may be inactivated by reaction with PMSF as described above for trypsin. In addition to these general methods for inhibiting digestion, inhibitors specific for various proteinases are available from commercial sources (*see* suppliers' literature).

11. The specificities of various enzymes are indicated in Table 1, but in addition, two common (but not generally applicable) rules may be applied:
 - a. Bonds involving prolyl residues may not be cleaved, e.g., endoproteinases Glu-C will commonly not cleave Glu-Pro bonds.

- b. Potentially labile bonds in a multiple sequence may not each be cleaved in good yield, e.g., Glu-Glu-X for endoproteinase Glu-C.
12. One of the most useful proteinases is endoproteinase Glu-C (or "V8 protease") isolated from *Staphylococcus aureus* strain V8, since like trypsin and endoproteinases Lys-C and Arg-C, it shows good specificity. It cuts to the COOH side of glutamyl residue. A lower frequency of cleavage to the COOH side of aspartyl residues may also occur at neutral pH, although at pH 4 this may not occur. Endoproteinase Glu-C functions in buffers containing 0.2% w/v SDS or 4M urea. Its sequence is known (3).
 13. Other proteinases are of broader specificity and may be affected by surrounding sequences. Their action is therefore difficult to predict. In particular instances, however, their observed action may prove beneficial—perhaps cleaving at one or a few particularly sensitive sites when incubated in suboptimal conditions (e.g., short duration digestion or nondenatured substrate).

Good examples of this come from work on preparation of F(ab)₂, antigen-binding fragments of immunoglobulin IgG, that are bivalent and lack the constant, Fc region of the molecule. Incubation of nondenatured IgG molecules with a proteinase of broad specificity can lead to proteolytic cleavage at a few sites or a single site in good yield. Pepsin has been used for this purpose [e.g., pH 4.2–4.5; enzyme/substrate, 1/33 w/w; 37°C (4)]. Different subclasses of mouse IgG were found to be digested at different rates, however, in the order IgG3 > IgG2a > IgG1. Different antibodies of the same

subclass may also be degraded differently, some rapidly and without formation of $F(ab')_2$ (4). Papain has been used to prepare $F(ab)_2$ fragments from the IgG1 subclass, which is the subclass that is most resistant to pepsin. The method described by Parham et al. (5) uses papain (which has been activated just before use by reaction with cysteine), at pH 5.5 (0.1M acetate, 3 mM EDTA) 37°C, with an IgG concentration of about 10 mg/mL. The enzyme is added at time 0, and again later (e.g., at 9 h) to an enzyme/substrate ratio of 1/20 w/w. Digestion can be halted by addition of iodoacetamide (30 mM) (5). Rousseaux et al. (6) also describe conditions for generating rat $F(ab')_2$, using papain (in the presence of 10 mM cysteine), pepsin, or endoproteinase Glu-C (V8 protease). Incubation of the IgG1 and 2a subclasses at pH 2.8 prior to digestion with pepsin improved the yields of $F(ab')_2$ fragments, presumably because the proteins thus denatured were effectively better substrates.

14. Two enzymes of broader specificities are worthy of further mention. The first is thermolysin, for its good thermostability, which may prove useful when keeping awkward substrates in solution. Thermolysin remains active at 80°C or in 8 M urea. The second enzyme is pepsin, which acts at low pH. Disulfide bonds rearrange less frequently in acid than in alkaline conditions, so use of low pH buffers may not only help solubilize a substrate, but may also help preserve naturally disulfide-bonded pairs of peptides. Endoproteinase Glu-C may also be used at low pH, having an optimum at pH 4.
15. For more information on the various proteinases available, the reader is guided to works such as

those by Allen (7), Wilkinson (8) (and references therein), as well as to the commercial suppliers of the enzymes.

16. Exoproteinases are not discussed here, since their use generally does not generate clean peptides. As an example of use of an exoproteinase, the reader is directed to the description of carboxypeptidase Y in vol. 1 of this series (9).

Acknowledgments

I would like to thank my colleagues for helpful discussions.

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Chapter 6

Chemical Cleavage of Proteins

Bryan John Smith

1. Introduction

There is a variety of chemical reactions known to result in the cleavage of the peptide bond. Some are nonspecific—for instance, 6M hydrochloric acid at 110°C for 24 h hydrolyzes a polypeptide to a mixture of single amino acids. Others show some discrimination, however, as to the precise nature of the amino acid residues around the bond to be broken. Some of these methods are sufficiently specific to be of use, for instance, for generating peptides for primary structure determination. These methods usefully augment those that use proteolytic enzymes (*see* Chapter 5), especially since they tend to act at positions occupied by less common amino acids and so generate large peptides.

The efficiency of the cleavage reaction is an important factor when considering which method to use for generating peptides from proteins. It is an unfortunate aspect of chemical methods of proteolysis that, in the main, they afford significantly less than 100% yields and commonly exhibit undesirable side reactions. One of the best current methods is cleavage of the bond to the COOH-side of methionyl residues by treatment with cyanogen bromide; yields are 90–100% with few side reactions. It is a well-established and well-used method. A newer technique for cleavage of the bond to the COOH side of tryptophanyl residues also involves treatment with cyanogen bromide, after an oxidation step using dimethylsulfoxide in acid. This method too is highly specific and efficient. Most other methods are less efficient, but still may be of some use for cleavage of small substrates in which problems of poor yields and side reactions are minimized. One such method is limited hydrolysis in dilute acid. This method, together with the two mentioned above, has the advantage of requiring reagents that are stable and pure enough as supplied commercially. The methods are also relatively simple and adaptable to large- and small-scale cleavages. These three methods are described below in some detail.

2. Materials

2.1. *Cleavage at Met-X Bonds*

1. Ammonium bicarbonate (0.4M) solution in distilled water. Stable for weeks in refrigerated stoppered bottle.

2. 2-Mercaptoethanol. Stable for months in dark, stoppered, refrigerated bottle.
3. Formic acid, minimum assay 98%. Aristar grade.
4. Cyanogen bromide. Stable for months in dry, dark, refrigerated storage. Warm to room temperature before opening. Use only white crystals, not yellow ones. *Beware of the toxic nature of this reagent.*
5. Sodium hypochlorite solution (domestic bleach).
6. Equipment includes a nitrogen supply, fume hood, and suitably sized and capped tubes (e.g., Eppendorf microcentrifuge tubes).

2.2. Cleavage at Tryp-X Bonds

1. Oxidizing solution:
Glacial acetic acid, 30 vol
9M HCl, 15 vol
Dimethylsulfoxide, 4 vol
Use best grade reagents. Though each of the constituents is stable separately, mix and use the oxidizing solution when fresh.
2. Ammonium hydroxide (15M).
3. Cyanogen bromide solution in formic acid (60%):
Formic acid (minimum assay 98%, Aristar grade) 6 mL. Make to 10 mL with distilled water. Use fresh. Add cyanogen bromide to a concentration of 0.3 g/mL. Use fresh.
Store cyanogen bromide refrigerated in the dry and dark, where it is stable for months. Use only white crystals. *Beware of the toxic nature of this reagent.*
4. Sodium hypochlorite solution (domestic bleach).
5. Equipment includes a fume hood and suitably sized capped tubes (e.g., Eppendorf microcentrifuge tubes).

2.3. Cleavage of Asp-X Bonds

1. Dilute hydrochloric acid (approximately 0.013M) pH 2 ± 0.04 : Dilute constant boiling (6M) HCl (220 μ L) to 100 mL with distilled water.
2. Pyrex glass hydrolysis tubes.
3. Equipment includes a blowtorch suitable for sealing the hydrolysis tubes, a vacuum line, and an oven for incubation of samples at 108°C.

3. Method

3.1. Cleavage at Met-X Bonds

1. Reduction
 - a. Dissolve the polypeptide in water to between 1 and 5 mg/mL, in a suitable tube. Add 1 vol of ammonium bicarbonate solution, and add 2-mercaptoethanol to between 1 and 5% v/v.
 - b. Blow nitrogen over the solution to displace oxygen, seal the tube, and incubate at room temperature for approximately 18 h.
2. Cleavage
 - a. Dry down the sample under vacuum, warming if necessary to help drive off all of the bicarbonate. Any remaining ammonium bicarbonate will form a nonvolatile salt on subsequent reaction with formic acid.
 - b. Redissolve the dried sample in formic acid (98%) to 1–5 mg/mL. Add water to make the acid 70% v/v finally.
 - c. Add excess white crystalline cyanogen bromide to the sample solution, to between 20- and 100-fold molar excess over methionyl residues. Practically,

this amounts to approximately equal weights of protein and cyanogen bromide. To very small amounts of protein, add one small crystal of reagent. Carry out this stage in the fume hood.

- d. Seal the tube and incubate at room temperature for 24 h.
- e. Terminate the reaction by drying down under vacuum. Store samples at -10°C or use immediately.
- f. Immediately after use, decontaminate spatulas, tubes, and so on, that have contacted cyanogen bromide, by immersion in hypochlorite solution (bleach) until effervescence stops (a few minutes).

3.2. Cleavage at Tryp-X Bonds

1. Oxidation: Dissolve the sample to approximately $0.5\text{ nmol}/\mu\text{L}$ in oxidizing solution (e.g., $2\text{--}3\text{ nmol}$ in $4.9\text{ }\mu\text{L}$ oxidizing solution). Incubate at 4°C for 2 h.
2. Partial neutralization. To the cold sample, add 0.9 vol of ice-cold NH_4OH (e.g., $4.4\text{ }\mu\text{L}$ of NH_4OH to $4.9\text{ }\mu\text{L}$ oxidized sample solution). Make this addition carefully so as to maintain a low temperature.
3. Cleavage. Add 8 vol of cyanogen bromide solution. Incubate at 4°C for 30 h in the dark. Carry out this step in a fume hood.
4. To terminate the reaction, lyophilize the sample (all reagents are volatile).
5. Decontaminate equipment, such as spatulas, that have contacted cyanogen bromide, by immersion in bleach until the effervescence stops (a few minutes).

3.2. Cleavage at Asp-X Bonds

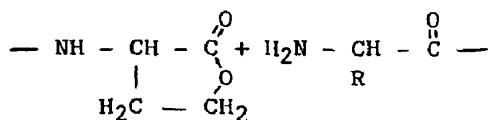
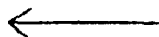
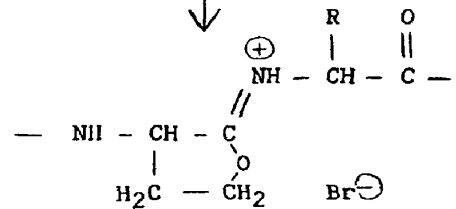
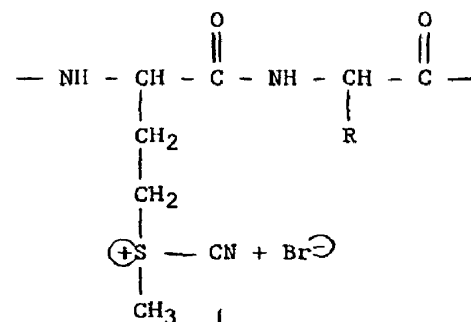
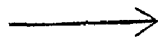
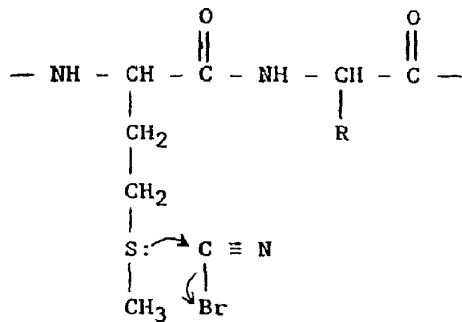
1. Dissolve the protein or peptide in the dilute acid to a concentration of $1\text{--}2\text{ mg/mL}$ in a hydrolysis tube.

2. Seal the hydrolysis tube under vacuum—i.e., with the hydrolysis (sample) tube connected to a vacuum line, using a suitably hot flame, draw out and finally seal the neck of the tube.
3. Incubate at 108°C for 2 h.
4. To terminate the reaction, cool and open the hydrolysis tube, dilute the sample with water, and lyophilize.

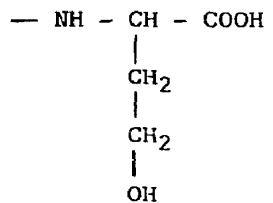
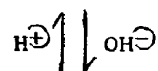
4. Notes

4.1. Cleavage at Met-X Bonds

1. The mechanism of the action of cyanogen bromide on methionine-containing peptides is shown in Fig. 1. For further details, see the review by Fontana and Gross (1). Methionine sulfoxide does not take part in this reaction. An acid environment is required to protonate basic groups and so prevent reaction there and maintain a high degree of specificity. Met-Ser and Met-Thr bonds may give less than 100% yields of cleavage because of the involvement of the β -hydroxyl groups of seryl and threonyl residues in alternative reactions, which do not result in cleavage (1).
2. Although the specificity of this reaction is excellent, some side reactions may occur. This is particularly so if colored (yellow or orange) cyanogen bromide crystals are used, when destruction of tyrosyl and tryptophanyl residues may occur. The acid conditions employed for the reaction may lead to small degrees of deamidation of side chains and cleavage of acid-labile bonds, e.g., Asp-Pro. A small amount of oxidation of cysteine to cysteic



Homoseryl lactone residue



Homoseryl residue

acid may occur, if these residues have not previously been reduced and carboxymethylated (*see* Chapter 7).

In great excess of cyanogen bromide (1000-fold or more) oxidation of methionine can occur. This phenomenon can be used to effect cleavage at tryptophanyl (but not methionyl sulfoxide residues). To promote this reaction, strong acid (heptafluorobutyric acid) is used, but the yield from this method is poor (*see* Table 1), and tyrosyl residues become brominated.

3. Acid conditions are required for this reaction. Formic acid is commonly used because it is a good protein solvent and denaturant, but other acids may be used instead if desired, e.g., 0.1M HCl; 70% HF. Stronger acid (e.g., 75% TFA; HFBA) may cause increased acid hydrolysis, and so on, as described above.
4. It is possible to cleave Met-X bonds of protein samples that have been applied to glass fiber discs, such as are used in automated protein sequencers of the gas-phase type. The filter is removed from the sequencer and saturated with a fresh solution of cyanogen bromide in 70% formic acid. After 24 h incubation in the dark at room temperature, the filter can be dried under vacuum and then replaced in the sequencer. Yields may be low (less than 50%), but may prove useful for obtaining protein sequence data from samples that have proven to be blocked, i.e., do not have a free NH_2 group at their N-terminus.
5. As described in (1) above, the peptide to the N-terminal side of the point of cleavage has at its C-terminus a homoserine or homoserine lactone residue. The lactone derivative of methionine can be

coupled selectively and in good yield (2) to solid supports of the amino type, e.g., 3-amino propyl glass. This is a useful technique for sequencing peptides on solid supports. The peptide from the C-terminus of the cleaved protein will, of course, *not* end in homoserine lactone (unless the C-terminal residue was methionine!) and so cannot be so readily coupled.

6. The reagents used are removed by lyophilization, unless salt has formed following failure to remove all of the ammonium bicarbonate. The products of cleavage may be fractionated by the various forms of electrophoresis and chromatography currently available, as described elsewhere in volumes of this series. It should be remembered, however, that since methionyl residues are among the less common residues, peptides resulting from cleavage at Met-X may be large. Accordingly, in HPLC, for instance, use of wide-pore column materials may be advisable (e.g., 30 μm pore size reverse-phase columns, using gradients of acetonitrile in 0.1% TFA in water).

4.2. Cleavage at Tryp-X Bonds

1. The method described is that of Huang et al. (3). Although full details of the mechanism of this reaction are not clear, it is apparent that tryptophanyl residues are converted to oxindolylalanyl residues in the oxidation step, and the bond to the COOH side at each of these is readily cleaved, in excellent yield (approaching 100% in ref. 3) by the subsequent cyanogen bromide treatment. The result is seemingly unaffected by the nature of the residues surrounding the cleavage site.

Table 1
Summary of Chemical Methods for Specific Cleavage of Peptide Bonds

Site of cleavage	Brief method details	Comments	Example ref
Asp↓X	Incubation in dilute acid (pH 2), high temperature (e.g., 108°C), short duration	Some X-Asp bonds cleaved in lesser yield. Rate of cleavage dependent on various factors (e.g., identity of X). Side reactions include deamidation. Yield moderate to good, but unpredictable	4
Asp↓Pro	Incubation of 7M guanidine HCl, dilute acid (e.g., 10% acetic acid adjusted to 2.5 with pyridine), moderate temperature (e.g., 37°C), prolonged incubation (e.g., 24–120 h)	Some Asp-Pro may remain resistant despite inclusion of guanidine HCl. Deamidation may occur. Yields moderate to good	6
Asn↓Gly	Incubation of 2M hydroxylamine, 6M guanidine HCl (brought to pH 9 with LiOH) 45°C, 4 h	Some Asn-Gly bonds may remain resistant despite inclusion of guanidine HCl. Specificity generally good, although some Asn-X bonds may cleave	7
X↓Cys	(a) Incubation in 6M guanidine HCl, 0.2M Tris-acetate, pH 8, with excess dithiothreitol (b) Add 2-nitro-5-thiocyanobenzoic acid (5-fold molar excess over thiols). Adjust pH to 8 (c) Acidify to pH 4. Dialyze. Lyophilize (d) To cleave, dissolve in 6M guanidine HCl, 0.1M sodium borate, pH 9, 12–16 h, 37°C	For cleavage at cysteine only, not cystine, delete DTT from (a). Cysteinyl residue converted to iminothiazolidinyl residue, which blocks sequencing unless converted to alanyl residues by treatment as in ref. 9. Yield good	8
His↓X	Incubation in N-bromosuccinimide (3-fold molar excess over His, Trp, and Tyr) pH 3–4 (e.g., pyridine acetic acid, pH 3.3) 1 h, 100°C	Cleavage at Trp-X and Tyr-X occurs preferentially (q, v). Met and Cys may be oxidized. N-Bromosuccinimide unstable in storage. Yields moderate to poor. His converted to lactone derivative.	10
Met↓X	Incubation in cyanogen bromide (20–100-fold molar excess over Met) acid (e.g., 70% formic acid), 18–20 h, room temperature in the dark	Specificity and yield excellent. Cys may be slowly oxidized, and Asp-Pro bonds cleaved. Excess reagent can cause degradation of Trp and Tyr side chains. Cyanogen bromide is toxic	1
X↓Ser	(a) Incubation in conc. anhydrous acid (e.g., H ₂ SO ₄ , HF or HCOOH) room temperature, few days (b) Controlled acid hydrolysis (e.g., 6M HCl, 18°C, 6 h)	Mechanism involves (reversible) N-O peptidyl rearrangement in Ser. Yield moderate. Thre-X bonds cleaved at lower yield, and others non-specifically at low yield. Tyr sulfonation (in H ₂ SO ₄) and Trp destruction may occur	11
X↓Thr	As for X-Ser (N→O shift mechanism)—q, v	See Ser-X (N→O shift mechanism). Yields moderate to poor	11

Tyr↓X	As for His-X (N-bromosuccinimide)— <i>q v</i>	Tyr-X cleaved more quickly than His-X, more slowly than Tryp-X (<i>q v</i>) Yields moderate to poor, or zero, if Tyr is at the N-terminus Met and Cys may undergo oxidation Tyr converted to lactone derivative	10
Tyr↓X	Electrolytic oxidation at Pt electrodes in pH 2.2–2.4 (e.g., 10% acetic acid), 4–5 h	Oxidation of other groups may occur Yields moderate to poor	12
Tryp↓X	Incubation in BNPS-skatole (100-fold molar excess over Tryp) in 50% acetic acid (v/v) room temperature, 48 h, in the dark	Reagent is unstable—fresh reagent required to minimize side reactions. Some reaction with Tyr may occur Met and Cys may be oxidized Yields moderate to good	13
Tryp-X↓	As for His-X (N-bromosuccinimide)— <i>q v</i>	Also less rapid cleavage at His-X and Tyr-X (<i>q v</i>) Yields moderate to poor Tryp converted to lactone derivative	10
Tryp↓X	Incubation in very large (up to 10,000-fold) molar excess of cyanogen bromide over Met, in heptafluorobutyric acid 88% H COOH (1/1 v/v), room temperature, 24 h, in the dark	To inhibit Met-X cleavage, Met is photooxidized irreversibly to Met sulfone Yield poor	14
Tyrp↓X	(a) Incubation in glacial acetic acid 12M HCl (2/1, v/v) with phenol and dimethylsulfoxide, room temperature, 30 min (b) Add HBr and dimethylsulfoxide, room temperature, 30 min	Cys and Met oxidized Some hydrolysis of acid-labile bonds around Asp may occur as may deamidation Fresh (colorless) HBr required Yields moderate Tryp converted to dioxindolalanyl lactone	15
Tryp↓X	(a) Incubation in glacial acetic acid 9M HCl (2/1, v/v) with DMSO, room temperature, 30 min (b) Neutralization (c) Incubation with cyanogen bromide in acid (e.g., 60% formic acid) 4°C, 30 h in the dark	Yields and specificity excellent Met oxidized to sulfoxide.	3
Tryp↓X	Incubation in 80% acetic acid containing 4M guanidine HCl, 13 mg/mL iodosobenzoic acid, 20 μL/mL <i>p</i> -cresol, for 20 h room temperature, in the dark	Specificity and yields good The <i>p</i> -cresol is used to prevent cleavage Tyr Tryp converted to lactone derivative.	16

During the oxidation step, methionyl residues are converted to sulfoxides, so bonds at these residues are *not* cleaved by the cyanogen bromide treatment. Cysteinyl residues will also suffer oxidation if they have not been reduced and alkylated beforehand (see Chapter 7).

The peptide to the C-terminal side of the cleavage point has a free N-terminus and so is suitable for sequencing.

2. Methionyl sulfoxide residues in the peptides produced may be converted back to methionyl residues by the action, in aqueous solution, of thiols (e.g., dithiothreitol, as described in ref. 3, or see use of 2-mercaptoethanol above).
3. The acid conditions used for this reaction seem to cause little deamidation (3), but one side reaction that can occur is hydrolysis of acid-labile bonds. The use of low temperature minimizes this problem. If a greater degree of such acid hydrolysis is not unacceptable, speedier and warmer alternatives to the reaction conditions described above can be used as follows:
 - a. Oxidation—at room temperature for 30 min, but cool to 4°C before neutralization.
 - b. Cleavage at room temperature for 12–15 h.
4. As alternatives to the volatile base, NH_4OH , other bases may be used (e.g., the nonvolatile potassium hydroxide or Tris base).
5. As an alternative to formic acid (used in the cleavage step), 5M acetic acid may be used.
6. Samples eluted from sodium dodecyl sulfate (SDS) gels may be treated as described, but for good yields of cleavage, Huang et al. (3) recommend that the sample solutions are acidified to pH 1.5 before lyophilization in preparation for dissolution in the

oxidizing solution. Any SDS present may help to solubilize the substrate and, in small amounts at least, does not interfere with the reaction. However, nonionic detergents that are phenolic or contain unsaturated hydrocarbon chains (e.g., Triton, nonidet P-40), and reducing agents, are to be avoided.

7. The method is suitable for large-scale protein cleavage, this requiring simple scaling up. Huang et al. (3) make two points, however:
 - a. The neutralization reaction generates heat. Since this might lead to protein or peptide aggregation, cooling is important at this stage. Ensure that the reagents are cold and are mixed together slowly and with cooling. A *transient* precipitate is seen at this stage. If the precipitate is insoluble, addition of SDS may solubilize it (but will not interfere with the subsequent treatment.)
 - b. The neutralization reaction generates gases. Allow for this when choosing a reaction vessel.
8. The results of cleavage may be inspected by polyacrylamide gel electrophoresis, HPLC, and so on, as after cleavage at Met-X bonds (see above). All reagents in the described method are readily removed during lyophilization.

4.3. Cleavage at Asp-X Bonds

1. The bond most readily cleaved in dilute acid is the Asp-X bond, by the mechanism outlined in Fig. 2(i). The bond X-Asp may also be cleaved, in lesser yields—see Fig. 2(ii). Thus, either of the peptides resulting from any one cleavage may keep the aspartyl residue at the point of cleavage, or neither

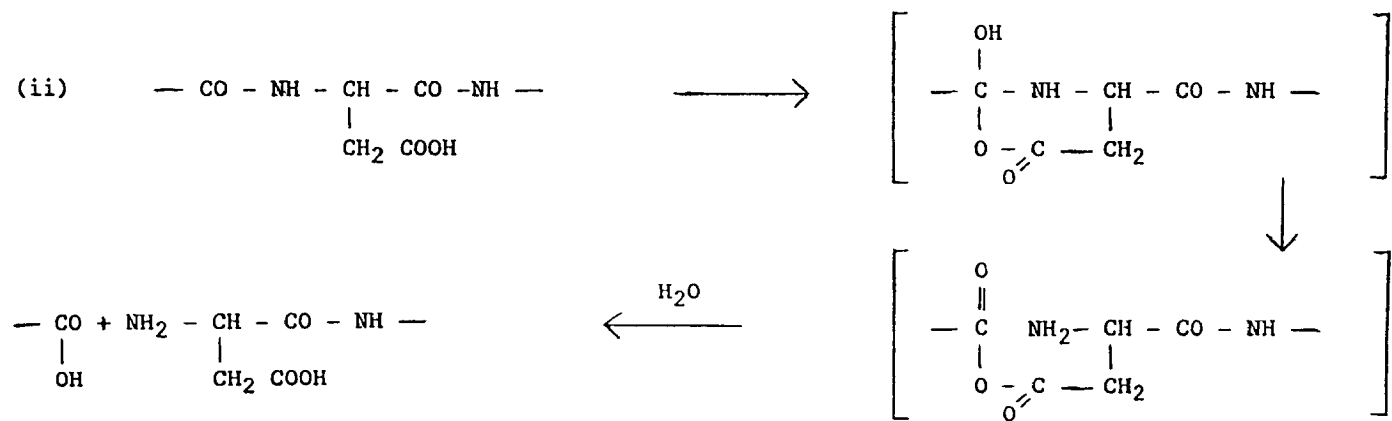
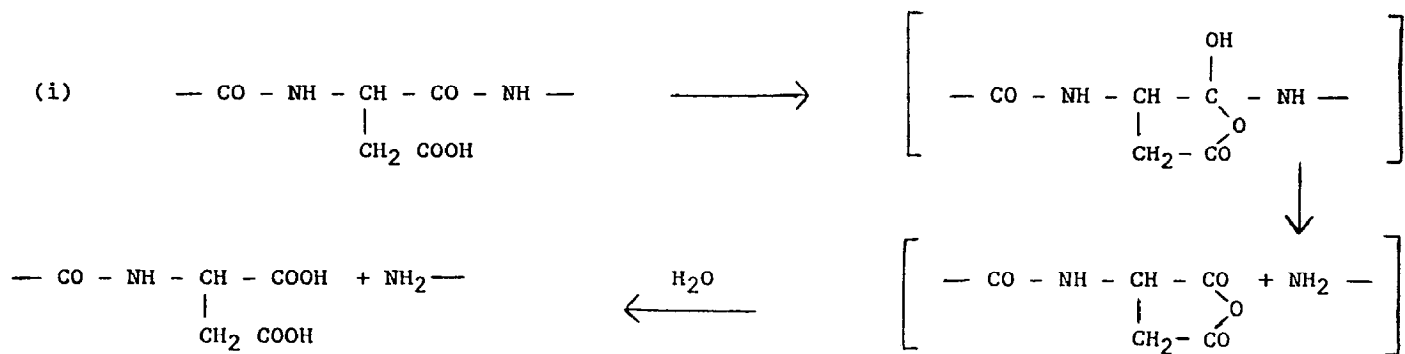


Fig. 2. Mechanisms of the cleavage of bonds to the COOH side (scheme i) and to the NH₂ side (scheme ii) of aspartyl residues in dilute acid.

might, if free aspartic acid is generated by a double cleavage event. Any of these peptides is suitable for sequencing.

The amino acid sequence of the protein can affect the lability of the affected bond. Thus, the Asp-Pro bond is particularly labile in acid conditions (*see* Table 1). Again, ionic interaction between the aspartic acid side chains and basic residue side chains elsewhere in the molecule can adversely affect the rate of cleavage at the labile bond. Such problems as these make prediction of cleavage points somewhat problematical, particularly if the protein is folded up (e.g., a native protein). The method may well prove suitable, however, for use in cleaving small proteins or peptides.

2. The conditions of low pH can be expected to cause a number of side reactions: cleavage at glutamyl residues; deamidation of (and subsequent cleavage at) glutaminyl and asparaginyl residues; cyclization of N-terminal glutaminyl residues to residues of pyrrolidone carboxylic acid; α - β shift at aspartyl residues, creating a blockage to Edman degradation; partial destruction of tryptophan. The short reaction time of 2 h is intended to minimize these side reactions. Loss of formyl and acetyl groups from N-termini (4) are other possible side reactions, but these may be advantageous in deblocking otherwise unsequenceable polypeptides.
3. A substrate that is insoluble in cold dilute HCl may dissolve during incubation at 108°C. Alternatively, formic acid, a good protein denaturant and solvent, may be used, as follows: dissolve the sample in formic acid, 98%, then dilute 50-fold to pH 2.
4. The comments above concerning the effect of the amino acid sequence and of the environment

around potentially labile bonds, and the various side reactions that can occur, indicate that the consequences of incubation of a protein in dilute acid are difficult to predict—they are best investigated empirically by monitoring production of peptides by electrophoresis, HPLC, and so on (as suggested for cleavage of Met-X, above).

Yields of cleavage of Asp-X bonds of up to about 70% have been reported (4). Such partial hydrolysis at just a few sites can give rise to a complicated mixture of peptides. For this reason this method may not be the one of choice for proteins, but may be suitable for small proteins or peptides containing few aspartyl residues, while also having the advantage of speed and simplicity. It is also readily adapted for large or small (subnanomolar) amounts of substrate.

5. Other Methods of Chemical Cleavage

1. A summary of various of common chemical methods of protein cleavage is given in Table 1. The literature contains descriptions of many other, less frequently used techniques (e.g., *see* ref. 1). Reaction mechanisms are not known for all methods. Yields are generally less than 100% and often vary according to the nature of the substrate, surrounding amino acid sequence, exact conditions of reaction, and so on, but for guidance are put into broad classes in Table 1.
2. The most useful and widely used techniques provide good specificity at good yield, but some of the less likely techniques may also prove useful in particular instances.

The simplicity of a method is also worthy of consideration. Thus cleavage at Met-X can be conducted in one vessel, and losses experienced in manipulation of the sample are minimized. On the other hand, cleavage at X-Cys bonds, as outlined in Table 1, involves dialysis and further treatment to deblock peptides for sequencing. Accumulative losses may limit the minimum size of samples required for this method.

3. Various of the techniques generate a lactone derivative at the C-terminus of the peptide to the NH_2 side of the cleavage point (*see* Table 1). This may be useful for the selective attachment of these peptides to amino supports such as 3-amino propyl glass (5), for the purposes of solid phase protein sequencing.
4. Because of the variety of chemical techniques available, it is not easy to generate general rules. One point for consideration, however, is denaturation of the substrate. It is as well to reduce and alkylate cystinyl residues (*see* Chapter 7) so as to minimize the problems caused by aggregation of proteins or peptides. This is obviously not recommended if positions of disulfide bonds are of interest. If this is the case, be wary of strong acid (e.g., concentrated HCl) or alkaline conditions in which disulfide interchange may occur. Again, folding of the polypeptide may affect reactivity by allowing interaction of side chains. Denaturation by inclusion of agents such as guanidine HCl can help to overcome this and to solubilize the substrate and peptides from it.

Acknowledgments

I wish to thank my colleagues for useful discussions.

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Chapter 7

Chemical Modification of Proteins

Alex F. Carne

1. Introduction

The two most widely used applications for chemical modification are in primary structure analysis and in the identification of essential groups involved in the binding and catalytic sites of proteins. The methods discussed here are those used frequently in primary structure analysis. Chemical modifications involving protein "active center" identification are the subject of a review by Pfeleiderer (1).

With the advent of the gas-phase sequencer (2), the amounts of protein necessary for sequence analysis are

in the low picomole range, and it is therefore necessary to modify some of the more well-known methods to suit the smaller amounts of protein.

Cysteine presents a problem both in amino acid and sequence analysis. It is subject to autooxidation, leading to a variety of products such as mixed disulfides (3), and is involved in the formation of protein disulfide bridges. The presence of disulfide bridges within a polypeptide chain leads to difficulties in peptide isolation after enzymic digestion and in sequence determination. Reductive alkylation with iodoacetic acid yields the more stable carboxymethyl derivative and also introduces a charged group that tends to enhance the solubility of the polypeptide in aqueous buffers at alkaline pH. This makes the protein more susceptible to digestion by proteases such as trypsin, chymotrypsin, and so on. Also, using [^{14}C]-iodoacetic acid as the alkylating agent enables a radioactive label to be introduced into the polypeptide chain.

For quantitative amino acid analysis involving cystine or cysteine residues, carboxymethylation is not ideal since it is difficult to assess the completeness of the reaction. A better method is to oxidize the S-S or SH groups using performic acid (3), hydrolyze the polypeptide, and identify the cystine/cysteine as cysteic acid.

A further useful modification is the succinylation of the ϵ -amino lysine groups. This is an irreversible reaction and hinders tryptic attack at lysine residues, thus giving rise to peptides with arginine at the carboxy-terminus. As with carboxymethylation, the use of [^{14}C]-succinic anhydride allows for the incorporation of a radiolabel, and proteins modified by dicarboxylic anhydrides are generally more soluble at pH 8, and the polypeptide chains are unfolded thus allowing for ready digestion with proteases.

Aminoethylation of cysteine residues, to the *S*- β -aminoethyl derivatives, can be used to introduce new points of tryptic cleavage into a polypeptide chain. Trypsin cleaves peptide bonds involving the aminoethyl derivative since it is an analog of lysine. Cleavage is much slower, however, than that at lysine or arginine residues (4). The positively charged derivative is prepared by reaction of the reduced thiol containing protein with ethyleneimine.

2. Materials

2.1. Reductive Alkylation

1. Denaturing buffer: 6M guanidinium chloride, 0.1M Tris-HCl, pH 8.5.
2. A source of oxygen-free nitrogen.
3. Dithiothreitol (DTT): 4 mM in distilled water. This solution should be freshly made.
4. Iodoacetic acid: 500 mM in distilled water, pH adjusted to 8.5 with sodium hydroxide. Iodoacetic acid is light-sensitive, and it is preferable to make up a fresh solution each time, though the solution can be stored in the dark at -20°C .
5. Ammonium bicarbonate: 50 mM in distilled water, pH approximately 8.0.
6. Microdialysis equipment (as supplied by BRL).
7. HPLC system with a Brownlee RP-300 "cartridge"-type column fitted (30 $\times$ 2.1 mm).
8. 0.1% TFA (HPLC grade).
9. Acetonitrile (HPLC, far UV grade).

2.2. Performic Acid Oxidation

1. Hydrogen peroxide: 30% w/v. Use care! This is a strong oxidizing agent.

2. Formic acid: 88% w/v.
3. Hydrobromic acid: 48% w/v. This acid gives off a caustic, irritating vapor, and its use should be confined to a fume hood.

2.3. Succinylation

1. Guanidinium chloride: 6M in distilled water.
2. Succinic anhydride.
3. Sodium hydroxide: 1M in distilled water.
4. Micro-pH probe.
5. Ammonium bicarbonate: 50 mM in distilled water, pH approximately 8.0.
6. Microdialysis or HPLC equipment.

2.4. Aminoethylation

1. Denaturing buffer: 6M guanidinium chloride, 1.0M Tris-HCl, 0.025% β -mercaptoethanol, pH 8.9.
2. Source of oxygen-free nitrogen.
3. Ethyleneimine (Use care—suspected carcinogen).
4. Ammonium bicarbonate: 50 mM in distilled water, pH approximately 8.0.
5. Microdialysis equipment.

3. Method

3.1. Reductive Carboxymethylation

1. Dissolve the protein in 50 μ L of denaturing buffer in a 1.5-mL Eppendorf tube.
2. Gently blow N_2 over the top of the solution for 15 min.
3. Add an equal volume of DTT solution to a concentration of 2 mM.

4. Blow N_2 over the solution for 60 min.
5. Wrap the Eppendorf tube in aluminium foil before the addition of iodoacetic acid.
6. 40 μ L of iodoacetic acid solution is added dropwise using a micropipet. During this addition the solution should be stirred. (It is possible to obtain microstirrer bars that will fit Eppendorf tubes.)
7. N_2 is again blown over the surface of the solution and the reaction tube sealed.
8. Incubate in the dark at 37°C for 30 min. The reaction is carried out in the dark to prevent the formation of iodine, from the iodoacetate, thus preventing the iodination of tyrosine residues.
9. Remove excess reagents. This may be performed in one of two ways:
 - a. Microdialysis: Here, the solution is pipetted into a 1–2 mL perspex chamber under which there is a sheet of dialysis membrane (mol. wt. cut-off, 6000–8000). The underside of the membrane is in contact with the dialysis buffer, usually 50 mM ammonium bicarbonate. Dialysis is carried out for about 24 h.
 - b. HPLC: The protein can be separated from the reaction byproducts by reverse-phase HPLC. To minimize product loss, a 30 x 2, 1-mm microbore "cartridge" column is employed. For proteins, a C3 or C8 matrix with a 10- μ m particle diameter and a 30-nm pore size can be used (e.g., Aquapore octyl RP-300). The column is equilibrated in 0.1% aqueous trifluoroacetic acid. The sample is injected onto the column and eluted using an acetonitrile gradient (1%/min, flow rate 0.2 mL/min). The excess reagents and reaction byproducts all appear in the breakthrough, i.e., they are not retained by the column packing material.

3.2. Performic Acid Oxidation

1. Add 100 μL of hydrogen peroxide to 900 μL of formic acid and allow to stand at room temperature for 1 h. This produces performic acid (HCOOOH).
2. Cool the performic acid on ice to approximately 0°C .
3. Dissolve the protein in 50 μL of performic acid (about 400–500 $\mu\text{g}/\text{mL}$), in a precooled tube.
4. Keep at 0°C for 4 h.
5. Add 7.5 μL of *cold* HBr to neutralize the performic acid. Care must be taken at this stage since bromine will be liberated, albeit in small amounts.
6. Dry the sample thoroughly *in vacuo* over NaOH to remove bromine and formic acid.
7. The sample can then be hydrolyzed and subjected to amino acid analysis.

3.3. Succinylation

1. Dissolve the protein or protein derivative in 200 μL of 6M guanidinium chloride, and using a micro-pH probe adjust the pH to 9.0 with the sodium hydroxide solution.
2. Crush 1–2 crystals (1–2 mg) of solid succinic anhydride and add to the stirred solution over a period of about 15 min, while maintaining the pH of the solution at 9.0 with 1.0M NaOH .
3. Excess reagents are removed by microdialysis or by HPLC.

3.4. Animoethylation

This method is similar to that used by Slobin and Singer for γ -globulin (5).

1. Dissolve the protein (~0.5 mg) in 100 μ L of denaturing buffer in a 1.5-mL Eppendorf tube.
2. Flush with N_2 , cap, and leave for 2 h.
3. Add 3 μ L of ethyleneimine.
4. Flush with N_2 , cap, and allow to alkylate for 2 h at room temperature.
5. To remove excess reagents, dialyze against 50 mM ammonium bicarbonate and then distilled water, using the microdialysis apparatus.

4. Notes

1. Reagents for all the methods should be of the highest quality available.
2. The iodoacetic acid used for reductive alkylation must be colorless; any iodine present, revealed by a yellow color, causes rapid oxidation of thiol groups, preventing alkylation and possibly modifying tyrosine residues.
3. If good quality iodoacetic acid is not available, it may be recrystallized from hexane.
4. The method described for reductive alkylation does not include the use of iodo 2- $[^{14}C]$ -acetic acid. The radiolabeled iodoacetic acid should be diluted before use with carrier iodoacetic acid to the required final specific activity, according to the requirements of the worker.
5. Continuous stirring during the addition of the iodoacetic acid is necessary to keep the pH constant throughout the solution, especially when dealing with small volumes.
6. Dialysis tubing should be avoided since the losses involved when handling small amounts of protein can be high. The microdialysis system that uses a

sheet of dialysis membrane is preferable, but losses can still occur. If the reader has access to an HPLC system this should be the method of choice.

7. During performic acid oxidation, the temperature of all the reactants must be at or near 0°C to minimize any side reactions, such as the oxidation of phenolic groups and the hydroxyl groups of serine and threonine.
8. Oxidation of methionine occurs during performic acid oxidation producing methionine sulfone, which can be quantified by amino acid analysis.
9. The addition of HBr in the performic acid method causes bromination of tyrosine residues, producing a mixture of mono- and di-substituted bromotyrosines. These derivatives have lower color values than tyrosine, and quantitation is generally not good after acid hydrolysis.
10. Since several residues are modified, performic acid oxidation is best used in conjunction with amino acid analysis. It is not recommended to use performic acid-oxidized samples for sequence analysis.
11. Succinylation is best carried out in proteins that have been carboxymethylated to prevent S-S or SH groups from reacting. This gives complete selective modification.
12. The addition of small amounts of succinic anhydride over a 15-min time period enables the worker to control the pH, so that it can be maintained at or near pH 9.0.
13. For the removal of excess reagents following succinylation, the same criteria apply as for carboxymethylation.
14. Succinylation is a better method than maleylation or citraconylation for the modification of lysine residues, since it prevents problems arising from

the still-reactive ethylenic groups. If there is a need for the removal of the blocking groups, however, for subsequent tryptic cleavage at lysine residues, maleylation or citraconylation should be used since the groups introduced by citraconic or maleic anhydride can be removed by treatment with dilute acid.

15. Both clostripain (E.C. 3.4.22.8) and endoproteinase Arg-C (6) can be used for cleavage at arginine residues, instead of succinylation followed by trypsin digestion. Clostripain may cleave at other sites (7), however, and endoproteinase Arg-C, although highly specific for the C-terminal side of arginine residues, does cleave some arginine peptide bonds slowly (8). Thus, succinylation is still of use to the protein chemist trying to achieve specific cleavages coupled with the proper of enhanced solubility.
16. If the aminoethylation reaction is allowed to proceed for a long time with an excess of ethyleneimine, alkylation of methionine (9) or of the α -amino group of the peptide (10) can occur.
17. S- β -Aminoethylcysteine can be detected by amino acid analysis, eluting with the basic amino acids from a cation-exchange column (4).

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Chapter 8

The Design, Preparation, and Use of Immunopurification Reagents

***Andrew Kenney, Lynne Goulding,
and Christopher Hill***

1. Introduction

Immunoaffinity chromatography using immobilized monoclonal antibodies has become one of the most powerful macromolecular separation methods available. In many cases, simple one-step immunoaffinity separations will produce a several thousand-fold purification with yields of better than 80%. Monoclonal antibodies are now available with specificities for many different antigens, and the technology for producing new monoclonal antibodies has become widespread throughout the world since its invention (1; see vol. 1 of this series). It has been estimated that a new monoclonal antibody is produced every 40 min worldwide.

The wide-ranging use of monoclonal antibodies for immunopurification (IMP) is best illustrated by way of example (Table 1).

Table 1
Examples of Antigens Purified Using Immobilized MAbs

Antigen	Example	Reference
Interferons	Interferon- α	2
Lymphokines	Interleukin-2	3
Enzymes	Tissue plasminogen activator	4
Plasma proteins	Factor I	5
Peptide hormones	Prolactin	6
Immunoglobulins	Rat MAb	7
Membrane-bound proteins	Murine MHC antigens	8

An examination of these examples will demonstrate that IMP is a versatile and powerful technique. It will also be apparent that a number of different strategies have been used for different systems. It is not possible here to explore all aspects of the development of an efficient immunopurification method; rather we have attempted to provide a method in detail that will allow investigators to discover the potential that immunopurification may have for their own application.

In order to exploit this technology successfully, it is necessary to know how best to prepare the reagent. Many different matrices and coupling chemistries have been developed that are suitable for the immobilization of monoclonal antibodies (Table 2). The choice for research laboratory applications will normally be dictated by the availability and ease of preparation, and in most cases this means cyanogen bromide-activated agarose. The method described here will in most cases prove reli-

Table 2
Matrices Available for Immobilization of Antibodies for Affinity Chromatography

Class	Commercial name	Support material	Coupling chemistry	Supplier	Pore size exclusion
Natural polymers	CNBr-Sepharose 4B	Agarose	CNBr	Pharmacia	20 x 10 ⁶
	CM Bio gel A	Agarose	EDAC	Bio Rad	—
	Affigel 102/202	Agarose	EDAC	Bio Rad	—
	CNBr-Sepharose CL4B	Crosslinked agarose	CNBr	Pharmacia	20 x 10 ⁶
	Epoxy-Sepharose CL4B	Crosslinked agarose	Epoxide	Pharmacia	4 x 10 ⁶
	Affigel 10/15	Crosslinked agarose	N-Hydroxysuccinimide	Bio Rad	5 x 10 ⁶
	Reacti gel (6x)	Crosslinked agarose	CDI	Pierce	4 x 10 ⁶
	Reacti gel (25 DF)	Crosslinked agarose	CDI	Pierce	5 x 10 ³
Synthetic polymers	Reacti gel (HW 65F)	Fractogel	CDI	Pierce	5 x 10 ⁶
	Avid gel	Trisacryl	FMP	Bio Probe	20 x 10 ⁶
	Eupergit C	Acrylic beads	Epoxide	RhonePharma	100–2500 nm
	Amino ethyl Bio gel P2/P150	Polyacrylamide	EDAC	Bio Rad	—
Inorganic	Glycophase	Controlled pore glass	CDI	Pierce	20 nm
	HP beads	Controlled pore glass	—	Chemicon	—
	Biomag	Magnetic oxide	Glutaraldehyde	Advanced	—
	Various	Silica	Various	Magnetics, Inc.	—
				Boehringer	200 nm
				Beckman	—
				Chemapol	—
				Diagen	50–1000 nm
Composite	Acrobeads	Agarose/polyacrolein	Glutaraldehyde	Galil	—
	Ultrogel AcA 22	Agarose/acrylamide	Glutaraldehyde	LKB	—
Membrane	Biodyne	Nylon membrane		PALL	—

able and reproducible and is easy and convenient to use. In some cases, however, other matrices and chemistries will produce much better results. When the antigen to be purified is of a very high molecular weight, the choice of a matrix with a very large pore size will enable more efficient use of the antibody. This is because antibody that becomes immobilized well within the matrix bead will not be available to an antigen that is too large to diffuse into the pores of the bead. In addition, for applications in which leakage of the antibody from the matrix into the product is undesirable, it may be advantageous to choose an alternative coupling chemistry to CNBr. It is not possible to produce rules for use in choosing the matrix or chemistry, but the method described here will allow investigators to begin studying the performance of their antibody for immunopurification. Later on it may be necessary to investigate alternatives in order to enhance the performance in individual applications.

The development of a method for using the prepared immunosorbent is usually an empirical process, especially with regard to the choice of eluant. The immunosorbent must first be characterized in terms of its capacity for antigen under the conditions to be employed. Failure to do this will either result in a marked under-utilization of the immunosorbent or in a loss of antigen in the unbound protein fraction. Depending on the affinity of the antibody for antigen, the eluant will either be a simple change in pH and ionic strength or may need to make use of denaturants and chaotropes. A list of eluants that have been found useful is given in Table 3. These have been presented in approximately the order in which they should be tried. Individual circumstances, however, particularly in respect to the stability of antibody and antigen in these solutions, will mitigate against the use of some of them.

Table 3
Typical Elution Condition Used
for Immunoaffinity Chromatography

Eluant	Reference
<i>Extremes of pH</i>	
0.1M Glycine/HCl, pH 2.5	9
1.0M Propionic acid	10
0.15M NH ₄ OH, pH 10.5	11
0.1M Na CAPS, pH 10.7	12
<i>Chaotrophic agents</i>	
4.0M MgCl ₂ , pH 7.0	13
2.5M NaI, pH 7.5	14
3.0M NaSCN, pH 7.4	15
<i>Protein-deforming agents</i>	
8M Urea, pH 7.0	16
6M Guanidine/HCl	17
<i>Organic solvents</i>	
50% v/v Ethanediol, pH 11.5	18,19
Dioxane/acetic acid	

2. Materials

1. Monoclonal antibody at greater than 80% purity.
2. CNBr activated Sepharose (Pharmacia).
3. Coupling buffer: 0.1M sodium hydrogen carbonate (NaHCO₃), pH 8.3, containing 0.5M sodium chloride—to be made fresh daily.
4. 1mM Hydrochloric acid.
5. Acetate buffer, pH 4: 0.1M sodium acetate containing 0.5M sodium chloride adjusted to pH 4.0 with glacial acetic acid—to be made fresh daily.
6. Ethanolamine solution: 1M ethanolamine adjusted to pH 8 with concentrated HCl. Note: Make up ethanolamine to about half the desired volume and

add conc. HCl dropwise. This evolves a considerable amount of heat and requires quite a large volume of HCl to bring the pH to 8. Ethanolamine is also an irritant, which should not be inhaled.

7. Phosphate-buffered saline (PBS): 20 mM sodium phosphate buffer pH 7.2 containing 0.9% sodium chloride.
8. Phosphate-buffered saline with preservative: PBS containing 0.1% sodium azide.
9. pH 2.5 buffer: 0.1M glycine/HCl pH 2.5.

3. Methods

All procedures carried out at room temperature unless otherwise stated.

3.1. Immobilization of Monoclonal Antibody

1. Prepare the purified monoclonal antibody (MAb) in coupling buffer at a concentration of 1–5 mg/mL. An estimate of mouse MAb concentration can be obtained by measuring absorbance at 280 nm and by calculation as follows:

$$(\text{MAb}) = A_{280} / 1.4 \text{ mg/mL}$$

The MAb solution should be prepared and chilled to 4°C before proceeding to the next step.

2. Weigh desired amount of CNBr-activated Sepharose 4B: As a guide 0.3 g swells to produce approximately 1.0 mL gel. Calculate the volume of gel required on the basis of 1.0 mL of gel for every 5 mg of antibody to be coupled.
3. Swell and wash the CNBr-activated Sepharose with 1mM HCl, using 200 mL of HCl/g of Sepha-

rose. Sprinkle the Sepharose onto a small volume of 1 mM HCl contained in a sintered glass funnel (G3), and very gently stir the gel to ensure even swelling. Allow the gel to stand for 15 min at room temperature in 1 mM HCl, ensuring that the gel does not dry out. Then wash through the remaining 1 mM HCl under gentle suction.

4. Wash the swollen gel with 10 gel vol of cold coupling buffer (4°C) to chill the gel, and at the same time equilibrate in coupling buffer. Preparation of the gel for coupling should take approximately 30 min (from weighing to adding the antibody). Coupling capacity of the gel can be reduced if it is allowed to stand for longer at room temperature.
5. Add the swollen, washed, and chilled CNBr-activated Sepharose to 5 vol of monoclonal antibody solution (*see* 1 above) also cooled to 4°C, and allow the mixture to roll gently overnight at 4°C (*see* Note 6 in section 4).
6. Filter the coupling mixture through a sintered glass funnel (G3) using a gentle suction. Add an equal volume of cold coupling buffer to the gel slurry and mix gently to wash away unbound antibody. Drain this through the sintered glass funnel under gentle suction and combine the washes to determine the total amount of unbound antibody.
7. Return the gel slurry to the mixing vessel and add 5 vol of 1M ethanolamine pH 8 to block the unreacted CNBr-activated sites. Roll the slurry for 2 h at room temperature.
8. Filter the blocking agent away from the gel and wash the gel with three cycles of pH 4 and pH 8 buffers to remove noncovalently bound proteins as follows:
 - a. Mix the blocked gel with 10 gel vol of pH 4 acetate

- buffer. Check that the pH of the mixture is close to 4 and then allow the mixture to stand for at least 5 min before removing the buffer under gentle suction.
- b. Add 10 gel vol of pH 8.3 coupling buffer to the slurry. Check that the pH is close to 8 and allow this to stand for at least 5 min and then remove using gentle suction.
 - c. Repeat steps (a) and (b) twice.
9. Finally wash the gel with pH 8.3 coupling buffer containing 0.1% sodium azide and store the gel in this solution at 2–4°C. *Do Not Freeze.*
 10. To facilitate measuring known volumes of gel, it is recommended that the gel be stored as a 50% (v/v) gel buffer mixture. This is best done by allowing the gel to settle in a suitable container (at least 2x larger than gel volume) at 4°C for 24 h. When the gel has settled, the volume of the storage buffer can then be corrected to the same volume as the gel. When taking samples of gel from the vessel, the slurry should be gently mixed to an even consistency before removing twice the desired gel volume.

3.2. Affinity Chromatography Using the Immobilized Monoclonal Antibody

1. The 50% (v/v) gel slurry should be allowed to warm to room temperature.
2. Mix the gel slurry gently to form an even suspension, and then remove twice the desired gel volume.
3. Pack the MAb-Sepharose gel into a suitable column using an appropriate running buffer. We suggest PBS (*see* Note 8 in section 4) using a flowrate of 10 mL/h.

4. Pass through the column 5 column vol of elution buffer (Table 3). In this protocol, we suggest a buffer of pH 2.5. This step will substantially reduce the amount of antibody that might otherwise leak into the eluted product.
5. Equilibrate the column with 10 column vol of running buffer.
6. Apply the crude sample to the column. Ideally the sample should be equilibrated in running buffer.
7. Wash the sample onto the column with 5 column vol of running buffer. If the Ag:Ab interaction is shown to be strong enough to withstand high salt (e.g., up to 1M KCl), then this can be included in the washing step to improve the purity of the eluted material.

Before applying elution buffer, check that the A_{280} has returned to base line. Elute the bound antigen with 2 column vol of elution buffer (Table 3). If using an extreme condition, then the eluate should be neutralized upon collection (e.g., collect pH 2.5 eluate into 2M Tris-HCl pH 8.3).

8. The column can be regenerated by washing with 10 column vol of PBS/0.1% sodium azide. If all the antigen does not elute, then it is necessary to remove this before re-equilibration using more extreme eluant (e.g., pH 2).
9. The column should be stored in a preservative at 4°C between runs, e.g., it should be left packed if possible. Avoid handling the gel any more than necessary.

4. Notes

1. The size of antigen can influence the nature of the matrix chosen. Here we have illustrated the use of

Sephacryl, which is suitable for the majority of proteins, but if large proteins or protein complexes are being purified, then pore size must be a consideration. The nature of the matrix can affect the flow properties.

2. Coupling chemistries: Many different matrix types are available commercially with a variety of available coupling chemistries (*see* Table 2).
3. CNBr Sepharose CL 4B is available from Pharmacia and is preferable to noncrosslinked material.
4. *Never* allow the gel to dry out, especially when under suction.
5. The activated gel is incubated at 4°C prior to adding antibody because at higher temperatures the antibody would couple more quickly and tend to be mopped up by CNBr sites on the outside of the beads. This wastes sites on the inside of the beads and could crowd the outside sites enough to reduce efficiency.
6. Agitation machines come in various forms. We recommend a roller that imparts a continuous movement in one direction, e.g., shakers that change direction can inflict unnecessary shear forces.
7. The blocking reagent should contain an amine group attached to a moiety that will not impart ion-exchange properties or a hydrophobic character to the support, which could result in enhanced non-specific binding of proteins. Ehanolamine is commonly used because the moiety that becomes immobilized is small and is hydrophilic without carrying a charge.
8. Most generally available biochemical buffers are suitable for use with immunopurification. We have found phosphate or Tris to be useful in many situ-

ations. In general the buffer should be chosen to maintain a pH at which the ligand to be purified is stable. Buffers that can interfere with subsequent analysis should be avoided when practicable.

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Chapter 9

Dye–Ligand Chromatography

Sarojani Angal

1. Introduction

The affinity of two kinases for Blue Dextran was discovered in 1968 during gel permeation chromatography (1) and agarose electrophoresis (2). The observed interaction was mediated by the chromophore of Blue Dextran, Cibacron Blue F3G-A (3), which was postulated to bind to the supersecondary structure of certain enzymes known as the dinucleotide fold (4). Immobilized Cibacron Blue F3G-A replaced more expensive biological group-specific ligands in the affinity chromatographic purification of a wide variety of nucleotide-binding enzymes (5). The adsorbent was easier to synthesize, biochemically more stable, and had much higher capacities than the traditional nucleotide-based affinity matrices, making it more suitable for large-scale purifications.

Although the dinucleotide fold hypothesis was under scrutiny in many laboratories worldwide, a large number of other fiber-reactive dyes became available and proved their usefulness as "general" ligands in enzyme and protein purification (6,7). It has now become evident that the special place afforded to Cibacron Blue in dye ligand chromatography is merely historical.

The dyes appear to interact with ligand binding pockets or grooves on proteins through both hydrophobic and ionic forces complementary to those of the protein. The structures of most of the dyes are undisclosed, and the three-dimensional structures of most of the proteins of interest to the purification chemist are unknown. Thus it is not possible at present to predict which dye will provide the "best fit" for a given protein. The selection of a dye for protein purification must of necessity be empirical. The inexpensiveness of the dyes makes this possible even for the most modestly equipped laboratory. This chapter details methods for the synthesis and systematic screening (8,9) of a range of dye-adsorbents, together with guidance for the design of a purification scheme and for the optimization of operational parameters.

2. Materials

1. A selection of reactive dyes (e.g., Procion MX, H, HE, and P series available from ICI, Organics Division, Blackley, Manchester, UK. For other manufacturers, consult ref. 10). Some dyes are available from suppliers such as Sigma. Table 1 shows a list of suitable dyes.
2. Beaded agarose (e.g., Sepharose 4B from Pharmacia). About 40 mL of slurry per dye (*see* section 3).
3. Sodium chloride, 20% w/v.

Table 1
Examples of Some Triazine Dyes Suitable
for Making a Screening Kit^a

Monochlorotriazines	Dichlorotriazines
Procion Blue H-B	Procion Brown MX-5BR
Procion Blue HE-RD	Procion Blue MX-4GD
Procion Red H-8BN	Procion Blue MX-R
Procion Green H4G	Procion Rubine MX-B
Procion Green HE-4BD	Procion Scarlet MX-G
Procion Brown H-2G	Procion Orange MX-G
Procion Red HE-3B	Procion Yellow MX-8G
Procion Violet H-3R	
Procion Turquoise H-A	
Procion Yellow H-A	

^aDyes near the top of the lists tend to bind more protein than those from the bottom of the lists. Consult refs. 9 and 10 for additional information.

4. Sodium carbonate, 5% w/v stock.
5. 10 mM Sodium phosphate (PB), pH 7.4.
6. 6M Urea.
7. Disposable plastic columns, 6 mm id, 1 mL bed volume and 5 mL reservoir volume (from, e.g., BioRad, Amicon).
8. 1M KCl in PB.
9. Crude extract containing protein of interest (e.g., human serum).
10. Materials required for assay of protein of interest.

3. Method

3.1. Preparation of Dye-Ligand Adsorbents

1. Wash the Sepharose repeatedly with a total of 10 vol of distilled water on a sintered glass funnel (porosity 3) under gentle suction pressure from a

water pump. Each time, allow surplus water to be removed to the point at which the surface of the gel is just beginning to "crack." This is referred to as moist gel.

2. Weigh out 20 g of moist gel into separate containers (one for each dye and one for undyed control). The container should have lids and wide necks for ease of transfer (e.g., 150 mL Sterilin plastic containers). Suspend in distilled water to a final volume of 50 mL.
3. Weigh ≥ 200 mg of each dye into a 25-mL container. Great care must be taken in weighing, since most of the dyes are extremely fine. Place a large sheet of white paper under the balance and moisten with water. Wear gloves and weigh into containers with lids. Wash the balance pan afterward, and wipe the surrounding area carefully.
4. Dissolve the dyes in distilled water to a concentration of 10 mg/mL.
5. Add 20 mL of the dye solution to the suspension of Sepharose. Add 20 mL of distilled water to the "control." Mix and leave at room temperature for 5 min.
6. Add 10 mL of the sodium chloride solution to each suspension. Incubate at room temperature for 30 min with occasional mixing.
7. Add 20 mL of the sodium carbonate stock solution to each suspension. Incubate the gels in a gently shaking water bath set at 45°C. Do not use magnetic stirrer bars, since these will damage bead structure. Dichlorotriazine-containing dyes (MX series) need to be incubated for 1 h; therefore the incubations should be staggered to allow for sufficient gel-washing time between preparation. Monochlorotriazines need to be incubated for 40 h.

8. Wash each dyed adsorbent with several liters of warm (45°C) distilled water followed by 10 vol of 6M urea and finally by distilled water alone until the washings are colorless. Some gel may get trapped in the sinter, particularly if the sinter is old, in which case it should be cleaned with a few milliliters of warm 6M HCl (or with chromic acid if unsuccessful).
9. Suspend the dichlorotriazine adsorbents in an equal volume of 1% (w/v) sodium carbonate and incubate at 45°C for 24–30 h to hydrolyze the second chlorine of the triazine ring. Wash once on a sintered glass funnel and suspend in 0.1% (w/v) sodium carbonate. Alternatively, decant the higher concentration of sodium carbonate and replace with water. Suspend the monochlorotriazine adsorbents in 0.1% (w/v) sodium carbonate.
10. Store the adsorbents at 4°C in the presence of 0.02% (w/v) sodium azide to inhibit microbial growth. Storage in mild alkali prevents acid hydrolysis of the dye from the matrix.

3.2. Screening of Dye Ligand Adsorbents

1. Pack 1 mL of adsorbent into each disposable column. It is best to code each adsorbent at this stage to save writing the full dye name on fractions later. The screening experiment is run at room temperature under gravity. Rest the "lip" or reservoir of the column above a 15-mL test tube, such that the surface of the gel is below the top of the tube. This arrangement can be engineered to prevent columns running dry. Place the test tubes in racks (LKB fraction collector racks from older models are best).

Each rack can accommodate 10 columns and the rack holders allow 20 columns to be run at once.) It is best to run all the selected columns at the same time, but since they will not all have identical flow properties, good organization is needed.

2. Prewash all columns with 1M KCl in PB (5 mL) and equilibrate with PB (10 mL). If any dye leaches from the columns, the prewash process will need to be repeated.
3. Adjust the protein concentration of the crude protein extract to about 20 mg/mL with equilibration buffer (previously dialyzed and clarified, if necessary). Apply 0.5 mL to each column, allow to run through, then collect the effluent. Add 0.5 mL of PB to the top of each column, allow to run through, and collect. This procedure allows proteins to reside in columns for similar amounts of time. Wash with a further 4 mL of PB and collect into the same tube.
4. Transfer the column to a new tube and elute with 1M KCl in PB (5 mL). Re-equilibrate in PB containing 0.02% sodium azide for storage.
5. Measure the absorbance at 280 nm of all the flow through and eluate fractions. Similarly measure the amount of the protein of interest. Compare both values with those of the "control" column, which represents the extent of nonspecific adsorption of proteins to underivatized Sepharose.
6. Calculate the purification factor offered by each column, as well as the yield. A good "positive" column is one that binds the protein of interest and little else, and gives a good yield on elution with KCl. A good "negative" column is one that allows the protein of interest to flow through, but binds many of the impurities.

4. Notes

1. All the fiber-reactive dyes are destined for the textile industry, and the crude preparations available from manufacturers contain various amounts of other solids (e.g., buffer salts, sodium chloride antidusting agents, surfactants). Thus the amount of chromophore in a given dry weight may vary from batch to batch or from one manufacturer to another. This implies that the ligand concentration of all the adsorbents prepared at the same time may not be comparable. Although the consequences of this variation may be minimal in a preliminary screening experiment, the possibility should be noted when synthesizing further batches (*see* Notes 5 and 10 in this section). Alternatively, the selected dyes can be purified before use (10), and dry weights standardized to give similar numbers of moles in a reaction mix.
2. Dye preparations are usually heterogeneous in composition and may contain other chromophores as minor impurities. The same dye (i.e., having the same Color Index number) when synthesized by two different manufacturers may contain different isomers. Whether these factors influence dye-ligand chromatography is not yet documented. It is therefore preferable to use the same manufacturer for a particular dye. It should be noted that some dyes are no longer being manufactured, and therefore availability should be checked before planning a scale-up.
3. Additional care needs to be taken with dichlorotriazinyl dyes, since the second chlorine may be available for reaction with proteins during chromatography. This complication is minimized by incubat-

ing the dyed adsorbents in alkaline solutions as described above, but quantitative hydrolysis of the second chlorine may not occur even under these conditions. It is therefore preferable to use monochlorotriazinyl dyes if possible, particularly as "positive" adsorbents.

4. Although agarose is the most commonly used matrix, other supports have been reported (11). Cross-linked agarose polymers support lower ligand concentrations, but the molar effectiveness (moles of protein bound per mole of dye) is comparable to uncrosslinked agarose. Copolymers of agarose and acrylamide exhibit lower molar effectiveness, whereas cellulose-based adsorbents are particularly ineffective in binding proteins, although they bind much more dye than agaroses. Dextran polymers are comparable to agarose. The use of micro-particulate silica has also been successful, but it is unlikely to be useful for screening experiments.

The choice of matrix can be crucial to the success of dye-ligand chromatography. It is recommended that initial screening experiments are carried out using agarose-based adsorbents. Once a dye is selected, further exploration can be carried out on matrices that may be more suitable to specific applications for reasons of physicochemical strength or stability.

5. The amount of dye attached to a matrix can be measured by several methods:
 - a. Spectrophotometric measurement of the unbound dye after coupling and comparison with total amount of dye at the start of synthesis. Dye absorbances are nonlinear at high concentrations because of ring-stacking effects, and solutions need to be well diluted. If the molar absorption coeffi-

cients are not known, an estimate can be based on dry weight (*see* Note 1 in this section).

b. Direct spectrophotometric measurement of dye released by acid hydrolysis (e.g., boiling for 10 min in 50% acetic acid or incubation at 37°C for 5 min in 6N HCl) of a known weight or settled volume of gel. Suitable for agaroses.

c. Direct spectrophotometric measurement of beads suspended in 50% v/v glycerol. This method is suitable for celluloses or acrylamide crosslinked gels that are not easily hydrolyzed by acid.

6. The number of dyes available at any one time may be 60–70. It is recommended that as many as possible be tried in an initial screen, particularly if the protein or source has not been evaluated before. Since dye adsorbents are simple to prepare and stable for many years, this is not a difficult task. If for some reason (e.g., difficulty of assay) this is not possible, however, it is advisable to start screening with dyes known to bind substantial amounts of protein and attempt selective elution.

Some dyes bind more proteins from crude extracts than others in a consistent manner, whereas the reverse is true of others (9). It therefore follows that the best positive and negative dyes may be found by screening dyes in the low-binding and high-binding groups, respectively.

7. The best outcome of a screening experiment is the discovery of a positive dye column that can generate the protein of interest in pure form with little or no modification of the elution conditions. This is likely to be rare, particularly if the initial abundance of the protein is low.
8. The second best outcome is the discovery of a pair of adsorbents, one negative and one positive, that

can be operated sequentially to give substantially pure protein. When scaling up, it is preferable to run the negative column first, since it then protects the positive column from fouling, and the two columns can be run in tandem with almost the ease of a single step (8). This reasoning applies to an even greater extent if a good positive column is not found and a bioaffinity step is used instead.

9. A more common outcome of a screening experiment is that several positive and negative columns generate reasonable purification factors and must be investigated further before a decision can be made on selection. This can be done using the fractions already available by use of analytical techniques such as polyacrylamide gel electrophoresis. The qualitative examination of contaminants, together with the known physical properties of the protein of interest (e.g., molecular size) may make the choice of the positive adsorbent obvious. This in turn should allow a logical choice to be made for the best negative adsorbent (i.e., the one that binds most of the contaminant proteins bound by the positive dye).
10. Occasionally, the binding of a desired protein may be very tight, and the protein is not found in the flow-through or eluate. Under these conditions, stronger chaotropic elution (e.g., 0.5M NaSCN) or affinity elution (e.g., substrate of effector) may be successful. Alternatively, a lower immobilized dye concentration can lower the affinity of the dye for the protein. This should not be done by dilution with blank Sepharose, but by variation of incubation time during synthesis.
11. If a qualitative examination after an initial screening does not give a clear result, further screening

should be carried out to modulate dye-protein interactions. The choice of buffer composition and pH has a significant effect on the binding of proteins to dye columns (12). A greater number of proteins will bind when the pH is lowered to 5.5. Thus adsorption at lower pH values can be used to improve the purification obtained on negative columns. Conversely, increasing pH to 8 and adding 100 mM NaCl to the adsorption buffer will minimize ionic interactions of positive dyes with proteins and result in improved purification factors. Another variable parameter is the inclusion of divalent cations that promote binding of proteins to dye matrices (10). Investigation of these parameters will allow for selection of the best positive and negative dyes from the secondary screening.

12. Once a decision is made about the number and identity of the dye-ligands to be used and the most significant parameters are optimized, a model purification should be carried out at the laboratory scale to fine tune the process.

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Chapter 10

Aminohexyl-Sepharose Affinity Chromatography

Purification of an Auxin Receptor

R. D. J. Barker and H. M. Bailey

1. Introduction

Affinity chromatography is a widely used technique that is based on specific binding, usually between proteins and low-molecular weight ligands. It is often found that the presence of a spacer arm between the supporting matrix and the affinity ligand is beneficial to binding capacity. A convenient starting material for affinity chromatography involving carboxyl-containing ligands is aminohexyl-Sepharose, which is commercially available and contains a spacer arm; any carboxyl-containing ligand can easily be coupled to the amino group of the resin. As an example of the use of ami-

nohexyl-Sepharose, this chapter describes the purification of a soluble auxin receptor using an affinity resin produced by derivatization of aminohexyl-Sepharose.

Plant-growth regulators, such as the auxin indole-3-acetic acid (IAA), exert considerable influence on the growth of higher plants (1). It is generally held that such effects are likely to be mediated by receptors whose function is to produce a cellular response upon perception of the plant-growth regulator. Such receptors may be expected to show the majority of the following properties: they are likely to be proteins, in that proteins possess the necessary specificity of binding; they must display reversible binding; they must possess an association constant that would result in saturation of the receptor at the low concentrations of plant-growth regulator present within cells; they are likely to be present in very low concentrations; finally, they must cause transduction of the plant growth regulator "signal": one such effect is regulation of expression of particular genes at the level of transcription (2). Plant hormone receptors have recently been extensively reviewed (3).

To fully characterize plant hormone receptors, purification is necessary. Since cytoplasmic receptors are present at extremely low concentrations [typically between 0 and 0.5 pmol binding sites per mg protein, (4,5)] this is a difficult task. The possession of a low dissociation constant (typical K_d 10^{-6} to 10^{-9} M) suggests that affinity chromatography could be successfully applied. Various attempts have been made to use this method for purification of IAA receptors. Venis (6) synthesized a derivative of the synthetic auxin, 2,4-dichlorophenoxyacetic acid (2,4-D); this derivative (2,4-D- ϵ -L-Lysine), when coupled to Sepharose, retained proteins from pea and wheat leaf extracts. Although a fraction, eluted with 2 mM KOH (pH 11.2), was found to stimulate

Escherichia coli RNA polymerase-dependent incorporation of ATP into 10% trichloroacetic acid (TCA) precipitable material, no binding assays were conducted. Further studies (7), using the same adsorbent and elution conditions, confirmed stimulation of UTP incorporation in a TCA-insoluble form, but no binding assays were carried out.

Tappeser et al. (8) synthesized a number of affinity adsorbents by coupling different auxins and anti-auxin analogs to epoxy-activated Sepharose 6B. The most useful adsorbent, when tested with solubilized auxin-binding proteins from corn membranes, was found to be 2-hydroxy-3,5-diiodobenzoic acid. Although 0.15M NaCl eluted all auxin-binding proteins, they concluded that elution with 10^{-4} M naphthalene-1-acetic acid (NAA) gave greater purification: unfortunately no binding assays were conducted to confirm this. Auxin binding proteins have also been purified from etiolated mung bean seedlings using 2,4-D- ϵ -L-Lysine coupled to CNBr-activated Sepharose 4B (9). These have been shown to possess high-affinity binding for 2,4-D (10) and to stimulate RNA synthesis (11). A soluble auxin-binding protein from tobacco callus has been partially purified by gel filtration: this showed high-affinity binding and stimulated RNA synthesis in vitro (4).

The procedure described here has been successfully applied (5) to an IAA-binding protein extracted from suspension cultured cells of tobacco (*Nicotiana tabacum*). The adsorbent used for affinity chromatography is prepared by coupling 2,4-D to aminohexyl-Sepharose (AH-Sepharose). When extracts of tobacco cells possessing high-affinity IAA binding, as demonstrated by Scatchard analysis (12), are passed through a column of this resin, the high-affinity binding is retained on the column. Desorption is achieved by addition of IAA to

the eluting buffer; after removal of IAA by gel filtration, high-affinity IAA binding may again be demonstrated. In the presence of IAA, this purified material stimulated transcription, as measured by the incorporation of UTP into TCA-insoluble material by isolated nuclei from tobacco cells (5).

2. Materials

2.1. Preparation of 2,4-D-AH-Sepharose

1. AH-Sepharose 4B may be obtained from Pharmacia.
2. 0.5M NaCl; 2 L is sufficient.
3. 900 mg of 2,4-dichlorophenoxyacetic acid (2,4-D) is dissolved in approximately 30 mL of distilled water; to facilitate this, 1M KOH is added dropwise to maintain pH in the range of 8–11, while stirring in a fume hood. It does not matter if the final volume exceeds 30 mL. *Caution: 2,4-D is toxic; appropriate precautions must be taken, including use of gloves and mask; whenever possible, procedures involving this reagent should be conducted in a fume hood.*
4. 900 mg of 1-ethyl-3-(3-dimethyldiaminopropyl) carbodiimide (EDC) is dissolved in approximately 30 mL of distilled water. EDC is toxic; take appropriate precautions, as described for 2,4-D.
5. 1M HCl.
6. 0.05M sodium acetate buffer, pH 4.0, containing 1.0M NaCl; 1 L is adequate.
7. 0.05M Tris-HCl, pH 9.0, containing 1.0M NaCl; 1 L is sufficient.
8. Sodium azide.
9. Sephadex G-50 (Pharmacia). A 2.4 × 38 cm column of Sephadex G-50, equilibrated in BB + 0.1M NaCl

(see below), is suitable for removal of IAA following desorption of proteins from the 2,4-D-AH-Sepharose column.

2.2. Extraction and Binding Assay

1. The buffer used for extraction of the soluble IAA receptor from tobacco cells contains:

0.2M H_3BO_4

0.13M KCl

2 mM $\text{Na}_2\text{B}_4\text{O}_7$

4 mM Sodium diethyldithiocarbamate (Sigma)
(added just before use)

This is adjusted to pH 6.8 with KOH (see Note 1 in section 4).

2. Ultrafiltration cell and filters. A capacity of 100 mL is required: membranes with a molecular weight cut-off of 10,000 are appropriate (e.g., Ulvac ultrafiltration membrane type G10T, Chemlab).
3. Binding assays are carried out in binding buffer (BB), which contains: 20 mM Tris, 50 mM H_3BO_4 , 1.5 mM $\text{Na}_2\text{B}_4\text{O}_7$, 100 mM KCl, 10 mM MgCl_2 , 25 mM NaMoO_4 , 0.1 mM Na_2 EDTA, and 0.1 mM dithiothreitol (DTT). The pH is adjusted to 7.5 with 1M HCl. For chromatography this buffer is supplemented as follows:
 - a. For equilibration of columns, by addition of NaCl to 0.1M (BB + 0.1M NaCl);
 - b. For elution of proteins retained by the affinity column, by addition of NaCl to 0.05M and IAA to 0.04M (BB + 0.05M NaCl + 0.04M IAA; IAA dissolves more readily if the pH is raised with 1M KOH; once in solution the pH should be adjusted to 7.5). The increase in ionic strength is to minimize nonspecific binding to the affinity column.

4. High specific activity IAA is required: ^3H -IAA (777 G Bq/mmol) from Amersham International is suitable. For use ^3H -IAA is diluted with BB. Appropriate safety precautions must be taken while using radioisotopes, and local rules must be observed.
5. Nonradiolabeled IAA is initially dissolved in methanol. A $3 \times 10^{-2}\text{M}$ stock solution is suitable and may be stored at -20°C for up to 1 mo. For use the stock solution is diluted with BB. IAA solutions are light-sensitive and should be protected from light.
6. Dextran-coated charcoal (DCC) suspension: 0.6% (w/v) charcoal and 0.06% (w/v) dextran (Sigma, average mol. wt., 80,000) in distilled water.
7. Protein assay: a means of estimating protein concentration is required. The method of Bradford (13), as modified by Spector (14), is suitable (see Chapter 1).

3. Method

Unless otherwise specified, all procedures are carried out at 4°C .

3.1. Preparation of 2,4-D-AH-Sepharose

The method used for the preparation of the affinity resin is based on the general procedure for coupling carboxyl-containing ligands to AH-Sepharose provided by the manufacturers (Pharmacia, Sweden). Coupling is carried out at room temperature.

1. Six grams of AH-Sepharose 4B is allowed to swell in an excess (1200 mL) of 0.5M NaCl and is then washed with 4×200 mL aliquots of 0.5M NaCl, followed by 4×200 mL distilled water. These washes are carried out on a sintered glass funnel

over a vacuum (a water pump is adequate). The resin is transferred to a round-bottom quickfit flask.

2. The 2,4-D and EDC solutions are added to the resin, and the pH is adjusted to pH 5.0. The pH is maintained between 4.5 and 5.5 for 90 min by the addition of drops of 1M HCl. The flask is gently swirled (by hand) during this period.
3. The slurry is then stirred overnight by rotation on a slowly moving rotavap (without vacuum; a Buchi Rotavapor is suitable); magnetic stirrers, which may destroy the bead structure, should be avoided.
4. After 16–20 h, the resin is transferred to a glass sinter and washed alternately with buffered solutions of alkaline and acidic pH (0.05M sodium acetate buffer, pH 4.0, containing 1.0M NaCl, and 0.05M Tris-HCl, pH 9.0, containing 1.0M NaCl). Four washes with each buffer (200 mL per wash) are adequate; the resin is finally washed with 200 mL of distilled water.
5. The resin is stored at 4°C in 0.05M sodium acetate buffer, pH 4.0, containing 1M NaCl and 0.02% (w/v) sodium azide as a preservative (stable for several weeks at 4°C.) For the protocol described here, a 6 x 1 cm column of 2,4-D-AH-Sepharose is adequate.

3.2. Binding Assay for IAA Receptor

A variety of strategies are available (see Note 2 in section 4). Assays utilizing Dextran-coated charcoal (DCC) to adsorb free IAA, leaving receptor-bound IAA in the supernatant, have been found satisfactory with the IAA receptor from tobacco cells. This procedure, based on that of Oostrom et al. (15), is described here, but it is recommended that workers use whatever assay

they find effective and convenient for soluble receptors from any given tissue.

1. Twelve tubes are set up as shown:

Tube	^3H -IAA, pmol	Nonradioactive IAA, pmol
1	2.5	0
2	2.5	2.5
3	2.5	5.0
4	2.5	10
5	2.5	25
6	2.5	40
7	2.5	80
8	2.5	120
9	2.5	180
10	2.5	240
11	2.5	2400
12	2.5	0

All tubes are made up to 0.5 mL with BB.

2. One half milliliter of sample is added to each tube 12: This tube is made up to 1 mL with BB and is used to check on the efficiency of binding of free IAA by the DCC.
3. The tubes are incubated at 25°C for 20 min and are then cooled in an ice bath for 20 min.
4. One milliliter of DCC suspension is added to each tube and the tubes are shaken for 20 min at 4°C and centrifuged at 2000g for 10 min.
5. An aliquot of the supernatant is removed for counting in a liquid scintillation counter. Dissociation constants and numbers of binding sites per mL of sample are calculated by Scatchard analysis (12).

3.3. Affinity Chromatography Using 2,4-D-AH Sepharose

1. Thirty grams (fresh weight) of cells are harvested and suspended in 50 mL of extraction buffer and

- homogenized in a Potter-Elvehjem glass vessel (5 strokes at 1500 rpm) (*see* Note 3 in section 4).
2. The homogenate is centrifuged at 100,000g for 45 min and the supernatant concentrated in an ultrafiltration cell to approximately 15 mL (*see* Note 4 in section 4).
 3. The concentrated extract is adjusted to pH 7.5 with 1M KOH, and solid NaCl is added to a final concentration of 0.1M; 6 mL is used in a prepurification binding assay (*see* Note 5 in section 4).
 4. The remainder is applied to a 6 × 1 cm column of 2, 4-D-AH-Sepharose, pre-equilibrated by the passage of 40–50 mL of BB + 0.1M NaCl. A flow rate of 0.25–0.3 mL/min is used throughout, and 3-mL fractions are collected. After the sample has been applied, the column is washed with 30 mL BB + 0.1M NaCl. The nonretained fraction, identified by absorbance at 280 nm, may be collected for assay of high-affinity IAA binding (*see* Note 6 in section 4).
 5. Desorption of protein bound to the 2,4-D-AH-Sepharose column is achieved with BB + 0.05M NaCl + 0.04M IAA (*see* note 7 in section 4). The first 4–5 fractions that contain IAA (detected by A_{280}) are pooled (12–15 mL), and a small sample is taken for protein assay.
 6. The desorbed fraction is applied to the Sephadex G-50 column to remove IAA from the protein. To ease identification of protein-containing fractions after gel filtration, an inert protein, such as hemoglobin, may be added (0.33 mg/mL) prior to application to the Sephadex G-50 column (*see* note 8 in section 4).
 7. The protein peak (void volume) eluted from the Sephadex G-50 column is pooled and assayed for high affinity IAA binding in a binding assay. This partially purified material is suitable for use in studies of receptor function (*see* Note 9 in section 4).

4. Notes

1. The extraction buffer described is based on that of van der Linde et al. (4). They found that the use of Tris buffer at pH 7.5 for extraction of soluble receptors from tobacco callus resulted in high levels of contamination by polyphenols. The use of borate buffers at pH 6.8 appreciably reduced this contamination. When polyphenols are not a problem, a 50 mM Tris-HCl buffer, pH 7.5, containing 2 mM Na₂ EDTA and 2 mM DTT, can be used.
2. Various methods are available for assessment of dissociation constants and numbers of binding sites. Equilibrium dialysis is the most reliable method, but requires relatively large numbers of binding sites; the same drawback applies to the use of gel-filtration columns. Dextran-coated charcoal has been widely used to adsorb free hormone; following centrifugation, receptor-bound hormone remains in the supernatant and may be quantified. Although this is not an equilibrium method, it can be used, providing the rate constant for dissociation of the receptor-hormone complex is slow. Alternative assays involve filtration through a nitrocellulose filter, the use of ultrafiltration (or centrifugal ultrafiltration), or precipitating with ammonium sulfate to separate receptor-bound hormone from free hormone. Methods of quantifying and analyzing plant hormone receptors have been reviewed (3, 16).
3. Any homogenization method that results in the majority of cells being broken, while avoiding denaturation of proteins, is acceptable.
4. This step is to ensure that receptor concentration is sufficient to be detected in the binding assay; binding assays with extracts of low protein concentra-

- tions are often unsuccessful. Concentration also serves to reduce the time taken to load the extract onto the 2,4-D-AH-Sepharose column.
5. Affinity chromatography has only been successful when high-affinity binding has been detectable in the concentrated extract.
 6. The nonretained fraction routinely contains more than 90% of the protein applied to the 2,4-D-AH-Sepharose column; no high-affinity binding is found in this fraction.
 7. The high concentration of IAA is to ensure complete desorption of bound receptor; the lower ionic strength of this buffer, compared to that of the equilibration buffer (also used to wash the column prior to desorption), is to minimize nonspecific desorption. If the 0.04M IAA is replaced by 0.04M acetate, high-affinity IAA binding is not recovered.
 8. The protein concentration of the fraction desorbed by BB + 0.05M NaCl + 0.04M IAA is usually in the range of 0.02–0.04 mg/mL; this fraction contains 1–2% of the protein initially applied.
 9. The procedure described typically results in a 35–40-fold purification of IAA receptor, with recovery exceeding 40% of that initially present. In all cases in which high-affinity binding has been demonstrable, the purified fraction has stimulated transcription *in vitro*, as measured by incorporation of UTP into TCA-insoluble material (5).

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Chapter 11

Purification of DNA-Dependent RNA Polymerase from Eubacteria

N. W. Scott

1. Introduction

In eubacteria, the structure of DNA-dependent RNA polymerase is conserved; the enzyme is composed of at least four different subunits: beta (β), beta-prime (β'), and alpha (α), which form the core ($\beta\beta'\alpha_2$), and sigma (σ), which is required for efficient promoter selection (1–3).

Most of the procedures available for the isolation of this enzyme follow a set format; they normally make use of a gross fractionation step, such as phase separation, followed by several forms of column chromatography that make use of the protein's affinity for DNA or heparin and its binding to compounds such as DEAE (1,4,5).

Another feature of the enzyme that is utilized for its purification is its size. The absolute sizes of the subunits vary between species, but the M_r values of the beta

subunits are usually within the range 140,000–180,000, the M_r values of the alpha subunit are about 40,000, and M_r values of the sigma subunits are in the range 20,000–100,000. This means the holoenzyme can have a M_r of greater than 400,000, so it can be separated from other proteins by molecular sieving or even centrifugation (1–3). The conservation of subunit structure also means that the enzyme can be easily identified on polyacrylamide gels; this is a useful property during the purification procedure.

Despite the similarity in enzyme structures among species, there has been a proliferation of isolation procedures available (see ref. 4 for a list of the 14 procedures used for *Bacillus subtilis* alone). This is because different species have intrinsic properties that can influence RNA polymerase isolation and because of the presence of minor forms of the enzyme. For most organisms, only a single form of the enzyme has been found (1–3), but studies of *B. subtilis* (4), *Escherichia coli* (6), and *Streptomyces coelicolor* (7) have indicated that these organisms contain multiple forms of the enzyme that contain different sigma factors, and as such have different promoter specificities. The difficulty for purification comes from the fact that the alternative forms of the enzyme can represent a very minor amount of the total holoenzyme pool, less than 5% of the total in the case of the four *B. subtilis* minor holoenzymes (8).

The procedure described in this chapter is one used successfully to purify RNA polymerase from several Gram negative bacteria (9). It can be used with other bacteria and should yield enzyme purified to an extent that makes it suitable for transcription studies or for analysis of subunit properties. The method has four steps: (1) cell lysis and removal of cell debris; (2) a series of ammonium sulfate precipitations; (3) Bio-Gel A-5m

chromatography; and (4) DNA-cellulose chromatography. The first two steps are modifications of methods first introduced by Burgess (10); the aim of these steps and step 3 is to separate the enzyme from the bulk of the cell protein and particularly to remove contaminating DNA-binding proteins and any remaining nucleic acids prior to affinity chromatography with DNA-cellulose. This column is very effective for the purification of RNA polymerase, particularly when a linear salt gradient is used to elute the protein (4). It has advantages over other materials in that it is easy to make, it allows the user the choice of DNA for optimum purification, it stores well, and it is comparatively inexpensive. The column can also be used for the isolation of DNA binding proteins.

2. Materials

1. The buffer used in the purification is TGED (11): 0.01M Tris-HCl, 5% (v/v) glycerol, 0.1 mM EDTA, 0.1 mM dithiothreitol. TGED containing various concentrations of NaCl are required; 1-L volumes of the following are usually adequate:

TGED

TGED + 0.15M NaCl

TGED + 0.5M NaCl

TGED + 1.0M NaCl

TGED + 1.5M NaCl

The purified enzyme is stored in TGED buffer containing 50% glycerol and 0.1M NaCl (11).

2. Phenylmethyl sulfonyl fluoride (PMSF) (use care, toxic). Make fresh prior to use. It is solubilized in ethanol prior to dilution with TGED + 1M NaCl buffer, and is then added to the lysate when re-

quired (see methods) to give a final concentration of 23 $\mu\text{g/mL}$.

3. Ammonium sulfate.
4. Bio-Gel A5-m (Bio-Rad). Use a 25 x 2 cm column pre-equilibrated with TGED + 0.5M NaCl.
5. DNA-cellulose. This is prepared by the method of Alberts and Herrick (12). Mix 160 g of cellulose with 500 mL of 2 mg/mL calf thymus DNA (or DNA of choice) in 0.01M Tris-HCl (pH 7.9) and 0.001M EDTA (TE buffer), and dry in large drying dishes (14-cm Petri dishes are suitable) at 50°C for 3–4 d followed by 24 h under vacuum. Powder the paste during the drying period with a glass rod. Suspend the powder in 20 vol of TE buffer and leave at 4°C for 24 h. Wash the DNA-cellulose twice with TE buffer to remove free DNA; use an acid-washed scintered glass funnel over a vacuum for this procedure. Suspend the slurry in TE buffer containing 0.15M NaCl and either divide into 20-mL aliquots and freeze at -20°C until required or load into a column. For the protocol given here, a 30 x 1 cm column is adequate. The amount of DNA absorbed to the cellulose can be ascertained by boiling the suspension at 100°C for 20 min then measuring $A_{260\text{nm}}$ of the liquid phase (50 $\mu\text{g/mL}$ DNA gives absorbance of 1). Approximately 50% of the DNA will absorb to the cellulose under the conditions described. Before and after use, wash the DNA-cellulose column with 10 mL of TGED + 1.5M NaCl and 60 mL of TGED + 0.15M NaCl at a flow rate of 15 mL/h.
6. The RNA polymerase assay used here has evolved from the assays used by Burgess (10) and Jaehning et al. (13). As with all radioactive assays, safety precautions must be taken and local rules must be

followed. Make the assay buffer as a 10x stock and keep at -20°C until required. 10x assay buffer:

400 mM Tris-HCl (pH 7.9)

1.0 mM Dithiothreitol

2.0 mM Spermidine

100 mM MgCl_2

Dissolve ribonucleoside triphosphates and DNA in 1x assay buffer. The ribonucleoside triphosphate stock solution contains ATP, CTP, and GTP at 2 mM, UTP at 0.5 mM, and 200 $\mu\text{Ci/mL}$ ^3H -UTP. Store the solution at -20°C . Make a 1 mg/mL calf thymus DNA (or DNA of choice) solution fresh for the assay.

The assays are carried out in acid-washed, oven-sterilized glass tubes. The components should be added to the tubes, held in ice, in the following order:

30 μL Column fraction

50 μL 1x Assay buffer

10 μL DNA solution

10 μL Ribonucleoside triphosphate solution

Start RNA synthesis by shifting the assay tubes to 30°C , and continue the incubation for 20 min. Stop the reaction by putting 40 μL duplicate samples onto labeled GF/C filters (Whatman), and dropping them into ice-cold 5% (w/v) trichloroacetic acid (TCA) containing 0.01M sodium pyrophosphate (reduces nonspecific binding). After 15 min, wash the filters twice with TCA and twice with absolute ethanol. Dry the filter, then measure the precipitated radioactivity.

7. SDS-Polyacrylamide gel electrophoresis (PAGE) (see vol. 1 of this series). For rapid analysis of column fractions use 7.5–10% (w/v) acrylamide gradient gels and stain the gel with Coomassie brilliant

blue, the fractions containing the enzyme can be identified from the presence of core subunits (9).

8. Dialysis tubing.

3. Method

All procedures carried out at 4°C.

3.1. Cell Lysis

1. Lyse 50 g of cells (wet weight) suspended in 200 mL TGED + 1M NaCl using a French pressure cell (*see* Note 4 in section 4).
2. Add 23 µg/mL PMSF to the lysate to reduce proteolytic activity (*see* Note 6 in section 4).
3. Centrifuge the suspension at 100,000g for 1.5 h to remove cell debris (*see* Note 8 in section 4).

3.2. Ammonium Sulfate Precipitations

1. Measure the volume of supernatant, then put it into a beaker on ice, and stir using a magnetic bar. The stirring rate should be fast enough to provide effective mixing, but not fast enough to cause frothing. Slowly add 23.1 g of solid ammonium sulfate per 100 mL supernatant, making sure there is no build up of undissolved material. Also add 0.05 mL of 1M NaOH per 10 g of ammonium sulfate to keep the pH above 7. Continue the stirring for 15 min, then centrifuge at 10,000g for 30 min, and discard the pellet.
2. Measure the volume of the supernatant and stir. Slowly add 10.75 g of solid ammonium sulfate per

100 mL of supernatant to give a 50% saturated solution. Again control the pH with NaOH. Centrifuge at 10,000g for 30 min, and discard the supernatant.

3. Suspend the pellet in TGED + 1M NaCl containing 25.6 g of solid ammonium sulfate (42% saturation). Stir for 20 min. Centrifuge at 10,000g for 30 min, discard the supernatant, and resuspend in the same buffer (42% ammonium sulfate), centrifuge as above, then discard the supernatant.

3.3. Bio-Gel A-5m Chromatography

1. Dissolve the pellet in 5 mL of TGED buffer and load onto the Bio-Gel column; use a flow rate of 20 mL/h. Elute the column with TGED + 0.5M NaCl and collect 10-mL fractions. The highest molecular weight proteins elute first, so analyze 10 fractions around $1.8 \times$ the void volume; this can be estimated from the $A_{280\text{nm}}$ of the fractions. Use PAGE and the enzyme assay to identify the fractions containing the enzyme.
2. Pool the enzyme containing fractions and measure the volume. Add 35 g of ammonium sulfate per 100 mL of eluate (plus required NaOH) to give a 50% saturated solution. Centrifuge at 10,000g for 30 min. Suspend the pellet in TGED such that the conductivity of the solution is equivalent to TGED buffer containing 0.15M NaCl.

3.4. DNA-Cellulose Chromatography

1. Load the dissolved enzyme onto the DNA-cellulose column; use a flow rate of 15 mL/h. Wash the column with 60 mL of TGED + 0.15M NaCl, then

elute the protein with a linear salt gradient (60 mL TGED + 0.15–1.5M NaCl). Collect 6-mL fractions, and analyze the salt gradient by PAGE and the RNA polymerase assay. The gradient fractions can be identified by $A_{280\text{ nm}}$ and by using a refractometer to measure salt concentration.

2. Pool the fractions containing the enzyme and dialyze into storage buffer. Store at -20°C .

4. Notes

1. If large amounts of DNA cellulose are made, it can either be stored at 4°C for up to 9 mo or frozen at -20°C . To avoid excessive freeze-thaw cycles, store in small aliquots. Commercially prepared DNA, such as calf thymus, can be used for this column. Alternatively, DNA prepared from the organism being studied can be used.
2. RNA polymerase assays may require modification depending upon the properties of the enzyme being studied. A larger volume of sample or a longer incubation time may be required for analysis of column fractions. To optimize assays with purified enzyme, salt concentration, pH, the DNA template, the specific activity of the label, and the type of labeled ribonucleoside can all be changed. The enzyme is particularly sensitive to rapid dilution, so to get maximum activity it is particularly important to add the assay components in the order indicated (14).

Simple modification of the assay can provide valuable information on the nature of the enzyme. For instance, it can be seen if the extract contains a holoenzyme: prebinding the enzyme to DNA for 10 min at 30°C prior to the addition of ribonucleo-

sides allows for the formation of a rifampicin-resistant complex only if the enzyme used is a holoenzyme (3). If species-specific bacteriophage or plasmids containing defined promoters are available, they can be used to study the promoter specificity of the enzyme. If they are not available, however, analysis of binding to T7 promoters is a simple system that can be used to provide some information (3).

3. The method described can be used for between 10 and 500 g of cells, if modifications are made to the size of the columns. It is not always possible to get such amounts of cells, for instance for stages of sporulation or for synchronized cell cultures. For smaller amounts of cells, the micropurification system of Gross et al. is suitable (15).
4. If a French pressure cell cannot be used for lysis, other methods can be tried. Some experimentation may be necessary to identify the best procedure for the cells being used. Some of the most popular techniques in use involve blending frozen cells in a Waring blender (10). Either blend 50 g of frozen cells with 150 g of Superbrite 100 glass beads and 50 mL of lysis buffer for 5 min at low speed and 10 min at high speed (10), or blend the cells in the presence of lysozyme and deoxycholate instead of the glass beads. For this latter method, blend pieces of frozen cells 2 cm or less in diameter, in grinding buffer (0.05M Tris-HCl) (pH 7.9), 5% (v/v) glycerol, 2 mM EDTA, 0.1 mM dithiothreitol, 1 mM 2-mercaptoethanol, 0.233M NaCl, 130 μ g lysozyme/mL, and 23 μ g PMSF/mL. For 50 g of cells, use 150 mL of buffer, blend at low speed for 2–3 min to resuspend the cells, then incubate the cells at 4°C for 20 min. Add 2.5 mL 4% (v/v) sodium deoxycholate and blend at

low speed for 30 s, and continue the incubation for another 20 min. Finally, blend at high speed for 30 s to shear the DNA. This method has been used most often with the Polymin P purification technique (Note 7 in this section). Grinding frozen cells with alumina or gentle lysis with deoxycholate and lysozyme (15) are techniques that can also be used.

5. No DNase is used in this method because it was reported that it can introduce proteases (11). More suitable grades of DNase may now be available.
6. PMSF is added to the lysate to reduce proteolytic activity. If this is not effective, the lysate may be treated with hemoglobin-Sepharose to absorb out the proteases (4). The effectiveness of enzyme inhibitors may be judged by PAGE analysis of protein degradation in extracts incubated in the presence and absence of inhibitors.
7. 1M NaCl is included in the lysis buffer to dissociate the enzyme from DNA; the core and σ factors have DNA-binding properties (9).

The ammonium sulfate concentrations used here were optimized for *E. coli*. They may require modification for use with other organisms, particularly because the use of high concentrations of NaCl and ammonium sulfate can be detrimental to enzyme activity and can dissociate subunits from the core (4). Avoid using high-salt treatments with purified enzyme, since this can inhibit enzyme activity (4).

Alternative procedures that could be used are PEG6000 (15) or Polymin P (11) precipitations, or it is possible to apply crude lysates directly to affinity columns without using a crude fractionation (7).

8. Centrifugations can be carried out with a cooled bench centrifuge, but longer spin times must be used.

9. The Bio-Gel A-5m column is prewashed with TGED + 0.5M NaCl, and can be reused indefinitely if kept in this buffer.
10. Although Mg^{2+} improves RNA polymerase stability, it is not used in the column buffer since it interferes with RNA polymerase binding to DNA-cellulose (4). The absence of Mg^{2+} also reduces any DNase activity in the extract and prevents fungal growth on the agarose column (11).
11. Some methods use higher concentration of glycerol [up to 20% (v/v)] in the buffers since it stabilizes the holoenzyme on DNA-cellulose (4).
12. Both the assay and PAGE are used to locate the enzyme, since some discretion is allowed as to which fractions are to be pooled. This is particularly important for the DNA-cellulose column, since fraction-containing contaminants can either be subjected to further purification steps or discarded (9).

To decide whether a protein that copurifies with the core is an enzyme subunit or a contaminant is not an easy task. Several different approaches can be used to investigate this. A physical association can be established by centrifugation (10), one-dimensional PAGE (4), or even two-dimensional PAGE (first-dimension, nondenaturing, second-dimension, denaturing) (9). To show a functional relationship involves purification of individual proteins, by either column chromatography (e.g., ref. 5) or excision from acrylamide gels, and addition to in-vitro transcription systems, where the effect of the protein on enzyme function can be evaluated (16).

13. If further purification steps are required, several different approaches are available. A second DNA-cellulose column with a different salt gradient may

resolve the enzyme from the contaminants. Alternatively a different affinity column could be used; Heparin-agarose columns are suitable (5). Ion exchange chromatography using DEAE sephadex (9) or DEAE-cellulose can also be effective (4), and finally glycerol gradient centrifugation can also be used (10).

If different columns are used, avoid changing the salt in the buffer, since this can lead to enzyme inactivation.

14. The purity and amount of enzyme that will be recovered by this method will be species-specific and will relate to the conditions used to grow the bacteria, but the amount of enzyme may well be in the order of 1–10 mg. Enzyme activity is also likely to be variable; enzyme activity can vary between 200 and 2000 units (per mg protein) (9) (where a unit is the amount of enzyme required to incorporate 1 nmol UMP into acid-insoluble material in 10 min).

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Chapter 12

Direct Immunoprecipitation of Protein

***Christopher F. Thurston
and Lucy F. Henley***

1. Introduction

Many biochemical experiments depend on the measurement of the amount of an enzyme (or other protein) independently of any enzymatic activity it may have. When working with cell-free extracts or other complex mixtures this is not simple to do, and most procedures that have general application involve the use of antibodies. Two distinct strategies have been used to good effect. The amount of antibody bound can be made to reflect amount of antigen, as in an ELISA assay (ref. 1 and Vol. 1 of this series) or in a "Western blot" (ref. 2 and Chapters 28 and 29 in this volume). Alternatively, the antibody can be used effectively to purify the antigen protein. The method described here

is of the latter sort and has advantages of simplicity and rapidity both over ELISA and blotting techniques, on the one hand, and indirect precipitation techniques on the other.

Direct immunoprecipitation is achieved by addition of a specific antibody to a sample at appropriate concentration to form a precipitable matrix with antigen (3). The precipitate so formed is contaminated with nonspecifically bound protein that is "scrubbed" off the precipitate by centrifugation through a viscous sucrose solution (4). The resultant contamination-free precipitate may be counted for radioactivity (of labeled antigen) or subjected to further analysis by electrophoresis or isoelectric focusing (*see* Fig. 1).

If the protein to be detected/quantitated is distinguishable from immunoglobulin polypeptides on the basis of size or charge, the method can be used with nonradioactive samples, but in practice its usefulness is for analysis of radiolabeled mixtures, in which very small amounts of labeled antigen can be isolated for analysis by coprecipitation with a relatively large amount of unlabeled "carrier" antigen.

2. Materials

1. **Antigen:** The protein to be analyzed must be available in a highly purified form as the immunogen for antibody production. Purified antigen is also required to titrate the antibody and, when samples containing very small amounts of antigen are to be analyzed, to act as carrier. The pure antigen is conveniently used at 1–2 mg/mL and must be dissolved in a buffer mixture that does not interfere with antibody binding, such as TEC described below.



Fig. 1. Analysis of isocitrate lyase synthesis in the green alga *Chlorella fusca*. Fluorograph of an SDS gel showing immunoprecipitated isocitrate lyase in tracks 1 and 2, obtained from the cell-free extracts run in tracks 3 and 4, respectively. Protein was labeled in vivo by ^{35}S -methionine incorporation for 2 h (tracks 1 and 3) and 5 h (tracks 2 and 4) after the removal of inducer (acetate) from the medium. Clearly there was significant residual enzyme synthesis after inducer removal, although this represents a small fraction (1–2%) of total incorporation and does not result in easily distinguishable bands in the total protein tracks. Kinetic analysis of these data are shown in ref. 5.

2. Antiserum: Polyclonal antibodies raised in rabbits are normally used in this procedure, but polyclonal antibodies raised in animals other than the rabbit could equally well be used. The IgG fraction is purified from serum as described below.

2.1. IgG Purification

1. 0.1M sodium phosphate, pH 7.5.
2. Ammonium sulfate. Use the grade supplied for protein purification (which has the least metal contamination).
3. Protein A Sepharose CL-4B from Pharmacia.
4. 0.1M Glycine-HCl pH 3.0.
5. 2M Potassium carbonate.
6. pH indicator papers (range 6–8).

2.2. Immunoprecipitation

1. TEC buffer: 15 mM Tris-HCl, 1 mM EDTA, 1 mM dithiothreitol, pH 7.6. Store at 4°C; add the dithiothreitol on the day of use.
2. Binding buffer: 10 mM sodium phosphate, 150 mM NaCl, 1% Triton X-100, 0.5% bovine plasma albumin pH 7.2. Store at –20°C.
3. 0.9M Sucrose. Make up in sterile distilled water and store at 4°C. Prior to use, centrifuge as 1-mL amounts in microfuge tubes for 15 min at 4°C (see Note 6 in section 4).

3. Method

3.1. IgG Purification

1. All steps should be done at 4°C or in containers on ice.

2. Measure 10 mL of serum into a small beaker or flask and stir magnetically while adding 2.26 g of ammonium sulfate (to bring the solution to 40% saturation). Maintain the pH at 7.5 by adding ammonia solution. This should be fairly concentrated, so as not to significantly increase the overall volume. Continue stirring for 10 min after all of the ammonium sulfate has been added.
3. Centrifuge for 10 min at 8000g, and discard the supernatant, allowing the pellet to drain thoroughly.
4. Dissolve the pellet in 10 mL of phosphate buffer.
5. Repeat steps 2 and 3.
6. Dissolve in 5 mL of phosphate buffer, and dialyze against the same buffer, using three successive 1-L amounts.
7. Centrifuge the dialyzed protein solution for 15 min at 12,000g to remove any particulate material. If the supernatant is not homogeneous, filter through a 0.45- μ m pore size membrane filter.
8. Load the clarified protein solution onto a 5-mL protein A-Sepharose column at 30 mL/h (*see* Note 1 in section 4). Reapply the eluate twice to ensure maximum retention of IgG.
9. Wash the column with phosphate buffer at 30 mL/h collecting samples from time to time for measurement of absorbance at 280 nm. Continue washing until the absorbance of the eluate is less than 0.05 (*see* Note 2 in section 4).
10. Elute the bound IgG with glycine buffer, collecting 1.5 mL fractions that are immediately neutralized with 2M potassium carbonate (*see* Note 3 in section 4).
11. Measure the absorbance of each fraction at 280 nm, and pool those fractions containing a significant

proportion of the eluted IgG. Store at -20°C in 200–500 μL amounts.

3.2. Immunoprecipitation

1. The amounts for the immunoprecipitation mix are as follows: A standardized amount of antigen must be used, 5 μg is sufficient (*see* Note 4 in section 4). The antigen is made up to 50 μL with TEC in a 400- μL microfuge tube. IgG fraction (diluted to give a twofold excess, *see* Note 5 in section 4) is added in 50 μL , followed by 25 μL of binding buffer.
2. Incubate the mixture for 2 h at 4°C to allow for formation of the immunoprecipitate.
3. Centrifuge for 15 min at 4°C to pellet the crude precipitate.
4. Carefully aspirate off the supernatant.
5. Add 50 μL of binding buffer to the precipitate and take up and expel with a pipet 5–10 times to disperse the precipitate as a fine suspension.
6. Overlay the suspended precipitate onto 200 μL of 0.9M sucrose in a 400- μL microfuge tube, and centrifuge for 15 min at 4°C (*see* Note 6 in section 4).
7. Carefully aspirate the supernatant and resuspend the pellet for analysis (*see* Note 7 in section 4).

4. Notes

1. Protein A Sepharose should be equilibrated and regenerated as recommended by the manufacturer. A 10-mL syringe is a convenient container in which to make the column. Control the system by pumping off the bottom of the column with a peristaltic pump for this and steps 9 and 10.

2. It is more convenient to monitor the absorbance of the column with a flow monitor if available, washing until there is no significant absorbance in the eluate (say less than 0.5% of the absorbance of the sample loaded on the column).
3. The stability of antibody at pH 3 is limited and varies from one antibody to another; there is no doubt that the neutralization step is the most crucial in this part of the procedure. In our experience it is quicker to use pH papers than a pH electrode, but whatever can be done quickest is best.
4. The total amount of antigen must be sufficient to give a visible immunoprecipitate, for which purpose 5 μ g is adequate for proteins that bind about twice their own weight of immunoglobulin. If the antigen is an abundant constituent of the sample mixture, e.g., 5% of total protein, then the total protein concentration of the sample must be adjusted to, for example, 2 mg/mL so that 50 μ L contains 5 μ g of antigen. In this situation it is not uncommon to find that some samples are too diluted and need to be concentrated rapidly. Amicon "Minicon-B" disposable clinical sample concentrators are very useful for this, both because they are quick and because the extent of concentration can be quite accurately estimated by eye at any stage in the process.

For many other antigens the concentration in cell-free extract or other sample (such as the products of *in vitro* translation) is much less, and the precipitation is achieved by using unlabeled carrier antigen. In this case use 5 μ L of 1 mg/mL carrier and 45 μ L of sample extract, containing not more than 1 μ g of antigen.

5. For each batch of IgG prepared, a calibration must be done. Set up tubes containing 5 μ g antigen, and

add a range of volumes of IgG fraction (say 10–200 μL) making all tubes up to the same volume with phosphate buffer and adding binding buffer to give a 2:1 v:v ratio of phosphate buffer to binding buffer. Stand for 2 h at 4°C, and centrifuge for 15 min. Thoroughly aspirate off the supernatants and dissolve the pellets for protein determination. If a suitable range of IgG amounts is tested, the pellets increase progressively to a plateau value of about 20 μg . The minimum amount of IgG to give the plateau value (i.e., complete precipitation of the antigen) is doubled in the routine procedure, which is what is intended by "a twofold excess" in step 1. If 50 μL of IgG fraction is insufficient to give this amount, concentrate the solution as described for protein samples in Note 4 above. As an example, one recent preparation of anti-isocitrate lyase IgG fraction required 280 μg to give twofold excess for precipitation of 5 μg of antigen protein. If the antigen of interest has an easily assayed activity that is not inactivated by the conditions of the precipitation reaction, the residual activity in supernatants from the precipitation step can be used to determine the amount of IgG required for complete precipitation more easily, and often more precisely, than measurement of precipitated protein.

6. The centrifugation of the sucrose solution prior to its use never produces a visible pellet, but the removal of nonspecific contamination works less well if it is not done. The immunoprecipitate may be colored prior to centrifugation through sucrose (for instance, when precipitating from reticulocyte lysates, the pellets are bright pink at this stage), but after passage through the sucrose, the pellets should be white.

In this and all centrifugation steps except those during the preparation of IgG fraction, in which larger volumes are involved, use a microcentrifuge giving at least 10,000g. Since microcentrifuges are usually not refrigerated, the centrifuge should be used in a cold room or cold cabinet. The standard 1.5-mL capped polythene "Eppendorf" microfuge tubes are suitable for precentrifuging the sucrose solution, but not for the immunoprecipitation reaction itself or the centrifugation of the immunoprecipitate through sucrose. The long, narrow 400- μ L tubes are used for these steps since their shape results in a more compact pellet that can be seen relatively easily. The pellets formed using 5 μ g of antigen are clearly visible once the procedure has become familiar, but they can be difficult to see in microfuge tubes that have been badly moulded and are uneven at the bottom.

7. The final immunoprecipitates can be dissolved in loading buffers for analysis by electrophoresis or isoelectric focusing, or may be counted for radioactivity without further processing, in which case the bottom 5 mm of each tube can be cut off and dropped directly into a scintillation vial containing a protein-solubilizing agent.

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Chapter 13

Detection of Proteins in Polyacrylamide Gels Using an Ultrasensitive Silver Staining Technique

***Ketan Patel, David J. Easty,
and Michael J. Dunn***

1. Introduction

Polyacrylamide gel electrophoresis is a versatile and powerful tool for the analysis of biological samples and is capable of good separation and high resolution of complex protein mixtures. Although Coomassie Brilliant Blue R250 has proved ideal as a general protein stain for the more traditional applications of this method, current trends toward thinner gels, decreased sample loading (to improve resolution), and the recent

developments of two-dimensional gel electrophoresis and peptide mapping techniques have necessitated increasingly sensitive detection methods. Electrophoretic separation of radioactively labeled proteins followed by autoradiography permits the detection of trace proteins (10^{-4} to 10^{-5} % of total protein) in a sample (1). However, problems inherent in radioactive methods include: (a) *in vitro* labeling may alter physical properties of proteins, and (b) *in vivo* experiments require excessively large quantities of isotopes that are prohibitively expensive in animals and unethical in human clinical studies. Alternative methods such as fluorescent staining (2) and heavy metal stains (3) are of less than, or at best of equivalent sensitivity to, Coomassie Brilliant Blue R250.

The property of silver to develop images was discovered in the mid-17th century. Subsequently, imaging techniques using silver were first applied to photography, followed closely by their use in histology. Recently, silver staining methods have been developed to detect proteins after polyacrylamide gel electrophoresis; first by employing histological principles (4) and then by photochemical methods (5). Silver staining techniques are reported to be between 20 and 200 times more sensitive than Coomassie Brilliant Blue R250 and represent a significant advantage in the range of protein detection methods in polyacrylamide gels. Silver staining can be used as a detection method for techniques other than protein polyacrylamide gel electrophoresis. Proteins transferred to nitrocellulose membranes can be visualized using silver staining procedures (6), as can DNA (7) and RNA molecules (8) in gels and on membranes.

The precise mechanism leading to the formation of a visible silver/protein complex has not been fully clarified, but both histological and photochemical stains

involve the reduction of ionic silver to the metallic form. It has been proposed that silver cations complex with protein amino groups in an alkaline environment and with cysteine and methionine sulfur residues (9). More recently, however, Gersten et al. (10) have implicated three-dimensional protein structure and, therefore, the steric presentation of reactive moieties in three-dimensional space as being of most consequence. They proposed that other factors such as amino acid composition are of secondary importance. Caution should be exercised in the interpretation of protein staining patterns since certain protein species, e. g., calmodulin (11), fail to respond to single-step silver staining.

After testing several current silver staining techniques (12–14), we have found that the most adequate procedure for our purposes in terms of sensitivity, resolution, time required, and expense was the method described by Wray et al. (15). This method has been assessed favorably in comparative methodological studies made by other workers (16). We describe the Wray technique for silver staining of gels, together with modifications and technical advice that will enable an experimenter to optimize results. An example of a 2-D gel stained by this procedure is shown in Fig. 1.

2. Materials

1. All water used must be distilled and deionized.
2. Gel fixation solution: Trichloroacetic acid (TCA) solution, 20% (w/v).
3. Silver diamine solution: 21 mL of 0.36% (w/v) NaOH is added to 1.4 mL of 35% (w/v) ammonia and then 4 mL of 20% (w/v) silver nitrate is added drop-wise with stirring. When the mixture fails to

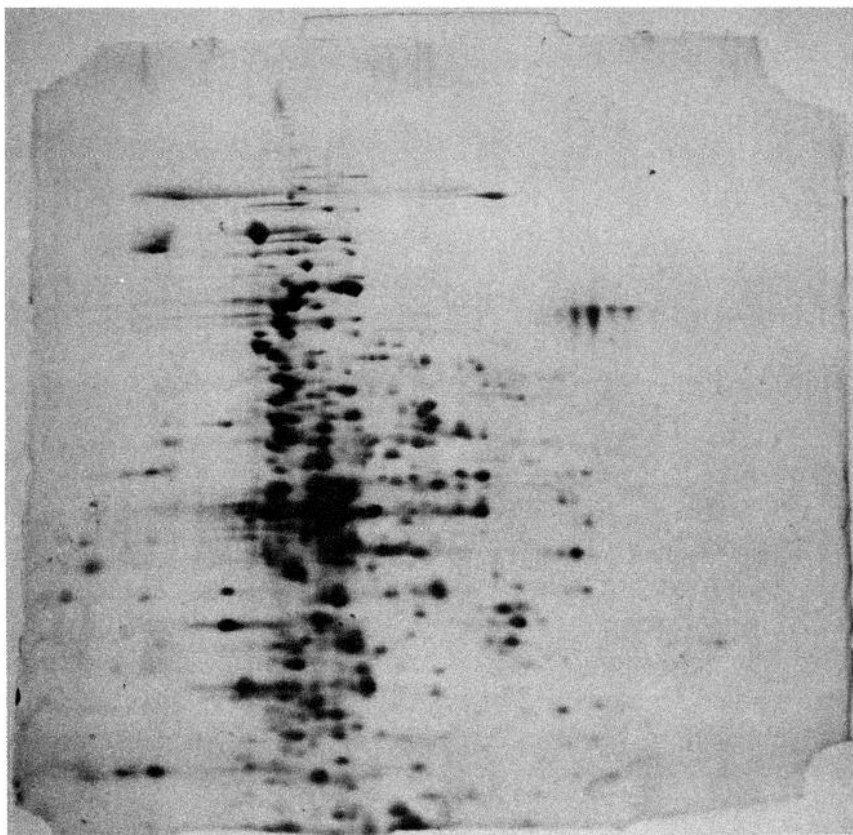


Fig. 1. Two-dimensional gel of bacterial proteins stained with the silver staining procedure. A horizontal, flat-bed pH 3–10 isoelectric focusing gel was used in the first dimension, and an 8–20% gradient polyacrylamide SDS-PAGE gel was employed in the second dimension.

clear with the formation of a brown precipitate, further addition of a minimum amount of ammonia results in the dissolution of the precipitate. The solution is made up to 100 mL with water. The silver diamine solution is unstable and should be used within 5 min.

4. Reducing solution: 2.5 mL of 1% citric acid, 0.26 mL of 36% (w/v) formaldehyde made up to 500 mL with water.
5. Farmer's reducer: 0.3% (w/v) potassium ferricyanide, 0.6% (w/v) sodium thiosulfate, 0.1% (w/v) sodium carbonate.
6. Clearing solution: 100 mL of 1.5M sodium thiosulfate mixed with an equal volume of a solution of 0.15M copper sulfate, 0.6M sodium chloride, and 0.9M ammonium hydroxide.

3. Method

3.1. Fixation

1. After electrophoresis the gel is immediately fixed in 200 mL of TCA for a minimum of 1 h at room temperature. High-percentage polyacrylamide and thick gels require an increased period for fixation, and overnight soaking is recommended.
2. The gel is placed in 200 mL of 50% (v/v) methanol and soaked for 2 x 30 min with agitation.
3. The gel is washed in excess water for 2 x 20 min, facilitating the rehydration of the gel and the removal of methanol. An indication of rehydration is the loss of the hydrophobic nature of the gel.

3.2. Staining

1. The gels are soaked in the silver diamine solution for 15 min with constant agitation. For thick gels (1.5 mm upward), it is necessary to use increased volumes so that the gels are totally immersed. Caution should be exercised in disposal of the

ammoniacal silver reagent, since it decomposes on standing and may become explosive. The ammoniacal silver reagent should be treated with dilute hydrochloric acid (1N) prior to disposal.

2. The gel is washed (2 x 5 min) in water.
3. The gel is placed in reducing solution. Proteins are visualized within 10 min, after which the background will gradually increase. It is important to note that this reaction displays inertia, and that staining will continue for 2–4 min after removal of the gel from the reducer. Proteins are visualized from 8 min until 15 min, after which background becomes unacceptable.
4. The gels are rinsed in water for 2 min.
5. Stain development is terminated by immersing the gel in 45% (w/v) methanol, 10% (v/v) acetic acid. It is important to note that staining displays inertia and will continue for a further 2–4 min after removal of the gels from reducing solution.

If staining is of low sensitivity, it can be intensified by recycling. Stained gels are washed in water (5 min) and returned to fixation in methanol (section 3.1, step 2), and the whole process repeated.

3.3. Destaining

1. Selective destaining: Gels are washed in water (5 min) to remove the stop solution. For the controlled removal of background staining that obscures proper interpretation of the protein pattern, partial destaining of gels using Farmer's reducing reagent (9) is recommended. The gels are placed in Farmer's reducer for a time dependent upon the intensity

of the background. Destaining is terminated by returning gels to the stop solution.

2. Complete destaining: Gels that are totally unusable because of overstaining can be efficiently destained prior to restaining by immersion in the clearing solution. Once the gel has cleared it must be washed extensively in water (3 x 30 min) prior to return to 50% (v/v) methanol (section 3.1, step 2) and recycled.

4. Notes

1. Gloves should be worn at all stages when handling gels, since silver staining will detect skin and sweat proteins present on fingers.
2. Volumes of the solutions used at all stages should be sufficient such that the gels are totally immersed. If the volume of solution is insufficient for total immersion, staining tends to be uneven and leads to gel surface drying.
3. All solutions for the staining and destaining steps should be freshly prepared, and overnight storage is not recommended. Fixing solutions can be stored.
4. All solutions must be prepared using clean glassware and deionized, distilled water.
5. An inherent problem with the staining of gradient SDS-PAGE gels is uneven staining along the concentration gradient. The less concentrated polyacrylamide region develops background staining prior to the more concentrated region. A partial solution to this problem is to increase the time of staining in silver diamine (*see* section 3.2, step 1).
6. Various chemicals used in one- and two-dimensional procedures will inhibit staining, whereas

others impair resolution or produce artifacts. Acetic acid will inhibit staining and should be completely removed prior to the addition of silver diamine solution (section 3.2, step 1). Glycerol, used to stabilize SDS gradient gels during casting, and urea, used as a denaturing agent in isoelectric focusing, are removed by water washes. Agarose, often used to embed rod IEF gels onto SDS-PAGE gels in 2-D PAGE procedures, contains proteins that are detected by silver staining as diffuse bands and give a strong background. Tris and glycine present in electrophoresis buffers complexes with silver (17) and must be washed out with water. The use of 2-mercaptoethanol as a disulfide bond reducer should be avoided since it leads to the appearance of two artifactual bands at 50 and 67 dalton on the gel (18).

7. Using this method, most proteins appear as dark brown to black spots or bands. Certain species, however, stain to give yellow spots regardless of protein concentration. The phenomenon of color variation in silver staining has been linked to the post-translational modification of the proteins. Glycosylation and the addition of lipid groups have been known to affect the staining intensities of proteins as well as their *pI* values.
8. Radioactively labeled proteins can be detected by silver staining prior to autoradiography or fluorography for the majority of the commonly used isotopes (^{14}C , ^{35}S , ^{32}P , ^{125}I). In the case of ^3H , however, silver deposition will absorb most of the emitted radiation.

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Chapter 14

Chromatofocusing

Ake Sidén and Paolo Gallo

1. Introduction

Investigations of different protein species in complex biological fluids, e.g., cerebrospinal fluid (CSF) and serum, are important tasks in clinical laboratory work as well as in biochemical research. One crucial part of such investigations is the application of a suitable method in order to separate the individual proteins and/or resolve a certain protein into subtypes or microheterogeneous components. There are two major categories of procedures that are used to achieve these separations. One is represented by electromigration techniques, such as electrophoresis or isoelectric focusing, which separate the sample molecules according to size or charge; an immunoelectrophoretic step may also be used, i.e., a separation according to antigenicity. The second major category is formed by the liquid chromatographies, which give a partition of the sample molecules on the

basis of size, charge, hydrophobicity, or binding to a biospecific ligand.

Chromatofocusing is a liquid chromatography that separates biomolecules on the basis of their respective isoelectric point (pI), i.e., it is a chromatographic analog of isoelectric focusing. This procedure was initially described in 1976 by Sluyterman and Wijdenes (1), and later more detailed basic data were published (2–4). The method is based on the interaction of an ion exchanger and a suitable buffer, which results in the formation of a pH gradient without the use of any gradient mixer. Accordingly, proteins that have been bound to the ion exchanger are eluted and focused in order of the pI values. The following steps form the basis of the procedure:

1. Choice of the optimal pH gradient for the protein(s) of interest.
2. Selection of start and eluent buffers on the basis of the above decision.
3. Sample preparation and application onto the ion exchanger, which has been equilibrated with start buffer.
4. Performance of the separation by running the eluent buffer through the ion exchanger.
5. Washing and regeneration of the ion exchanger followed by equilibration with start buffer.

The present description of chromatofocusing is based on experiences with the FPLC separation media and instruments. References 5–7 give further basic theoretical and practical data for the methodology, as well as different applications. The chromatofocusing column contains an anion exchanger prepared from a monodisperse beaded hydrophilic resin carrying tertiary and quaternary amines as charged groups. The sample

is applied onto the column at conditions giving a negative charge to the molecules of interest, i.e., at a pH value above the pI of these latter components. Consequently, these sample molecules are bound to the positive charges of the ion exchanger. The eluent contains a mixture of amphoteric buffers with different pK_a values and has been adjusted to a pH value below the pI of the molecules that are to be separated. During the elution, these amphoteric species interact with the anion exchanger that results in the formation of a pH gradient within the column and a gradually decreasing pH of the eluate. The lowering of the pH at a certain level of the column is more pronounced than at a level further down, i.e., the pH increases down the length of the column until the whole column is in equilibrium with the eluent buffer. The net result of these conditions will be an elution and focusing of the sample molecules in order of their respective pI values (components with a higher pI being eluted before those with a lower pI).

The following sections will give basic methodological data for the use of chromatofocusing in studies of CSF and serum proteins, predominantly immunoglobulin G (IgG). The results from such separations have already been presented in more detail (8–12). Because of the application for CSF protein separations, these methods were designed to be used for samples with relatively low protein amounts.

2. Materials

1. All chemicals and solvents, including the water, should be of HPLC or analytical grade, and their compatibilities with the system should be checked. The buffers and other solutions run through the column have to be filtered through a sterile 0.2- μ m

filter and degased. Furthermore, they should be kept at 3–8°C when not used and prior to use should be carefully inspected with regard to microbial contamination.

2. Buffers:

Start buffer: 0.025M diethanolamine, pH 9.5.

Eluent buffers:

10% (v/v) Polybuffer 96 (Pharmacia, Uppsala, Sweden), pH 6.0.

10% (v/v) Polybuffer 74 (Pharmacia), pH 4.0.

The buffers are titrated to their respective pH values with hydrochloric acid prior to the addition of the final few milliliters of water up to the total volume. The Polybuffer eluents should be protected from light.

3. Solutions for column washing and regeneration:

2M Sodium chloride

70% (v/v) Acetic acid

4. Chromatofocusing columns and equipment (Pharmacia):

1 Mono P HR 5/20 (5 mm x 20 cm) column

1 Small chromatofocusing column^a

2 Prefilters

1 GP-250 gradient programmer^b

2 P-500 Pumps

1 0.6-mL Mixer

1 V-7 Valve

2 V-8 Valves^c

1 UV-1 Single-path monitor

1 HR Flow cell

1 280 nm Filter kit

1 REC-482 Two-channel recorder

1 FRAC-100 Fraction collector

1 PSV-100 Valve^c

1 pH Monitor^d

- 1 Standard pH electrode^d
- 1 Flow-through pH electrode^e
- 1 Chromatography rack
- 1 10-mL Superloop
- Sample loops 0.5 and 1.0 mL
- Cable and flanging/start up kits

^dNot commercially available, it was prepared by the manufacturer of the equipment according to instructions from the authors. a HR 5/5 (5 mm x 5 cm.) glass column packed with 1 mL of the same medium as the Mono P column.

^eAlternatively, a LCC-500 control unit; besides carrying out the same functions as the GP-250, the LCC-500 may also be used to control automated multisample/multidimensional separations. Several further components are necessary, however, in order to make full use of this control unit

^fOptional, the V-8 valves make it possible to connect several columns to the system, the PSV-100 valve increases the flexibility of the FRAC-100 fraction collector, and the flow-through pH electrode simplifies the registration of the pH gradient.

^gCheck whether suitable components are already available in your laboratory.

3. Method

The method is based on the following procedures: equipment preparation, sample preparation, sample loading and running, column washing and regeneration, and result evaluation. Instrument set-up, calibration, and programming, as well as column preparation for first-time use, are performed according to instructions from the manufacturer. The prefilters are connected to the system in such a way that they protect the column(s). A schematic system design is given in Fig. 1.

Three different chromatofocusing procedures will be described in section 3.3. The first of these methods (section 3.3.1) is based on the use of a commercially available column and a pH gradient suited for most IgG components. The second technique (section 3.3.2) util-

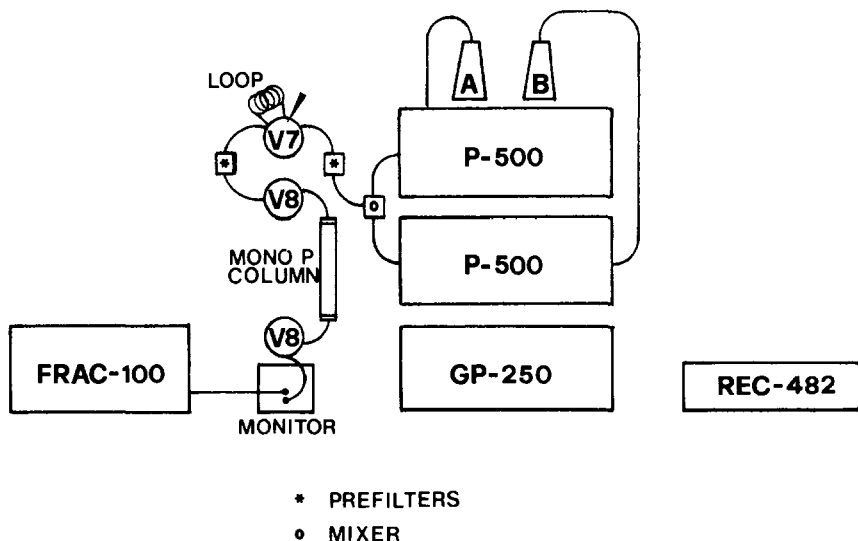


Fig. 1. Instrument set-up when two V-8 valves are used. In systems without such valves, the column is introduced (with the second prefilter) between the V-7 valve and the UV-1 monitor. A and B: reservoirs for the buffers. The sample loop is connected to the V-7 valve and loaded with a syringe via a 0.5-mm outer diameter, blunt-tipped needle (arrowhead).

izes the same type of pH gradient, but the separations are run on a small column. Finally, a method for the use of a gradient covering alkaline, as well as acid pH values, is described (section 3.3.3).

3.1. Equipment Preparation

1. Program the chromatofocusing method(s) as well as the recording and fraction collection modes into the equipment. The basic parameters are specified in section 3.3.
2. Fill the reference cell of the UV-1 single-path monitor with the Polybuffer eluent (10% Polybuffer 96,